

Date: 5/18/16

**PROFESSIONAL SERVICES AGREEMENT
BETWEEN
THE CITY OF CHEYENNE AND SMITH ENVIRONMENTAL AND ENGINEERING
FOR LOWER CAPITOL BASIN SEDIMENT TRAP/WETLAND
AKA "PUMP HOUSE PARK"**

This Agreement is made this _____ day of _____, 2016, by and between the City of Cheyenne, a municipal corporation organized under the laws of the State of Wyoming, 2101 O'Neil Avenue, Cheyenne, Wyoming 82001 ("City"), and Smith Environmental and Engineering, a Colorado corporation, 1490 West 121st Avenue, Suite 101, Westminster, Colorado 80234 ("Firm").

WITNESSETH

WHEREAS, the City, acting through the City Engineer, is responsible for the management of storm water drainage within the City of Cheyenne; and

WHEREAS, on April 1, 2016, the City issued Request for Proposals (#12-16), attached hereto as Exhibit "A", for construction documents, bidding assistance and construction administration services for the Lower Capitol Basin Sediment Trap/Wetland aka "Pump House Park"; and

WHEREAS, the Firm is qualified to provide the professional services desired by the City pursuant to the Request for Proposals; and

WHEREAS, the Firm has submitted a technical proposal dated April 22, 2016, attached hereto as Exhibit "B", that describes the scope of work to be performed by the Firm and a price proposal dated April 22, 2016. The Firm has submitted an updated Project Schedule, attached hereto as Exhibit "C", that specifies the work plan for the performance of work as described in Exhibits "A" and "B". The Firm has also submitted an Exhibit "D" that amends the price proposal originally submitted to the City as part of Exhibit "B". Exhibits "A", "B", "C", and "D" are incorporated into this Agreement by this reference; and

WHEREAS, the City desires to employ the Firm under the terms and conditions hereinafter set forth.

NOW, THEREFORE, in consideration of the mutual promises and covenants herein contained, it is agreed by and between the City and the Firm as follows:

1. SCOPE OF WORK:

The Firm agrees to provide professional services for the City in accordance with RFP-12-16, attached hereto as Exhibit "A", and in accordance with its Response to said RFP, attached hereto as Exhibit "B". The Project Schedule is attached hereto as Exhibit "C". The Scope of Work to be performed by Firm shall consist of Exhibits "A", "B", and "C". The City may order major changes

in the scope or character of work, either decreasing or increasing the amount of the Firm's services. In the event that such changes are ordered, the Firm shall be entitled to additional compensation for additional work performed. Additional compensation for major changes shall be determined in accordance with Paragraph 2 of this Agreement, but in no event shall the City be liable for payment unless the amount of such additional compensation shall first have been agreed to in writing by the City. In the event major changes are ordered pursuant to this Section, the schedule for progress and completion in Section 3 of this Agreement and the compensation to be paid pursuant to Section 2 shall be adjusted by negotiation between the Firm and the City pursuant to Section 4 of this Agreement.

2. COMPENSATION:

The Firm shall perform the services set forth in Exhibits "A", "B", and "C" for a cost not to exceed Three Hundred Ninety-Two Thousand One Hundred Seventy-Six Dollars and Zero Cents (\$392,176.00) as set forth in Exhibit "D" attached hereto. The Firm shall bill the City on a monthly basis. The Firm's invoices, and supporting documentation, shall contain all of the information necessary to enable the City to comply with the record keeping, billing, quarterly reporting, and environmental monitoring requirements of the cooperative agreement between the City and the Department of Environmental Quality of the State of Wyoming, dated October 14, 2013 (City Contract No. 6070). Upon acceptance and approval of work performed in accordance with the Scope of Work described in Exhibits "A", "B", and "C", the City shall make payment to the Firm within forty-five (45) days of receipt of said invoice. The Firm shall maintain hourly records of the time worked by its personnel and records of direct project expenses to support any audits the City may require and make those records available to the City at the City's request. Such records shall be maintained by the Firm for five (5) years following the date of final payment. The Firm shall ensure that its invoices and supporting documentation clearly and specifically identify fees, costs, and charges that are not properly reimbursable under City Contract No. 6070 for this project.

3. TERM AND EFFECTIVE DATE:

This Agreement shall commence on the date the last required signature is attached hereto and shall continue in full force and effect until the Scope of Work is completed in accordance with Exhibits "A", "B", and "C", unless this Agreement is earlier terminated pursuant to its provisions.

4. MODIFICATIONS:

Any changes, modifications, revisions or amendments to this Agreement which are ordered pursuant to Section 1 herein or which are mutually agreed upon by the parties to this Agreement shall be incorporated by written instrument, executed and signed by all parties to this Agreement.

5. INDEMNIFICATION/HOLD HARMLESS/TRADE SECRETS/COPYRIGHT:

a. The Firm agrees to indemnify, hold harmless and defend the City from and against any and all liabilities, claims, penalties, forfeitures and suits, and the cost and expenses incident thereto, including reasonable attorney's fees, which may hereafter arise as

a result of death or bodily injury to any person, destruction or damage to any property, contamination of or adverse effects on the environment, or any violation of governmental laws, regulations or orders to the extent caused by (1) the Firm's breach of any term or provision of this Agreement; or (2) any negligent or wrongful act, error or omission by the Firm, or its employees or subcontractors in the performance of this Agreement. The Firm acknowledges that it may incur a financial obligation to the City pursuant to the terms of this paragraph.

b. The Firm shall also indemnify, defend and hold the City harmless from any and all claims, demands and causes of action that are asserted against the City as the result of the Firm's failure, neglect or refusal to clearly and specifically identify fees, costs, and charges that are not properly reimbursable under City Contract No. 6070 for this project. This sub-section is intended to impose a duty on the Firm to maintain accurate and complete records and to prepare its invoices in such a manner as to protect the City from liability that may result from having sought, based on the Firm's representations or warranties, any reimbursement that is denied at any time, in whole or in part, by the Wyoming Department of Environmental Quality.

c. The Firm agrees to indemnify, pay the defense costs of, and hold the City harmless from any and all claims, demands, costs, liabilities, losses, expenses and damages (including attorneys' fees, costs, and expert witnesses' fees) arising out of or in connection with this Agreement that sound in an intellectual property claim (including, but not limited to, patent, copyright, trademark, trade name, or trade secret infringement). This Section specifically survives the expiration or termination of this Agreement.

6. GOVERNMENTAL IMMUNITY:

The City and its officials and employees do not waive governmental immunity by entering into this Agreement and specifically retain all immunities and defenses available to them as Governmental Entities pursuant to Wyo. Stat. § 1-39-101, *et seq.*, and all other applicable laws, except to the extent necessary solely for the enforcement of the terms and conditions of this Agreement as between the parties. Further, the City and its officers and employees fully retain all immunities and defenses provided by law with regard to any action, whether in tort, contract or any other theory of law, based on this Agreement. Designation of venue, choice of law and similar provisions should not be construed as a waiver of governmental immunity.

7. INSURANCE:

The Firm must provide proof of the following insurance coverages:

a. Commercial General Liability Insurance:

For claims arising out of bodily injury, illness or death, or from damage to or destruction of property of others, including loss or use thereof, with minimum limits of \$1,000,000 per occurrence, \$2,000,000 aggregate for the entire term of the Agreement.

b. Business Automobile Insurance:

Including owned, non-owned and hired vehicles with minimum limits for bodily injury and property damage of \$1,000,000 per accident for the entire term of the Agreement.

c. Workers' Compensation:

Workers' Compensation coverage shall be in effect for the entire term of the Agreement, as required by Wyoming law, for all employees or agents providing services under this Agreement. Firm shall provide the City with proof of workers' compensation or employer's liability insurance coverage.

d. Professional Liability Insurance:

The Firm shall provide proof of professional liability insurance or errors and omissions liability insurance in an amount not less than \$500,000 to protect the City from any and all claims arising from the Firm's negligence in the performance of duties under this Agreement.

The City prefers that this liability insurance coverage be provided pursuant to an "occurrence" policy. If this coverage is provided pursuant to a "claims made" policy:

(i) Firm, shall, concurrently with the execution of this Agreement, provide the City with a certificate of insurance demonstrating that such coverage is or shall be in effect at the time the Firm begins the provision of services under this Agreement; and

(ii) In the event the Firm's services extend into a future policy period, the Firm shall, prior to the policy expiration date, provide the City with a new certificate of insurance demonstrating that such coverage is or shall be in effect during all periods of time that Firm will provide services under this Agreement; and

(iii) Firm shall maintain said "claims made" coverage for a period of five (5) years following the last date that Firm has provided services under this Agreement; and

(iv) In the event the Firm or the Firm's professional liability insurer terminates "claims made" coverage prior to the expiration of the periods provided in subparagraphs (i), (ii), or (iii) of this paragraph, the Firm shall provide to the City advance written notification of the termination of said coverage and shall provide the City with an endorsement for an extended reporting period ("tail coverage") which shall be in effect for a period of time not less than five (5) years following the last date that Firm has provided services under this Agreement.

e. Additional Insurance Information:

The Firm shall name the City of Cheyenne as an Additional Insured by endorsement on its insurance policies, with the exception of workers' compensation and professional liability insurance, and shall provide the City with a copy of the endorsements.

Firm shall provide the City with certificates of insurance acknowledging the above-stated coverages prior to beginning any work under this Agreement.

It is understood and agreed that these policies are primary and not contributory. All policies required under this Agreement shall be in effect for the duration of the Agreement. It shall be an affirmative obligation upon Firm to immediately notify in writing the city risk manager, city clerk, and city attorney of any fact, circumstance, or occurrence that has resulted in or may result in the cancellation or substantive change of any insurance coverage required by this Agreement, and failure to do so shall be construed to be a breach of this Agreement.

In addition, Firm shall provide the City with copies of insurance policies and policy endorsements listing the City of Cheyenne as an additional insured. The City's failure to request or review such policies, endorsements, or certificates shall not affect the City's rights or Firm's obligation hereunder.

Any insurance company providing coverage under this agreement shall have a minimum A. M. Best rating of A- (excellent).

8. GOVERNING LAW/JURISDICTION/VENUE:

The construction, interpretation and enforcement of this Agreement shall be governed by the laws of the State of Wyoming. The courts of the State of Wyoming shall have jurisdiction over any action arising out of this Agreement and over the parties, and the venue shall be the First Judicial District, Laramie County, Wyoming.

9. COMPLIANCE WITH LAWS:

The Firm shall comply with all applicable federal, state and local laws, rules and regulations in the performance of this Agreement, including federal and state anti-discrimination laws. The Firm shall be responsible for all such non-compliance and shall defend, hold harmless and indemnify the City therefrom.

10. CONTRACT WORK HOURS AND SAFETY STANDARDS ACT:

The Firm agrees to comply with Section 102 and 107 of the Contract Work Hours and Safety Standards Act (40 U.S.C. 327-330) as supplemented by Department of Labor Regulations (29 C.F.R. Part 5).

11. NONDISCRIMINATION:

The Firm shall comply with the Civil Rights Act of 1964, the Wyoming Fair Employment Practices Act (Wyo. Stat. § 27-9-105, *et seq.*), the Americans With Disabilities Act (ADA), 42 U.S.C. § 12101, *et seq.*, and the Age Discrimination Act of 1975 and any properly promulgated rules and regulations thereto, and shall not discriminate against any individual on the grounds of age, sex, creed, color, race, religion, national origin, ancestry, pregnancy or qualifying disability in connection with the performance under this Agreement.

12. EQUAL OPPORTUNITY EMPLOYER:

The Firm shall comply with Executive Order 11246 of September 24, 1965, entitled "Equal Employment Opportunity," as amended by Executive Order 11375, of October 13, 1967, and as supplemented in Department of Labor regulations (41 C.F.R. Chapter 60).

13. PATENT RIGHTS:

The Firm agrees to comply with Environmental Protection Agency requirements and regulations for grantees and sub-recipients pertaining to patent rights with respect to any discovery or invention which arises or is developed in the course of or under this Agreement.

14. ALCOHOL AND CONTROLLED SUBSTANCE POLICY:

In compliance with the Drug-Free Workplace Act of November 1988, the City has established an Alcohol and Controlled Substance Policy that pertains to alcohol and drug usage by City employees. All parties under contract with the City, including the Firm and its employees and subcontractors, are required to comply with the provisions of the City's Alcohol and Controlled Substance Policy that pertains to drug and alcohol usage on City property and other sites occupied by the Firm while performing duties and responsibilities of this Agreement. It is the responsibility of the Firm to familiarize itself with the requirements of this policy and to inform all its employees and subcontractors of those requirements and to insure their compliance therewith. If the Firm, its employees or subcontractors are found to be in violation of this policy, this Agreement may be terminated.

15. INDEPENDENT CONTRACTOR:

At all times during the term of this Agreement, the Firm shall be considered an independent contractor. Neither the Firm nor any one employed by it shall represent, act, purport to act or be deemed to be the agent, representative, employee or servant of the City.

16. PERSONNEL:

The Firm agrees that it will employ at its own expense, all personnel necessary to perform the services required by this Agreement and in no event shall such personnel be employees of the City. All of the services required hereunder shall be performed by the Firm and all personnel

engaged therein shall be fully qualified under applicable federal, state and local law to undertake the work performed by them. The Firm assumes full and sole responsibility for the payment of all compensation and expenses of such personnel and for all state and federal income tax, unemployment insurance, Social Security, disability insurance and other applicable withholdings.

17. CONFIDENTIALITY:

To the extent allowed by law, the City and the Firm shall treat as confidential and not disclose to others information (including technical information, experience or data) regarding either party's plans, programs, plants, processes, products, costs, equipment, operations, or customers which come within the knowledge of the parties, without in each instance securing the prior written consent of the other party, unless such disclosure is required by law or legal process. However, nothing shall prevent either the Firm or the City from disclosing to others or using in any manner information which either party can show (a) has been published or has become part of the public domain other than by acts of the Firm or the City; (b) has been furnished or made known to the Firm or the City by third parties without restrictions on its disclosure; or (c) was in either party's possession prior to the disclosure thereof by the City or the Firm to each other. Neither the Firm nor the City shall be restricted from releasing information in response to a subpoena, court order, or legal process, but, in the event such disclosure becomes necessary, the disclosing party shall notify the other party of the demand for information at least seven (7) days prior to disclosing such information so that the other party may take any and all steps provided by law to prevent the release of such information.

18. FIRM'S RESPONSIBILITIES:

The Firm shall be responsible for all terms and conditions as set forth in this Agreement and the professional quality, technical accuracy, and timely completion of all services as set forth in Exhibits "A", "B", and "C", and all changes in the scope or character of the work made pursuant to Sections 1 or 4 herein. The Firm shall, without additional compensation, promptly remedy and correct any errors, omissions, or other deficiencies. The Firm shall perform all work in a professional manner in conformance with industry standards. The Firm shall assume the risk of any loss of state or federal funding, either administrative or program dollars, due to its failure to comply with state or local requirements. The City shall notify the Firm of any state or federal determination of noncompliance.

19. ENVIRONMENTAL POLICY ACTS:

The Firm agrees all activities under this Agreement will comply with the Clean Air Act, the Clean Water Act, the National Environmental Policy Act, and other related provisions of federal environmental protection laws, rules or regulations. The Firm shall perform all environmental monitoring and reporting required of the City under City Contract No. 6070.

20. HUMAN TRAFFICKING:

As required by 22 U.S.C. § 7104(g) and 2 C.F.R. Part 175, this Agreement may be terminated without penalty if a private entity that receives funds under this Agreement:

- a. Engages in severe forms of trafficking in persons during the period of time that the award is in effect;
- b. Procures a commercial sex act during the period of time that the award is in effect; or
- c. Uses forced labor in the performance of the award or sub-awards under the award.

21. KICKBACKS:

Firm certifies and warrants that no gratuities, kickbacks, or contingency fees were paid in connection with this Agreement, nor were any fees, commissions, gifts, or other considerations made contingent upon the award of this Agreement. If the Firm breaches or violates this warranty, City may, at its discretion, terminate this Agreement without liability to City, and deduct from the agreed upon price or consideration, or otherwise recover, the full amount of any commission, percentage, brokerage, or contingency fee.

22. MONITORING ACTIVITIES:

The City shall have the right to monitor all activities related to this Agreement that are performed by the Firm or its subconsultants. This shall include, but not be limited to, the right to make site inspections at any time and with reasonable notice; to bring experts and Firms on site to examine or evaluate completed work or work in progress; to examine the books, ledgers, documents, papers, and records pertinent to this Agreement; and to observe personnel in every phase of performance of the related work.

23. SUSPENSION AND DEBARMENT:

By signing this Agreement, the Firm certifies that neither it nor its principals are presently debarred, suspended, proposed for debarment, declared ineligible, or voluntarily excluded from participation in this transaction nor from federal financial or non-financial assistance, nor are any of the participants involved in the execution of this Agreement suspended, debarred, or voluntarily excluded by any federal department or agency in accordance with Executive Order 12549 (Debarment and Suspension) or 44 C.F.R. Part 17, or are on the debarred vendors list at www.sam.gov. Further, the Firm agrees to notify City by certified mail should it or any of its officers, employees or agents become debarred, suspended, or voluntarily excluded during the term of this Agreement.

24. NO GOVERNMENT OBLIGATION TO THIRD PARTIES:

The Firm acknowledges and agrees that, notwithstanding any concurrence by the Federal Government in or approval of the solicitation or award of this Agreement, absent the express written consent by the Federal Government, the Federal Government is not a party to this Agreement and shall not be subject to any obligations or liabilities to the Firm, or any other party (whether or not a party to this Agreement) as to any matter resulting from this Agreement.

25. PROGRAM FRAUD AND FALSE OR FRAUDULENT STATEMENTS AND RELATED ACTS:

a. The Firm acknowledges that the provisions of the Program Fraud Civil Remedies Act of 1986, as amended, 31 U.S.C. § 3801, *et seq.*, and "Program Fraud Civil Remedies," 49 C.F.R. Part 31, apply to its actions under this Agreement. Upon execution of this Agreement, the Firm certifies or affirms the truthfulness and accuracy of each statement it has made, it makes, it may make, or causes to be made, pertaining to the performance of this Agreement. In addition to other penalties that may be applicable, the Firm further acknowledges that if it makes, or causes to be made, a false, fictitious, or fraudulent claim, statement, submission, or certification, the Federal Government reserves the right to impose the penalties of the Program Fraud Civil Remedies Act of 1986 on the Firm to the extent the Federal Government deems appropriate.

26. LOBBYING:

In accordance with Public Law 101-121, payments made from a federal grant shall not be utilized by the City or the Firm in connection with lobbying Congressmen, or any other federal agency in connection with the award of a federal grant, contract, cooperative agreement or loan. Further, the Firm may not use any federal funds, either directly or indirectly, to support the enactment, repeal, modification or adoption of any law, regulation or policy at any level of government, without the express prior written approval of U. S. Department of Homeland Security.

27. PRIVACY ACT:

5 U.S.C. § 552 (*Applicability: The Federal Privacy Act requirements flow down to each third party contractor and their contracts at every tier*). The following requirements apply to the Firm and its employees that administer any system of records on behalf of the Federal Government under any contract:

a. The Firm agrees to comply with, and assures that its employees will comply with the information restrictions and other applicable requirements of the Privacy Act of 1974, 5 U.S.C. § 552a. Among other things, the Firm agrees to obtain the express consent of the Federal Government before the Firm or its employees operate a system of records on behalf of the Federal Government. The Firm understands that the requirements of the Privacy Act, including the civil and criminal penalties for violations of that Act, apply to all individuals involved, and that failure to comply with the terms of the Privacy Act may result

in termination of the underlying contract.

b. The Firm also agrees to include these requirements in each subcontract to administer any system of records on behalf of the Federal Government financed in whole or in part with Federal assistance.

28. MINORITY AND DISADVANTAGED BUSINESS ENTERPRISES (49 C.F.R. Part 26):

All parties to this Agreement assure that no person will be excluded from participation in, denied the benefits of, or otherwise discriminated against, in connection with the award and performance of this Agreement on the grounds of age, race, color, disability, national origin or sex.

29. NEPA:

The Firm agrees to comply with the National Environmental Policy Act (NEPA). Grant activities requiring specific documentation of NEPA compliance (see DHS Management Directive 5100.1) must be approved by the Federal Emergency Management Agency of the U.S. Department of Homeland Security prior to obligation or expenditure of grant funds by the Firm.

30. NOTICES:

Any notice, correspondence or billing required to be given by the terms of this Agreement shall be delivered by hand, or delivered by mail, postage prepaid, to the addresses of the respective parties stated below:

City: Brandon Cammarata, Director
Development Department
2101 O'Neil Avenue
Cheyenne, WY 82001

Firm: Smith Environmental and Engineering
1490 West 121st Avenue, Suite 101
Westminster, CO 80234
ATTN: Peter L. Smith, Vice-President

31. INTELLECTUAL PROPERTY AND WORK PRODUCT:

a. All work (preliminary, draft, and final) performed by the Firm under this Agreement is the property of the City. The City will own any and all data, documents, working papers, computer programs, photographs, and other material produced by the Firm pursuant to this Agreement, and the Firm hereby assigns and transfers to the City any and all intellectual property rights for such materials. The Firm will provide the City with copies of all such materials including pending disputes between the City and the Firm. Under no circumstances, will the Firm fail to deliver possession of said documents and materials to the

City upon demand.

b. The Firm agrees that all documents, designs, plans, specifications, and other publications created under this Agreement shall bear the following statement: "This document was prepared under a grant from the U.S. Department of Homeland Security (DHS) administered by the Wyoming Department of Homeland Security (WOHS). Points of view or opinions expressed in this document are those of the authors and do not necessarily represent the official position or policies of DHS, the State of Wyoming, or WOHS."

32. CONFLICT OF INTEREST:

In entering this Agreement, the Firm represents and warrants that its past work in connection with the City of Cheyenne "West Edge" project does not constitute a conflict of interest that would render any of its services under this Agreement non-reimbursable under City Contract No. 6070 or the federal regulations that are incorporated into said contract. The Firm further represents and warrants it presently has no interest, and shall not acquire any interest, direct or indirect, financial or otherwise, which would conflict in any manner or degree with performance of the services hereunder. In addition, the Firm covenants that in the performance of this Agreement, no subcontractor or person having such an interest shall be employed. The Firm certifies that no one who has or will have any financial interest under this Agreement is an officer or employee of the City.

33. ACCEPTANCE NOT WAIVER:

The City's approval of drawings, plans, specifications, reports, and incidental work or materials furnished hereunder shall not in any way relieve the Firm from responsibility for the technical accuracy of the work. The City's approval or acceptance of, or payment for, any services shall not be construed to operate as a waiver of any of the City's rights under this Agreement or any of its legal rights under statute and common law arising out of the performance of this Agreement.

34. DEFAULT:

Each and every term and condition herein shall be deemed a material element of the Agreement. In the event either party shall fail or refuse to perform according to the terms of this Agreement, such party may be declared in default.

35. REMEDIES:

In the event a party declares the other party in default hereof, said party declaring default shall notify the defaulting party in writing and such defaulting party shall be allowed a period of fifteen (15) days to cure said default. In the event that the default remains uncorrected, the party not in default may elect to: (a) terminate the Agreement and seek damages; (b) treat the Agreement as continuing and require specific performance; or (c) avail itself of any other remedy at law or equity. In the event the Firm fails to strictly perform in accordance with this Agreement, the City may elect

to make good such deficiencies and charge the Firm therefore.

36. TERMINATION:

The City may, by written notice to Firm, terminate this Agreement in whole or in part by giving Firm five (5) calendar days' written notice. The effective date of termination shall be five (5) calendar days from the date of the notice of termination. Upon receipt of such notice of termination, the Firm shall:

a. Discontinue all services affected by the notice (unless the notice directs otherwise); and

b. Deliver to the City, within five (5) calendar days from the date of the notice of termination, all data, designs, drawings, specifications, reports, estimates, summaries and all other work that has been performed or is to be performed under this Agreement, whether completed or in process. In the event of termination, the City will pay the Firm for all labor and services supplied and accepted as of the effective date of termination.

37. WAIVER:

The waiver by either party of any term, condition or covenant, or breach of any term, condition or covenant, shall not constitute a waiver of any other term, condition or covenant, or breach thereof.

38. SEVERABILITY:

If any provision, section, subsection, sentence, clause, or phrase of this Agreement is invalidated by any court of competent jurisdiction, such holding shall not affect the validity of the remainder of this Agreement, which shall continue in full force and effect.

39. SUCCESSORS AND ASSIGNS:

All the terms, conditions, and provisions herein shall inure to the benefit of and be binding upon the parties hereto and their respective permitted successors and assigns.

40. ASSIGNMENT:

Neither party shall assign this Agreement without prior written consent of both parties. Any delegation or assignment shall not operate to relieve either party of its responsibilities hereunder. However, nothing in this Agreement shall prohibit the Firm from subcontracting to a qualified subcontractor.

41. GRANT NOT USED AS COLLATERAL:

The Firm may not use this Grant, or any portion thereof, for collateral for any financial obligation without the prior written permission of Homeland Security.

42. CONFIDENTIALITY OF INFORMATION:

Notwithstanding the release of records as required by the Wyoming Public Records Act, Wyo. Stat. § 16-4-201, *et seq.*, all documents data compilations, reports, computer programs, photographs and any other work provided to or produced by the Firm exclusive to the performance of the Grant, shall be kept confidential by any Firm unless written permission is granted by the City and Homeland Security for its release.

43. ETHICS:

Firm shall keep informed of and comply with the Wyoming Ethics and Disclosure Act (Wyo. Stat. § 9-13-101, *et seq.*) and any and all ethical standards governing the City and the Firm.

44. FORCE MAJEURE:

Neither party shall be liable for failure to perform under this Agreement if such failure to perform arises out of cases beyond the control and without fault or negligence of the nonperforming party. Such causes may include, but are not limited to, acts of God or the public enemy, fires, floods, epidemics, quarantine restrictions, freight embargoes, and unusually severe weather. This provision shall become effective only if the party failing to perform immediately notifies the other party of the extent and nature of the problem, limits delay in performance to that required by the event, and takes all reasonable steps to minimize delays. This provision shall not be effective unless the failure to perform is beyond the control and without the fault or negligence of the nonperforming party.

45. THIRD PARTY RIGHTS:

The Parties do not intend to create in any other individual or entity the status of third party beneficiary, and this Agreement shall not be construed so as to create such status. The rights, duties and obligations contained in this Agreement shall operate only between the parties to this Agreement, and shall inure solely to the benefit of the parties to this Agreement. The parties to this Agreement intend and expressly agree that only the parties signatory to this Agreement shall have any legal or equitable right to seek to enforce this Agreement, to seek any remedy arising out of a party's performance or failure to perform any term or condition of this Agreement, or to bring action for breach of this Agreement. This paragraph is not intended nor shall it be construed to waive any of the parties' immunities.

46. ENTIRE AGREEMENT OF THE PARTIES:

This Agreement, including Exhibits "A", "B", "C", and "D", contains the entire understanding of the parties. There are no other terms or conditions, written or oral, concerning or controlling this matter.

IN WITNESS WHEREOF, the parties to this Agreement, through their duly authorized representatives, have executed this Agreement on the days and dates set out below and certify that they have read, understood and agreed to the terms and conditions of this Agreement.

This Agreement is not binding on either party until approved by A&I Procurement and the Governor or the State of Wyoming or his designee as required by Wyo. Stat. § 9-2-1016(b)(iv).

The effective date of this Agreement is the date of the last signature affixed to this page.

CITY OF CHEYENNE

Date

By: _____
Richard L. Kaysen, Mayor

(SEAL)
ATTEST:

Carol Intlekofer, City Clerk

SMITH ENVIRONMENTAL AND ENGINEERING

Date

By: _____

Printed Name: _____

Title: _____

Approved as to
form only

D. White
Date: April 1, 2016

REQUEST FOR PROPOSAL (RFP-12-16)

for

**Construction Documents, Bidding Assistance and Construction Administration Services
for Lower Capitol Basin Sediment Trap/Wetland aka "Pump House Park" (rebid)**



CITY OF CHEYENNE, WYOMING

April 1, 2016

Issued by:

**City of Cheyenne, Wyoming
West Edge Project Management Team**

**Proposal Statements due:
3:00 PM – April 22, 2016**

at:

**City Purchasing Office, Room 104
Attn: Sara Vasquez, Purchasing Manager**

**2101 O'Neil Avenue
Cheyenne, WY 82001
(307) 637-6328**

EXHIBIT

tabbies

A

REQUEST FOR PROPOSALS No.: RFP-12-16

PROJECT NAME: Lower Capitol Basin Sediment Trap/Wetland

The City of Cheyenne ("CITY"), Wyoming, hereby requests proposals for construction engineering, sustainable ecological design, environmental remediation expertise, landscape design, construction observation, and project administration for the above named project. Proposers shall be professional firms with Engineers licensed by the Wyoming State Board of Examining Engineers and Landscape Architects licensed to provide services in the State of Wyoming. If the proposer is a non-resident business entity, the firm shall also be registered with the Secretary of State of Wyoming to do business in Wyoming.

Proposer shall submit, in writing, information outlined under Section B - PROPOSAL FORMAT section. The submittal shall address the selection criteria as listed herein and such other data as may be useful in evaluating the ability of the firm to perform the work required. Proposals should be short and concise.

Information contained in the Request for Proposal and in the proposal submitted by the winning consultant shall be incorporated by reference into and be considered part of the contract between the City and the team selected.

A. PROPOSAL SUBMITTAL

To be considered, proposals must be received by the CITY at the following location before 3 p.m. MST on April 22, 2016 at:

City of Cheyenne Purchasing Office
2101 O'Neil Avenue, Room 104
Cheyenne Wyoming 82001
Attn: Sara Vasquez, Purchasing Manager

Any proposal received after that time shall not be considered, but shall be returned unopened to the proposer. Electronic and fax submittals shall not be considered.

Proposers are advised that, consistent with provisions of the CITY's policy for the procurement of professional architectural and engineering services, the CITY shall use a two-step process for selecting a consultant through this solicitation. This process dictates that proposers divide their submittals in two parts: a *technical proposal* and a *price proposal*. Proposers shall submit their technical proposals and price proposals simultaneously on the date and time specified in this RFP, sealed in separate, clearly marked envelopes as follows:

Technical Proposal: One(1) PDF electronic copy of the technical proposal, included all required attachments, on a CD or flash drive; one (1) original of the technical proposal, bound together with all required attachments, and nine (9) copies shall be submitted in a sealed envelope and clearly marked with the following information: (a) the phrase "Lower Capitol Basin Sediment Trap/Wetland Technical Proposal"; (b) the date specified for receipt of proposals; and (c) the name and address of the proposer.

Price Proposal: Price proposals are to be submitted in accordance with the requirements of this section and as follows: Submit one (1) original copy, and nine (9) additional copies in a separate sealed envelope and clearly marked with the following information: (a) the phrase "Lower Capitol Basin Sediment Trap/Wetland Price Proposal"; (b) the date specified for receipt of proposals; and

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(c) the name and address of the proposer.

Proposers are advised that proposal content and completeness are most important and that clear and effective presentations are preferred, with unnecessarily elaborate, decorative, or extraneous materials discouraged. Each section within the proposal shall be clearly defined for easy reference. All blank spaces must be filled in (e.g., the number "0", the word "None", or the words "Not Applicable" or "NA" are to be inserted, where appropriate, on the forms).

B. PROPOSAL FORMAT

The proposer shall use the following format.

The proposer shall use the following format when preparing his proposal:

Technical Proposal (Envelope #1)

1. **Project Team** - Identify the project team members and the position of each member on the team, briefly outlining the responsibilities of each member. The offices of project team member not located in Cheyenne should be identified. Detailed resumes should be attached to the back of this section to identify the experience and qualifications of the individual team members.

If the proposer intends to use one or more sub-consultants to complete a portion of the project, all such firms must be listed on Attachment B-3. The proposer shall describe the relationship of each firm to the Project and explain past relationships between the proposer (the legally responsible entity) and each sub-consultant, and indicate which sub-consultants, if any, are DBE certified.

2. **Schedule** - The anticipated time to perform the required work using the tasks outlined under the SCOPE OF WORK. Identify the team members and the percentage of time that they will dedicate to each task. A bar graph is suggested for this section. Additional tasks or duties not addressed in "Exhibit A - Scope of Work", which, in its final form will become part of the Professional Service Agreement, should be discussed in paragraph form, under Section 7, Other Information. Attention is directed to "Exhibit C - Project Schedule" of the proposed Professional Services Agreement, draft copy attached.
3. **Project Area** - The proposer should show briefly and concisely his familiarity with the Lower Capitol Basin and the West Edge District. The proposer should have a thorough understanding of the West Edge vision for Pump House Park along with the various public infrastructure components that make up the West Edge and how they interact.
4. **Scope of Work** - All proposers will commit at a minimum to the services outlined in "Exhibit A - Scope of Work", which, in its final form will become part of the proposed Professional Services Agreement, draft copy attached. The City and proposer may supplement the draft Scope of Work with any additional work or clarification of work that is determined necessary for the successful completion of the project.

5. **References** - List at least three (3) similar projects that were completed during the preceding five

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(5) years by the members of the project team. Use additional sheets if more than three references are provided. Provide the following information for each reference: Client Name, Address, Contact Person, Telephone number, Project Dates, Project Description, Original Project Budget, Final Project Cost, and an explanation of any variation from original budget to final project cost.

6. **Conflict of Interest** - Please state whether the proposer, or any sub-contractor or sub-consultant, is currently performing professional services for, or has an on-going professional relationship with, any other person or legal entity whose interests may be adverse to the City of Cheyenne with respect to the scope of work to be performed by the proposer if selected pursuant to this RFP process. Please also disclose all other facts and circumstances relating to the proposer's past, present, and future work that create, or may appear to create a conflict of interest, between the proposer's performance of services for the City and the proposer's current professional obligations or relationships with any other person or legal entity.
7. **Other Information** - Any additional information the proposer feels would be useful to the committee in evaluating the proposal should be placed in this section.
8. **Project Specific Information** - The proposer should carefully review "Exhibit A - Scope of Work," which, in its final form, will become part of the Professional Services Agreement. This exhibit provides background information as well as directions necessary for proposal preparation.
9. **Proposal Submittal Checklist** - This form (Attachment B-1) must be completed and signed by the proposer and attached to the outside of the technical proposal envelope to ensure completeness of submittal.
10. **Consultant Eligibility Form** - Any individual or firm whose name appears on the U.S. Comptroller General's list of ineligible firms for federally financed and assisted projects shall be considered ineligible to submit a proposal. Each proposer is required to certify that it is not on the Comptroller General's list of ineligible firms, using Attachment B-4. This form must be completed and signed by the proposer and attached to the proposal.

The proposer must also require the entities it contracts with to comply with 49 CFR 29, subpart C, and include this requirement in subsequent covered transactions (i.e., the requirement flows down to subcontracts at all levels.)

11. **Non-Collusion Certification** - No proposer shall offer any gratuity, favor, or anything of monetary value to any officer, employee, agent, or the CITY for any purpose whatsoever. Each proposer is required to certify that it has not engaged in any conduct or practice with any other proposer(s) so as to restrict or eliminate competition or otherwise restrain trade, using Attachment B-5. This form must be completed and signed by the proposer and attached to the proposal. The CITY reserves the right to reject any proposal and to remove from mailing lists for future proposals the name of any individual or firm engaging in collusive or other unlawful practices.
12. **Restrictions on Lobbying Certification** - In the event that the proposal exceeds \$100,000, the certification regarding lobbying (Attachment B-6) must be completed and signed by the proposer,

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and attached to the proposal.

Any proposal not submitted in this format shall be deemed to be non-responsive and shall not be considered further by the CITY. The signer of the proposal must declare that the person(s), companies, or parties interested in the contract as principals are named therein; that it is, in all respects, fair, and in good faith without collusion or fraud; and that the signer of the proposal has the authority to bind the principal proposer.

Price Proposal (Envelope #2)

1. **Cover Letter** - Identify the proposer's name, mailing address, telephone number, and contact person(s). The letter must be signed by the proposer and should be limited to two (2) pages. The proposal shall be signed by a principal of the legal entity submitting the proposal and shall be considered responsible for the performance of any contract awarded pursuant to this RFP. A proposal shall be considered non-responsive if the proposal is submitted by any business entity that has limited financial resources and that is a wholly owned subsidiary or affiliate of another business entity.
2. **Provide a complete line item budget** for all Phases of the work listed in "Exhibit B COMPENSATION SCHEDULE," of the Scope of Work with a breakdown of each phase, as necessary to complete the Project as specified in the RFP, its attachments, and other documents referenced. Each proposer shall certify that its price proposal shall remain valid for a period of at least one hundred and twenty (120) days. In the event that a contract award by the CITY is delayed beyond the 120 day period, such award shall be conditioned upon the proposer's acceptance.

C. PRE-PROPOSAL CONFERENCE

A pre-proposal conference will be at the Municipal Building, 2101 O'Neil Ave., Room 122, on April 8, 2016, beginning at 9:30 a.m. MST, for the purpose of responding to questions or request for clarification regarding the contents of this RFP. Proposers are strongly encouraged to attend the pre-proposal meeting.

D. ADDENDA

Any and all proposers may make inquiries in writing via email at any time prior to noon on April 11, 2016. Any written question regarding the meaning or interpretation of the RFP, work scope, specifications, etc., must be submitted prior to the above specified date. The City shall make all reasonable efforts to provide clarifications to all proposers in summary form as one or more addenda to this RFP. No technical assistance shall be given by the CITY to any proposer in preparation of its proposal.

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PROJECT NAME: Lower Capitol Basin Sediment Trap/Wetland

Written inquiries shall be directed to:

Sara Vasquez,
City of Cheyenne Purchasing Manager
2101 O'Neil Avenue,
Cheyenne, WY 82001
svasquez@cheyennecity.org.

The City shall make reasonable efforts to provide clarifications in summary form as one or more addenda to this RFP. Each proposer must sign the "Acknowledgment of Receipt of RFP Packet, and Addenda" form "Attachment B-2" and submit the executed form with its proposal. Oral comments, responses and representations shall not be binding.

E. DISCLOSURE OF PROPOSAL CONTENTS

All proposals and supporting documents, except such information that discloses proprietary or financial information submitted in response to qualification statements, becomes public information held in custody of the CITY after the proposal submittal date given in this RFP. The CITY assumes no liability for the use or disclosure of technical or cost data submitted by any proposer.

Nevertheless, if a proposal contains information that the respondent does not want disclosed to the public, or used for any purpose other than the evaluation of the proposal, all such information must be indicated with the following or similar statement:

"The information contained on pages _____ shall not be duplicated, used in whole or in part for any purpose other than to evaluate the proposal provided; that if a contract is awarded to this firm as a result of the submission of such information, the City of Cheyenne shall have the right to duplicate, use or disclose this information to the extent provided in the contract or to the extent provided by law. This restriction does not limit the City of Cheyenne's right to use the information contained herein if obtained from another source."

All such nondisclosure items specified in the proposal shall be subject to disclosure as provided in "The Wyoming Public Records Act" or as otherwise provided by law.

F. PROPOSAL PROTEST PROCEDURES

Proposers are advised that the CITY shall administer any protest initiated by a protesting party concerning this solicitation in the following manner:

1. The protesting party may submit a written protest prior to the due date for Proposals stated elsewhere in this RFP, unless the protesting party did not know or could not have known of the facts giving rise to such protest prior to the Proposal opening. The protest shall be submitted within three (3) calendar days after the protesting party knew or should have known of the facts giving rise thereto.
2. Upon receipt of a timely written protest, the CITY shall not proceed further with the solicitation or contract award until all administrative remedies have been exhausted, or until the CITY's Purchasing Manager determines that a contract award without delay is necessary to protect the

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interests of the CITY.

3. Timely protests shall be reviewed by the CITY's Director of Purchasing, who shall consider all facts, technical support, and justification submitted by the protesting party, and in a timely manner render a decision on behalf of CITY as to the validity of the protest and the disposition thereof. The CITY's Director of Purchasing may request additional information from the protesting party to clarify or support its protest, in which case such information shall be supplied in writing by the protesting party within three (3) calendar days following the CITY's request.
4. Following the decision of the CITY's Purchasing Manager, the matter shall proceed to the CITY's governing body for contract award or other action.

G. OMISSION OF DETAILS

No advantage shall be taken by any proposer as a result of the omission of any parts or details that are necessary for the provision of detailed design and engineering services even though such details may not be mentioned in this RFP.

H. SELECTION PROCESS

Each responsive proposal received by the CITY shall be evaluated in a consistent manner as described in this subsection. A responsive proposal is one that complies with all material aspects of the solicitation, both as to the method and timeliness of submission and as to the substance of any resulting contract. Prospective proposers are advised to submit all required forms, attachments, and other information as requested in the RFP. Proposals that do not comply with all the terms and conditions of the solicitation may be rejected as non-responsive.

As noted above, the CITY shall use a two-step evaluation process for selecting a Consultant through this solicitation. The two-step process dictates that proposers make submittals in two parts: a *Technical Proposals* and a *Price Proposal*. It is the intent of the CITY to first evaluate all Technical proposals to establish a quantitative ranking of the proposals exclusive of price. The CITY shall use this process to identify one or more proposers who are considered to be best qualified to perform the work required by the CITY.

All responsive technical proposals will be reviewed by an evaluation committee comprised of the West Edge Project Management Team representing the City of Cheyenne, Laramie County, and the Cheyenne Downtown Development Authority and other personnel to be selected by the City's Development Director. A weighted evaluation methodology shall be used to rank each proposal, using the evaluation criteria and relative weighting indicated on "Attachment A-1".

After review of the technical proposals, the best qualified firms may be invited to participate in oral interviews with the evaluation committee for the purpose of clarifying and confirming the assertions made in their proposals. A weighted evaluation methodology shall be used to rank each interview, using the evaluation criteria and relative weighting indicated on "Attachment A-2".

REQUEST FOR PROPOSALS No.: RFP-12-16

PROJECT NAME: Lower Capitol Basin Sediment Trap/Wetland

At the end of the interview process, the CITY shall then unseal the highest ranked proposer Price Proposal and initiate negotiations toward a final contract containing a detailed scope of work and budget. If negotiations with the first selected proposer are unsuccessful, negotiations with the second highest ranked proposer will be pursued, followed by the third proposer, if necessary to secure a contract. In the event the CITY is unable to secure an Agreement with any of the qualified proposers, this procurement may be terminated at the CITY's option.

The CITY reserves the right to delay making an award as necessary to permit proper study and analysis of all proposals received, to split award on proposals, to reject any or all proposals received, and to make a pre-award survey to determine the capability of any or all proposers. The CITY reserves the right to reject any or all proposals if it believes there is a sound business reason for doing so, and to negotiate matters of Consultant performance, project schedule and other issues not directly related to the proposal price submitted prior to awarding a contract. The CITY further reserves the right to cancel the procurement process at any time during the evaluation process if it appears that the proposals received are not suitable for any reason whatsoever.

I. DISADVANTAGED BUSINESS ENTERPRISE PROGRAM

It is the policy of the CITY that disadvantaged business enterprises (DBE) as defined in 49 CFR Part 23, shall have the maximum opportunity to participate in the performance of contracts financed in whole or in part with Federal funds. Accordingly, the CITY has adopted a goal of awarding a minimum of sixty-six hundredths of one percent (0.66%) of the value of any contract resulting from this solicitation to qualified DBE firms or individuals.

All proposers who are not themselves qualified DBE firms or individuals are encouraged to involve qualified DBEs to participate in this Project. To ensure that a fair proportion of the purchases and contracts for supplies and services for the CITY is placed with DBEs, the proposer agrees to take actions to the fullest extent practicable to identify qualified disadvantaged firms, to solicit proposals and quotations from them for supplies and services related to this proposal, and in making awards and purchases, to give equitable consideration to DBEs.

As part of its submittal, each proposer shall describe any contract activities to be performed by Disadvantaged Business Enterprises (DBEs), using Attachment B-3: Use of Sub-Consultants. In the event that no DBE participation is proposed, the proposer shall describe its good faith efforts to involve DBEs in this Project.

(Proposer must present the information required above with initial proposals (see 49 CFR 26.53(3)).

Additionally, prospective proposers are encouraged to note all requirements stated in the Sample Professional Services Agreement, which are incorporated into this RFP. The selected Consultant shall be required to comply with all such requirements.

REQUEST FOR PROPOSALS No.: RFP-12-16**PROJECT NAME: Lower Capitol Basin Sediment Trap/Wetland****J. SELECTION PROCESS SCHEDULE**

Apr. 1 and Apr. 8, 2016	Legal Notice posted in paper
April 8, 2016	Pre-Proposal meeting
April 11, 2016	Noon – Deadline for questions/changes
April 22, 2016	Technical Proposals and Price Proposals due
April 29, 2016	Review Committee selects proposers for Interviews
May 6, 2016	Interviews
May 6, 2016	Review Committee completes final selection
May 16, 2016	Professional Services Agreement finalized with successful proposer
May 18, 2016	Agreement placed on City Council Agenda
May 23, 2016	City Council Meeting No.1: Agreement referred to Finance Committee
June 6, 2016	Agreement discussed at Finance Committee
June 13, 2016	City Council Meeting No. 2: Agreement approved/disapproved
June 15, 2016	Notice-to-Proceed issued by City if Agreement approved

REQUEST FOR PROPOSALS No.: RFP-12-16**PROJECT NAME: Lower Capitol Basin Sediment Trap/Wetland****"ATTACHMENT A-1" FIRST ROUND CONSULTANT PROPOSAL EVALUATION CRITERIA WEIGHTING**

CRITERIA	DESCRIPTION	WEIGHT
<u>Applicable Experience</u>	Quantity and type of experience with design and Construction Management of large-scale constructed wetlands projects; demonstrated evidence of past project success coordinating multidiscipline solutions integrating engineered and biological systems; experience coordinating landscape requirements with stormwater management objectives; successful management of state and federal funding on past projects; environmental remediation experience; treatment wetlands and sediment trap design experience.	25
<u>Approach to Project</u>	Demonstrated methods/approach for coordinating and integrating engineered and biological systems solutions into the underlying West Edge vision within the available budget and project schedule. Approach to leading a successful multidiscipline team of professionals.	15
<u>Cost Saving and Project Innovations</u>	Ideas or suggestions to improve schedule, cost, feasibility, or scope of the project.	5
<u>Technical Expertise</u>	Expertise in large scale system hydraulics, master drainage planning, and numeric modeling; demonstrated expertise using EPA SWMM hydraulic modeling software; expertise in the design and construction management of treatment wetlands; experience with multifunctional park design; expertise with sustainable ecological design; environmental clean-up expertise/familiarity with EPA, Region 8 crosswalk linkage between SAP & QAPP as approved for the City's EPA RLF & Assessment grants; expertise managing state and federal funding.	15
<u>Adequate Resources</u>	Sufficient available staff and equipment to complete the proposed work within the requested timeframe.	20
<u>Project Familiarity</u>	Understanding of City's West Edge vision and how it is integrated within the Pump House Park and the underlying economic drivers for public infrastructure related to the West Edge District.	5
<u>Performance on Past Projects and/or Reference Check</u>	Success on previous projects in the way of project deadlines, cost control, quality, innovative deadline driven design approaches, public participation/outreach, examples of leading successful multidiscipline teams, and general cooperative nature of the firm.	5
<u>Jobsite Proximity</u>	Physical location relative to project site in order to minimize response time to project needs.	5
<u>Proposal Quality</u>	Readability, completeness, brevity, and organization of the proposal.	5
TOTAL		100

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PROJECT NAME: Lower Capitol Basin Sediment Trap/Wetland

"ATTACHMENT A-2"
SECOND ROUND CONSULTANT PROPOSAL EVALUATION
(INTERVIEWS) CRITERIA WEIGHTING

INTERVIEW		SCORE	COMMENTS
<u>Completeness</u>	10		
<u>Method of presentation</u>	10		
<u>Professionalism</u>	10		
<u>Enthusiasm for project</u>	10		
<u>Innovative Ideas/Creative Project Approach</u>	10		
<u>Applicable Experience/Technical Expertise</u>	25		
<u>Familiarity with West Edge</u>	25		
TOTAL		100	

List reasons you have for recommending a particular firm:

.....

.....

.....

.....

.....

List reasons you have for rejecting a particular firm:

.....

.....

.....

.....

General comments/clarifications/questions:

.....

.....

.....

.....

.....

Date: _____

REQUEST FOR PROPOSALS No.: RFP-12-16**PROJECT NAME: Lower Capitol Basin Sediment Trap/Wetland****"ATTACHMENT B-1"
PROPOSAL SUBMITTAL CHECKLIST***This form must be completed and returned with the proposal submittal.*

Attachment	Description	Proposer's Initials	CITY's Initials
B-1	Proposal Submittal Checklist		
<u>Envelope #1</u>	<u>Technical Proposal</u> Sealed Envelope w/required information on front Signed Cover Letter Project Team Schedule Project Area Scope of Work References Project Specific Information		
B-2	Acknowledgment of Receipt of RFP Packet, Sample Agreement and Addenda		
B-3	Use of Sub-consultants		
<u>Envelope #2</u>	<u>Price Proposal</u> Sealed Envelope w/required information on front Signed Cover Letter Project Fee Proposal with breakdown		

Signature of Proposer: _____

Date: _____

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"ATTACHMENT B-2"
ACKNOWLEDGMENT OF RECEIPT
OF RFP PACKET, SAMPLE AGREEMENT AND ADDENDA

The undersigned hereby acknowledges receipt of the City of Cheyenne Request for Proposals (RFP) packet for _____ including: Sample Professional Services Agreement, Scope of Work, Compensation Schedule, Project Schedule, and the following addenda issued during the procurement process.

Addendum #1: _____ dated _____

Addendum #2: _____ dated _____

Addendum #3: _____ dated _____

Failure to acknowledge receipt of the proposal packet, and all addenda may cause the proposal to be considered non-responsive to this solicitation.

Name of Proposer: _____

Signature of Proposer: _____

Date: _____

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**"ATTACHMENT B-3"
USE OF SUB-CONSULTANTS LIST**

Sub-consultant Name/City/State	% of Value of Work	Work Tasks to be Assigned
Total Value of Subcontracts		

Name of Proposer: _____

Signature of Proposer: _____

Date: _____

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"ATTACHMENT B-4"
CERTIFICATION REGARDING DEBARMENT, SUSPENSION
AND OTHER RESPONSIBILITY MATTERS

This contract is a covered transaction for purposes of 49 CFR Part 29. As such, the consultant is required to verify that none of the consultant, its principals, as defined at 49 CFR 29.995, or affiliates, as defined at 49 CFR 29.905, are excluded or disqualified as defined at 49 CFR 29.940 and 29.945.

The consultant is required to comply with 49 CFR 29, Subpart C and must include the requirement to comply with 49 CFR 29, Subpart C in any lower tier covered transaction it enters into.

By signing and submitting its bid or proposal, the bidder or proposer certifies as follows:

The certification in this clause is a material representation of fact relied upon by the City of Cheyenne. If it is later determined that the bidder or proposer knowingly rendered an erroneous certification, in addition to remedies available to the City of Cheyenne, the Federal Government may pursue available remedies, including but not limited to suspension and/or debarment. The bidder or proposer agrees to comply with the requirements of 49 CFR 29, Subpart C while this offer is valid and throughout the period of any contract that may arise from this offer. The bidder or proposer further agrees to include a provision requiring such compliance in its lower tier covered transactions.

Date: _____

Signature: _____

Title: _____

REQUEST FOR PROPOSALS No.: RFP-12-16

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**"ATTACHMENT B-5"
NON-COLLUSION AFFIDAVIT**

The undersigned duly authorized official of the proposer hereby certifies, to the best of her/his knowledge and belief, that:

That I am an officer or employee of the _____ (proposing entity)
having the authority to sign on the behalf of the corporation; and,

That the prices in the attached proposal were arrived at independently by

(proposing entity) without collusion, consultation, communication, or any agreement, for the purpose of restricting competition as to any matter relating to such prices with any other proposer or with any other competitor regarding an understanding, or planned common course of action with any other vendor of materials, supplies, equipment, or service described in the RFP designed to limit independent proposals or competition; and

That unless otherwise required by law, the contents and prices contained in the proposal have not been communicated by _____ (proposing entity) or its employees or agents to any person not an employee or agent of _____ (proposing entity), or its surety on any bond furnished with the proposal, and will not be communicated to any such person prior to the official opening of the proposal; and,

That I have fully informed myself regarding the accuracy of the statements made in this affidavit.

Proposer: _____

Signature: _____

Title: _____

Date: _____

REQUEST FOR PROPOSALS No.: RFP-12-16

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"ATTACHMENT B-6"
RESTRICTIONS ON LOBBYING CERTIFICATION

Byrd Anti-Lobbying Amendment, 31 U.S.C. 1352, as amended by the Lobbying Disclosure Act of 1995, P.L. 104-65 [to be codified at 2 U.S.C. § 1601, et seq.] - Contractors who apply or bid for an award of \$100,000 or more shall file the certification required by 49 CFR part 20, "New Restrictions on Lobbying." Each tier certifies to the tier above that it will not and has not used Federal appropriated funds to pay any person or organization for influencing or attempting to influence an officer or employee of any agency, a member of Congress, officer or employee of Congress, or an employee of a member of Congress in connection with obtaining any Federal contract, grant or any other award covered by 31 U.S.C. 1352. Each tier shall also disclose the name of any registrant under the Lobbying Disclosure Act of 1995 who has made lobbying contacts on its behalf with non-Federal funds with respect to that Federal contract, grant or award covered by 31 U.S.C. 1352. Such disclosures are forwarded from tier to tier up to the recipient.

APPENDIX A, 49 CFR PART 20--CERTIFICATION REGARDING LOBBYING

Certification for Contracts, Grants, Loans, and Cooperative Agreements

Pursuant to United States Public Law 101-121, Section 319, the undersigned duly authorized official of the proposer hereby certifies, to the best of her/his knowledge and belief, that:

1. No Federal appropriated funds have been paid or will be paid, by or on behalf of the undersigned, to any person for the purpose of influencing or attempting to influence an officer or employee of any agency, a Member of Congress, an officer or employee of Congress, or an employee of a Member of Congress in connection with the awarding of any Federal contract, the making of any Federal grant, the making of any Federal loan, the entering into of any cooperative agreement, and the extension, continuation, renewal, amendment, or modification of any Federal contract, grant, loan, or cooperative agreement.
2. If any funds other than Federal appropriated funds have been paid or will be paid to any person for making lobbying contacts to an officer or employee of any agency, a Member of Congress, an officer or employee of Congress, or an employee of a Member of Congress, in connection with this Federal contract, grant, loan, or cooperative agreement, the undersigned shall complete and submit a Standard Form-LLL, "Disclosure Form to Report Lobbying", in accordance with its instructions [as amended by "Government wide Guidance for New Restrictions on Lobbying," 61 Fed Reg. 1413 (1/19/96). Note: Language in paragraph (2) herein has been modified in accordance with Section 10 of the Lobbying Disclosure Act of 1995 (P.L. 104-65, to be codified at 2 U.S.C. 1601, et seq.)]
3. The undersigned duly authorized official shall require that the language of this certification be included in the award documents for all subawards at all tiers (including subcontracts, subgrants, and contracts under grants, loans, and cooperative agreements) and that all subrecipients shall certify and disclose accordingly.

REQUEST FOR PROPOSALS No.: RFP-12-16

PROJECT NAME: Lower Capitol Basin Sediment Trap/Wetland

This Certification is a material representation of fact upon which reliance was placed when this transaction was made or entered into. Submission of this Certification is a prerequisite for making or entering into this transaction, imposed by 31, USC Section 1352 (as amended by the Lobbying Disclosure Act of 1995). Any proposer (person) who fails to file the required certification shall be subject to a civil penalty of not less than ten thousand dollars (\$10,000) and not more than one hundred thousand dollars (\$100,000) for each such failure to file.

[Note: Pursuant to 31, USC Section 1352(c)(1)-(2)(A), any person who makes a prohibited expenditure or fails to file or amend a required certification or disclosure shall be subject to a civil penalty of not less than ten thousand dollars (\$10,000) and not more than one hundred thousand dollars (\$100,000) for each such failure to file.

The Proposer, _____, certifies or affirms the truthfulness and accuracy of each statement of its certification and disclosure, if any. In addition, the Proposer understands and agrees that the provisions of 31 USC A 3801, et seq., apply to this certification and disclosure, if any.

Proposer: _____

Signature: _____

Title: _____

Date: _____

REQUEST FOR PROPOSALS No.: RFP-12-16

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PRELIMINARY DRAFT

**PROFESSIONAL SERVICES AGREEMENT BETWEEN
THE CITY OF CHEYENNE AND _____ FOR LOWER CAPITOL BASIN
SEDIMENT TRAP/WETLAND ENGINEERING DESIGN SERVICES**

This Agreement is made this _____ day of _____, 20____ by and between the City of Cheyenne, Wyoming ("CITY"), and _____ ("Firm").

WITNESSETH

WHEREAS, the City, acting through the City Engineer is responsible for the management of storm water drainage within the City of Cheyenne; and

WHEREAS, on _____, 2016, the City issued Request for Proposals (#_____) for design services for the Lower Capitol Basin Sediment trap/Wetland Project, aka "Pump House Park, in connection with the West Edge project in downtown Cheyenne; and

WHEREAS, the Firm is qualified to provide the professional services desired by the City pursuant to the Request for Proposals; and

WHEREAS, the Firm has submitted a proposal dated _____, 2016 that describes the scope of work and to be performed by the Firm and the total cost thereof, a copy of which is attached hereto as Exhibit "A" and incorporated by this reference; and

WHEREAS, the City desires to employ the Firm under the terms and conditions hereinafter set forth.

NOW, THEREFORE, in consideration of the mutual promises and covenants herein contained, it is agreed by and between the City and the Firm as follows:

1. SCOPE OF SERVICES:

Except as modified by this Agreement, the Firm agrees to provide professional services for the City for the contract price as set forth in Exhibit "A" attached hereto and incorporated herein by this reference. Subject to the provisions of this section, the City may order major changes in the scope or character of work, either decreasing or increasing the amount of the Firm's services. In the event that such changes are ordered, the Firm shall be entitled to additional compensation for additional work performed. Additional compensation for major changes shall be determined in accordance with Paragraph 2 of this Agreement, but in no event shall the City be liable for payment unless the amount of such additional compensation shall first have been agreed to in writing by the City. In the event that major changes are ordered pursuant to this Section, the schedule for progress and completion in Section 3 of this Contract and the compensation to be paid pursuant to Section 2 shall be adjusted by negotiation between the Firm and the City.

2. COMPENSATION:

The Firm shall perform the services set forth in Exhibit "A" for a cost not to exceed

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_____ Dollars (\$_____). The Firm shall bill the City on a monthly basis. Upon acceptance and approval of work performed in accordance with the Scope of Work described in Exhibit "A", the City shall make payment for the Firm within forty-five (45) days of receipt of said invoice. The Firm shall maintain hourly records of the time worked by its personnel and records of direct project expenses to support any audits the City may require and make those records available to the City at the City's request. Such records shall be maintained by the Firm for one year following the date of final payment.

3. TERM AND EFFECTIVE DATE:

This Agreement shall commence on the date the last required signature is attached hereto and shall continue in full force and effect until the Scope of Work is completed in accordance with Exhibit "A", unless this Agreement is earlier terminated pursuant to its provisions.

4. MODIFICATION:

Any changes, modifications, revisions or amendments to this Agreement which are ordered pursuant to Section 1 herein or which are mutually agreed upon by the parties to this Agreement shall be incorporated by written instrument, executed and signed by all parties to this Agreement.

5. INDEMNIFICATION / HOLD HARMLESS/TRADE SECRETS/COPYRIGHT:

a. The Firm agrees to indemnify, hold harmless and defend the City from and against any and all liabilities, claims, penalties, forfeitures and suits, and the cost and expenses incident thereto, including reasonable attorney's fees, which may hereafter arise as a result of death or bodily injury to any person, destruction or damage to any property, contamination of or adverse effects on the environment, or any violation of governmental laws, regulations or orders to the extent caused by (1) the Firm's breach of any term or provision of this Agreement; or (2) any negligent or wrongful act, error or omission by the Firm, or its employees or subcontractors in the performance of this Agreement. The Firm acknowledges that it may incur a financial obligation to the City and the City pursuant to the terms of this paragraph.

b. The Firm agrees to indemnify, pay the defense costs of, and hold the City harmless from any and all claims, demands, costs, liabilities, losses, expenses and damages (including attorneys' fees, costs, and expert witnesses' fees) arising out of or in connection with this Agreement that sound in an intellectual property claim (including but not limited to patent, copyright, trademark, trade name, or trade secret infringement). This Section specifically survives the expiration or termination of this Agreement.

6. GOVERNMENTAL IMMUNITY:

The City and its officials and employees do not waive governmental immunity by entering into this Agreement and specifically retain all immunities and defenses available to them as Governmental Entities pursuant to Wyo. Stat. § 1-39-101, *et seq.*, and all other applicable laws, except to the extent necessary solely for the enforcement of the terms and conditions of this Agreement as between the parties. Further, the City fully retains all immunities and defenses provided by law with regard to any action, whether in tort, contract or any other theory of law, based on this Agreement. Designation of venue, choice of law and similar provisions should not

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be construed as a waiver of governmental immunity.

7. INSURANCE:

The Firm must provide proof of the following insurance coverages:

Commercial General Liability Insurance:

For claims arising out of bodily injury, illness or death, or from damage to or destruction of property of others, including loss or use thereof, with minimum limits of \$1,000,000 per occurrence, \$2,000,000 aggregate for the entire term of the Agreement.

Business Automobile Insurance:

Including owned, non-owned and hired vehicles with minimum limits for bodily injury and property damage of \$1,000,000 per accident for the entire term of the Agreement.

Workers' Compensation

Workers' Compensation coverage shall be in effect for the entire term of the Agreement, as required by Wyoming law, for all employees or agents providing services under this Agreement. Firm shall provide the City with proof of workers' compensation or employer's liability insurance coverage.

Professional Liability Insurance

The Firm shall provide proof of professional liability insurance or errors and omissions liability insurance in an amount not less than \$500,000 to protect the City from any and all claims arising from the Firm's negligence in the performance of duties under this Agreement.

The City prefers that this liability insurance coverage be provided pursuant to an "occurrence" policy. If this coverage is provided pursuant to a "claims made" policy:

(1) Firm, shall, concurrently with the execution of this Agreement, provide the City with a certificate of insurance demonstrating that such coverage is or shall be in effect at the time the Firm begins the provision of services under this Agreement; and

(2) In the event the Firm's services extend into a future policy period, the Firm shall, prior to the policy expiration date, provide the City with a new certificate of insurance demonstrating that such coverage is or shall be in effect during all periods of time that Firm will provide services under this Agreement; and

(3) Firm shall maintain said "claims made" coverage for a period of

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five (5) years following the last date that Firm has provided services under this Agreement; and

(4) In the event the Firm or the Firm's professional liability insurer terminates "claims made" coverage prior to the expiration of the periods provided in subparagraphs (1), (2), or (3) of this paragraph, the Firm shall provide to the City advance written notification of the termination of said coverage and shall provide the City with an endorsement for an extended reporting period ("tail coverage") which shall be in effect for a period of time not less than five (5) years following the last date that Firm has provided services under this Agreement.

Additional Insurance Information

The Firm shall name the City of Cheyenne as an Additional Insured by endorsement on its insurance policies, with the exception of worker's compensation and professional liability insurance, and shall provide the City with a copy of the endorsements.

Firm shall provide the City with certificates of insurance acknowledging the above-stated coverages prior to beginning any work under this Agreement.

It is understood and agreed that these policies are primary and not contributory. All policies required under this Agreement shall be in effect for the duration of the Agreement. It shall be an affirmative obligation upon Firm to immediately notify in writing the city risk manager, city clerk, and city attorney of any fact, circumstance, or occurrence that has resulted in or may result in the cancellation or substantive change of any insurance coverage required by this Agreement, and failure to do so shall be construed to be a breach of this Agreement.

In addition, Firm shall provide the City with copies of insurance policies and policy endorsements listing the City of Cheyenne as an additional insured. The City's failure to request or review such policies, endorsements, or certificates shall not affect the City's rights or Firm's obligation hereunder.

Any insurance company providing coverage under this agreement shall have a minimum A. M. Best rating of A- (excellent).

8. GOVERNING LAW/ JURISDICTION/ VENUE:

The construction, interpretation and enforcement of this Agreement shall be governed by the laws of the State of Wyoming. The courts of the State of Wyoming shall have jurisdiction over any action arising out of this Agreement and over the parties, and the venue shall be the First Judicial District, Laramie County, Wyoming.

9. COMPLIANCE WITH LAWS:

The Firm shall comply with all applicable federal, state and local laws, rules and

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regulations in the performance of this Agreement, including federal and state anti-discrimination laws. The Firm shall be responsible for all such non-compliance and shall defend, hold harmless and indemnify the City therefrom.

10. ALCOHOL AND CONTROLLED SUBSTANCE POLICY:

In compliance with the Drug-Free Workplace Act of November 1988, the City has established an Alcohol and Controlled Substance Policy that pertains to alcohol and drug usage by City employees. All parties under contract with the City, including the Firm and its employees and subcontractors, are required to comply with the provisions of the City's Alcohol and Controlled Substance Policy that pertains to drug and alcohol usage on City property and other sites occupied by the Firm while performing duties and responsibilities under this Agreement. It is the responsibility of the Firm to familiarize itself with the requirements of this policy and to inform all its employees and subcontractors of those requirements and to insure their compliance therewith. If the Firm, its employees or subcontractors are found to be in violation of this policy, this Agreement may be terminated.

11. INDEPENDENT CONTRACTOR:

At all times during the term of this Agreement, the Firm shall be considered an independent contractor. Neither the Firm nor any one employed by it shall represent, act, purport to act or be deemed to be an agent, representative, employee or servant of the City.

12. CONFIDENTIALITY:

To the extent allowed by law, the City and the Firm shall treat as confidential and not disclose to others information (including technical information, experience or data) regarding either party's plans, programs, plants, processes, products, costs, equipment, operations, or customers which come within the knowledge of the parties, without in each instance securing the prior written consent of the other party, unless such disclosure is required by law or legal process. However, nothing shall prevent either the Firm or the City from disclosing to others or using in any manner information which either party can show (a) has been published or has become part of the public domain other than by acts of the Firm or the City; (b) has been furnished or made known to the Firm or the City by third parties without restrictions on its disclosure; or (c) was in either party's possession prior to the disclosure thereof by the City or the Firm to each other. Neither the Firm nor the City shall be restricted from releasing information in response to a subpoena, court order, or legal process, but, in the event such disclosure becomes necessary, the disclosing party shall notify the other party of the subpoena, order or other legal process at least seven (7) days prior to disclosing responsive information so that the other party may take any and all steps provided by law to prevent the release of such information.

13. FIRMS'S RESPONSIBILITIES:

The Firm shall be responsible for all terms and conditions as set forth in this Agreement and the professional quality, technical accuracy, and timely completion of all services as set forth in Exhibit A and all changes in the scope or character of the work made pursuant to Sections 1 or 4 herein. The Firm shall, without additional compensation, promptly remedy and correct any and all errors, omissions, or other deficiencies. The Firm shall perform all work in a professional manner in conformance with industry standards.

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14. NOTICES:

Any notice, correspondence or billing required to be given by the terms of this Agreement shall be delivered by hand, or delivered by mail, postage prepaid, to the addresses of the respective parties stated below:

City: Brandon Cammarata
Development Director
2101 O'Neil Avenue
Cheyenne, WY 82001

Firm: _____

15. INTELLECTUAL PROPERTY AND WORK PRODUCT:

All work (preliminary, draft, and final) performed by the Firm under this Agreement is the property of the City. The City will own any and all data, documents, working papers, computer programs, photographs, and other material produced by the Firm pursuant to this Agreement, and the Firm hereby assigns and transfers to the City any and all intellectual property rights for such materials. The Firm will provide the City with copies of all such materials. Under no circumstances, including pending disputes between the City and the Firm, will the Firm fail to deliver possession of said documents and materials to the City upon demand.

16. CONFLICT OF INTEREST:

In entering this Agreement, the Firm covenants that it presently has no interest, and shall not acquire any interest, direct or indirect, financial or otherwise, which would conflict in any manner or degree with performance of the services hereunder. In addition, the Firm covenants that in the performance of this contract, no subcontractor or person having such an interest shall be employed. The Firm certifies that no one who has or will have any financial interest under this contract is an officer or employee of the City.

17. ACCEPTANCE NOT WAIVER:

The City's approval of drawings, plans, specifications, reports, and incidental work or materials furnished hereunder shall not in any way relieve the Firm from responsibility for the technical accuracy of the work. The City's approval or acceptance of, or payment for, any services shall not be construed to operate as a waiver of any of the City's rights under this Agreement or any of its legal rights under statute or common law arising out of the performance of this Agreement.

18. DEFAULT:

Each and every term and condition herein shall be deemed a material element of the Agreement. In the event either party shall fail or refuse to perform according to the terms of this

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Agreement, such party may be declared in default.

19. REMEDIES:

In the event a party declares the other party in default hereof, said party declaring default shall notify the defaulting party in writing and such defaulting party shall be allowed a period of fifteen (15) days to cure said default. In the event the default remains uncorrected, the party not in default may elect to: (a) terminate the Agreement and seek damages; (b) treat the Agreement as continuing and require specific performance; or (c) avail itself of any other remedy at law or equity. In the event the Firm fails to strictly perform in accordance with this Agreement, the City may elect to make good such deficiencies and charge the Firm therefore.

20. TERMINATION:

The City may, by written notice to the Firm, terminate this Agreement in whole or in part by giving the Firm fifteen (15) days written notice. Upon receipt of such notice, the Firm shall discontinue all services affected (unless the notice directs otherwise) and deliver to the City representative, within five (5) days after the effective date of termination, all data, drawings, specifications, reports, estimates, and summaries, accumulated by the Firm in performing this Agreement, whether completed or in process. In event of termination, the City will pay the Firm for accepted work done as of the effective date of termination.

21. WAIVER:

The waiver by either party of any term, condition or covenant, or breach of any term, condition or covenant, shall not constitute a waiver of any other term, condition or covenant, or breach thereof.

22. SEVERABILITY:

If any provision, section, subsection, sentence, clause, or phrase of this Agreement is invalidated by any court of competent jurisdiction, such holding shall not affect the validity of the remainder of this Agreement, which shall continue in full force and effect.

23. SUCCESSORS AND ASSIGNS:

All the terms, conditions, and provisions herein shall inure to the benefit of and be binding upon the parties hereto and their respective permitted successors and assigns.

24. ASSIGNMENT:

Neither party shall assign this Agreement without prior written consent of both parties. No other delegation or assignment shall operate to relieve either party of its responsibilities hereunder. However, nothing in this Agreement shall prohibit the Firm from subcontracting to a qualified subcontractor.

25. THIRD PARTY RIGHTS:

The Parties do not intend to create in any other individual or entity the status of third party

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beneficiary, and this Agreement shall not be construed so as to create such status. The rights, duties and obligations contained in this Agreement shall operate only between the parties to this Agreement, and shall inure solely to the benefit of the parties to this Agreement. The parties to this Agreement intend and expressly agree that only the parties signatory to this Agreement shall have any legal or equitable right to seek to enforce this Agreement, to seek any remedy arising out of a party's performance or failure to perform any term or condition of this Agreement, or to bring action for breach of this Agreement. This paragraph is not intended nor shall it be construed to waive any of the parties' immunities.

26. ENTIRE AGREEMENT OF THE PARTIES:

This Agreement, to include Exhibit A, contains the entire understanding of the parties. There are no other terms or conditions, written or oral, concerning or controlling this matter.

IN WITNESS WHEREOF, the parties to this Agreement, through their duly authorized representatives, have executed this Agreement on the days and dates set out below and certify that they have read, understood and agreed to the terms and conditions of this Agreement.

The effective date of this Agreement is the date of the last signature affixed to this page.

SIGNATURE PAGES TO FOLLOW

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EXHIBIT "A"
SCOPE OF WORK

This project will be partially funded with federal funds from an EPA Section 319 Grant administered through the Wyoming DEQ, Wyoming Clean Water State Revolving Fund, 6th Penny Sales Tax, and potential EPA Revolving Loan Fund monies and is therefore subject to the federal and state laws and regulations associated with these corresponding programs.

Compliance with all applicable standards, orders, or requirements issued under section 306 of the Clean Air Act (42 U.S.C. 1857(h)), section 508 of the Clean Water Act (33 U.S.C. 1368), Executive Order 11738, and Environmental Protection Agency regulations (40 CFR part 15) is required.

A.1 Project Description

Overview

The City of Cheyenne located in Laramie County, Wyoming proposes to construct a sediment trap/wetland BMP for the Lower Capitol Basin to control sediment and reduce fecal coliform bacteria (E. coli) populations. A primary contributor to the sediment and E. coli is believed to be urban runoff from the City's downtown and near west side neighborhoods.

The goal of this project is the design and construction of a wetland BMP to reduce sediment and E. coli from stormwater runoff generated in the City's downtown and near west side area; incorporation of a greenway path connecting to the Greater Greenway System along Crow Creek either as funding permits or as a future consideration; design considerations of pedestrian amenities adjacent to the constructed wetland, again, as funding permits or as a future consideration; updating the Lower Capitol Basin EPA SWMM model to accurately reflect the Lower Capitol Basin Sediment Trap/Wetland and the soon-to-be-completed 26th Street storm sewer project; an accelerated design timeline and bid process; and efficient management of the various funding sources in place to fund this project. The design phase of the project is the development of a complete set of construction documents, including Reports, Plans, Specifications, Cost Estimates, Bid Documents, and the preparation of the necessary environmental documentation to fulfill NEPA requirements, comply with the WYDEQ's VRP program, and provide an effective environmental remediation strategy for the Lower Capitol Basin Sediment Trap/Wetland Project if necessary. Construction of the project is expected to begin in the fall of 2016 and be completed by May 2017. Completion includes installation of all landscaping and vegetation elements. This is a very firm date in order to meet grant requirements.

Presently, runoff from the City's downtown and near west side area discharges directly to Crow Creek. This project is expected to develop an approximately 2 acre shallow wetland with sediment trap in an unimproved portion of the Lower Capitol Basin in an initial phase to passively treat stormwater runoff before it is discharged to Crow Creek. The goal of this project is to target sediment removal; E. coli removal is expected to be a secondary water quality benefit of the design. The wetland is expected to reduce sediment from the Lower Capitol Basin and reduce sediment from the baseflow discharged to the creek by 30-50%.

The investment area has a history of severe flooding and drainage problems. Presently, runoff from the

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City's downtown and near west side area discharges directly to Crow Creek. The Capitol Basin watershed in Cheyenne, Wyoming, encompasses approximately 11.7 square miles. Of that, 2.4 square miles is in the highly developed downtown area. Land use in the downtown portion of the basin is primarily old residential housing neighborhoods, commercial, and light industrial. There is a significant network of storm sewers and inlets. Although the extent of the infrastructure is considerable, historic flooding problems have proven it is vastly undersized and inadequate to prevent flood damages. At present, there is no allowance for storm water quality.

The Lower Capitol Basin Sediment Trap/Wetland project is funded, in large part, by federal and state funding. The successful design team will need to demonstrate past experience with administering state and federal grant dollars on behalf of their clients. Additionally, the successful design team should demonstrate past project experience managing highly technical large-scale capital improvement projects, keeping those projects within or below budget and on schedule. An innovative final design and construction approach, not jeopardizing state and/or federal project funding, may be warranted in meeting project objectives and the available budget.

The scope of work for the Lower Capitol Basin Sediment Trap/Wetland includes the development and design of a complete set of construction documents, including Plans, Specifications, and Estimates. The successful design team shall be responsible for providing a public participation process coordinated with the West Edge Project Management Team and other key project stakeholders; property boundary surveying; design of a wetland BMP to reduce sediment and E. coli from stormwater runoff generated in the City's downtown and near west side area in the approximately 2 acre site located east of Ames Avenue between the Union Pacific Rail corridor and the 15th Street Right-of-Way.

The selected consulting firm shall:

- a. Design a sediment trap within a wetland complex that can be easily and reasonably maintained;
- b. Identify and estimate anticipated E. coli removal within the designed wetland system;
- c. Prepare a detailed hydrologic analysis/water budget characterization of the contributing watershed;
- d. Design a functional wetland treatment system based on models found in nature with appropriate native vegetation;
- e. Obtain written approval from the WDEQ at the 65% and 95% design levels; and
- f. Provide construction oversight and management of the project during construction.

Tentative Project Schedule

RFP Issued	Early April, 2016
Questions Due	April 11, 2016
Responses to questions posted on-line	April 15, 2016
Proposal Due	April 22, 2016
Consultant Selection	May 6, 2016
Project Design Start	June 15, 2016
Final Deliverables	Fall 2016, sufficient to allow a bid process, construction and vegetation schedule that will be complete May 30, 2017

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Qualifications

The selected consultant firm will demonstrate experience in:

- a. Identifying and modeling sediment and e. coli removal;
- b. Designing and developing cost-effective remediation techniques;
- c. Modeling and designing appropriate sediment bed loads;
- d. Preparing sampling analysis plans and conducting verification sampling;
- e. Designing treatment wetlands;
- f. Designing and sizing large storm sewers and detention ponds in highly urbanized areas;
- g. Overseeing the construction of treatment wetlands and sediment traps;
- h. Adjusting and modifying construction documents to fit unknown field conditions;
- i. Managing state and federal funding;
- j. Working with the City of Cheyenne; and
- k. Working in and familiarity with the West Edge of the City of Cheyenne.

The City does have to comply with both Federal and State (WDEQ) requirements for storm drainage quality, particularly in the lower Capitol Basin. The constructed wetland must be designed with consideration of pedestrian paths and amenities that may be built concurrently or budgeted to follow at some point in the future. The successful design team is encouraged to incorporate any innovative design and/or construction solutions that are cost effective and meet project goals while not jeopardizing project funding.

The successful design team shall be responsible to develop preliminary construction sequencing/phasing and traffic routing plans and special provisions. The design team shall provide surface utility locates and non-destructive subsurface utility excavation locates of potential utility conflicts.

Public Outreach

Two public meetings will be required for this project. As envisioned in the grant application to DEQ, one public meeting will occur toward the end of the design phase to explain the purpose of the project, its benefits to the Crow Creek watershed, and present general stormwater quality information. The second public meeting will include a project tour at the end of the project construction. The design team will coordinate the public process with the consultant team currently engaged in the City's EPA Planning Grant. Public notice and media releases; securing a meeting site, preparation of exhibits, handouts, and comment cards; and correlation of public comments will be the responsibility of the design team. The successful design team is encouraged to incorporate any innovative outreach ideas that are cost effective and meet project goals.

Right-of-Way Surveying

It is the intent of the City to construct the proposed improvements within the existing platted right-of-way and/or entirely on City-owned properties. Acquisition of new permanent right-of-way is not anticipated. A temporary construction easement will be required on adjacent Union Pacific right-of-way. The City will negotiate with the Union Pacific.

Utility Design

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The design team shall investigate and coordinate designs with all known utilities. The design team shall be responsible for resolving any conflicts with other utilities, either active or abandoned. The City has already completed a geophysical investigation to identify possible underground utilities, drums, storage tanks, or other anthropogenic items that could interfere with proposed excavation operations. This information will be provided at the pre-proposal meeting. The design team shall include any additional investigation of these elements in their proposal. The design team shall provide surface utility locates and non-destructive subsurface utility excavation locates of potential utility conflicts as deemed necessary. An incomplete list of utilities includes Cheyenne Light, Fuel, and Power (natural gas, electricity and street lighting); Charter Communications (cable television); Century Link (telephone and fiber optics); City of Cheyenne Public Works (traffic signals and signs); and Cheyenne Board of Public Utilities (water and sanitary sewer).

Coordination with Other Agencies and Projects

The design team shall anticipate extensive coordination with the Department of Environmental Quality (DEQ) and obtain DEQ permits to construct proposed water and sewer improvements. In addition to the typical involvement of the DEQ (stormwater, water, and sanitary sewer) for projects of this nature, potential contamination at the site may require remediation with potential funding through the City's EPA Revolving Loan Fund. The design team will coordinate environmental remediation efforts with the City's environmental consultant currently administering remediation efforts funded through the City's revolving loan account. Any environmental remediation will be incorporated into the design and specifications.

A.2 General Conditions

The following items are considered to be included as part of the basic compensation for this project:

- Normal business expenses - phone, postage, etc.
- Cost of insurance
- In-house computer time and service
- Word processing, accounting, and man-hour records
- Permits and license fees
- Mileage
- Travel fees, room and board, per diem
- Printing costs for all standard review, bidding, and as-constructed plans and other correspondence and contract documents
- Film and processing
- Soils testing
- Surveying
- Acceptance testing
- Utility investigation
- Overtime engineering and inspection
- Special inspections, including vacuum excavation pot-holing
- Survey materials, stakes, etc.
- Additional required services

Any changes in the Consultant's or Subconsultant's staff or fee structure shall be presented in writing to

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the City for approval prior to initiating any changes or performing any work.

A.3 Preliminary Design Phase

Upon receipt of Notice to Proceed with this project, the Consultant shall proceed with the following tasks:

1. Finalizing of the storm sewer alignment into and out of; and the Lower Capitol Basin Sediment Trap/Wetland facility footprint and will include:
 - a. Prepare conceptual Plan and Profile sheets for the alignment and corresponding area for the Lower Capitol Basin Sediment Trap/Wetland;
 - b. Field survey horizontal and vertical location of any utilities critical to the alignment and facilities;
 - c. Verify the appropriateness of the site with respect to adequate water flow, underlying soils, and capacity to support newly created wetland vegetation;
 - d. Review existing information regarding any environmental contamination, including meeting with the Wyoming Department of Environmental Quality and continuing the VRP process initiated by the City;
 - e. Prepare conceptual cost estimates for the Lower Capitol Basin Sediment Trap/Wetland including the potential inclusion of any required environmental remediation;
 - f. Prepare a detailed hydrologic analysis/water budget characterization of the contributing watershed;
 - g. Prepare a report detailing the findings;
 - h. Meet with the City to present the report; and
 - i. Obtain WDEQ's written approval of the report.
2. Conduct subsurface investigations including borings, subsurface explorations, sampling, laboratory and field tests as necessary, and professional interpretation of all of the foregoing subsurface data. Each bore shall be drilled to depth of 20 feet from existing grade. Coordinate with consultant's surveyor to get boreholes surveyed and onto the project mapping.
3. Perform all field surveys necessary for the design of the project, including but not limited to cross-sections, utilities, property, and boundary surveys. Establish project control points as needed. All surveying, design, and temporary control points shall be referenced into the City control network. The vertical datum shall be the NAVD 1988.
4. Identify and evaluate alternate solutions available to the City, and after consultation with City Representative, recommend to the City those solutions which in consultant's judgment meet the City's requirements for the project.
5. On the basis of the above selection of a recommended solution, prepare preliminary design documents consisting of final design criteria, preliminary drawings, specifications, special conditions, design data, and an opinion of probable construction cost including allowances for contingencies.

The services will be completed and the data submitted within the stipulated period indicated in Exhibit C "Project Schedule."

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Consultant shall coordinate with the City's Representative and the consultant team currently engaged in the City's EPA Planning Grant to determine the timing for the public outreach for the Lower Capitol Basin Sediment Trap/Wetland and Pump House Park during the course of preliminary and final design for this project.

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A.4 Final Design Phase

After acceptance by City Representative of the Preliminary Design Phase documents and opinion of probable project cost, indicating any specific modifications or changes in the extent of the project desired by the City, and upon written authorization from City Representative to proceed with the Final Design Phase, Consultant shall:

1. Prepare final drawings showing the character and extent of the Project and Specifications incorporating any modifications or changes indicated during review of preliminary documents.
2. Advise City Representative of any adjustments to the latest opinion of probable cost caused by changes in extent or design requirements of the project. The Consultant shall provide a summary of the probable cost based on the Final Design Drawings and Specifications showing estimated construction quantities, bid prices, and items of work. The bid tabulation shall be laid out using standard numbering for work items and basic pay units.
3. Prepare for review and approval by City Representative the Project Manual including the following as appropriate:
 - Instruction to Bidders
 - Bid Forms
 - Contract Agreements
 - Performance and Payment Bond
 - Notice of Award and Notice to Proceed
 - Change Order, Payment Certification, and Other Forms
 - General and Supplemental Conditions
 - EPA Requirements and/or Provisions
 - Special Provisions
 - Construction Specifications
 - Supplementary Specifications
 - Phasing Schedule
 - Testing Plan

The City of Cheyenne and Board of Public Utilities Construction Specifications and Standard Drawings 2014 with all approved amendments shall be used on this project. The City will provide the Standard Construction boiler plate and other pertinent data in electronic form for use in final bid documents.
4. Furnish one (1) electronic (pdf) and five (5) paper copies of the above final design documents and present and review them in person with the City Representative.
5. Obtain WDEQ's written approval of the final design documents.
6. Prepare Bid Package incorporating any modifications or changes indicated during review of Final documents and furnish three copies and one reproducible set of the construction Contract Documents.

The services will be completed and the data submitted within the stipulated period indicated in Exhibit C "Project Schedule."

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A.5 Bidding Phase

The Consultant will assist the City through all the bidding process, including:

1. Prepare a minimum of one (1) electronic (pdf) and five (5) paper copies of the construction documents, including addenda.
2. Organize and conduct a pre-bid conference with the City and Contractors and record this meeting (take notes and attendance, etc.).
3. Prepare and publish any necessary addendums to the Contract Documents.
4. Give recommendations to the City Representative in selecting a responsible Contractor, Subcontractors, and/or Suppliers.
5. Consultant may be required to attend and provide information at City Council meetings, Finance Committee meetings, or other meetings during the award process.

A.6 Construction Engineering

Used in this Exhibit, the term Consultant's Engineer denotes final responsibility for the project; tasks may be performed by the Resident Project Representative, or Inspector, as appropriate.

Consultant's Engineer will act under the general direction of the City's Representative and will confer with City's Representative regarding engineering, ecological, landscape design, environmental, and geomorphic matters. The Consultant's Engineer shall act as an independent contractor and shall control and supervise the details of the work. In so doing the Consultant's Engineer shall deal solely with the General Contractor and dealings with Subcontractors shall be only through or with the full knowledge of General Contractor.

Price proposals for construction management shall be based on an estimate of 1000 hours being required to administer the project, including 50 hours of after-hours (nights/weekends).

To promote a level playing field in construction management price proposals, an estimate of 1000 hours of construction management time was provided. The following should be considered in developing fee proposals:

- The 1000 hours will need to be split as the consultant feels appropriate into time for Project Managers, Field Engineers, Inspectors, Technicians, Survey Teams, and whatever other personnel categories the consultant feels are necessary.
- The Construction Staking and Materials Testing will be the responsibility of the construction contractor (not the Consultant). The Consultant (including subcontractors) will not be allowed to provide construction staking; even under separate contract so as to be available to the City during construction should an outside check be required on any surveying or staking issue.

CONSULTANTS ENGINEER WILL:

REQUEST FOR PROPOSALS No.: RFP-12-16

PROJECT NAME: Lower Capitol Basin Sediment Trap/Wetland

1. **Contractor Selection:**
Advise City's Representative regarding recommendation for the selection of a Contractor and any subcontractors.
2. **Inspection Plan:**
Develop an overall inspection plan, a Traffic Control Inspection Plan, and testing plan to be submitted and approved by City's Representative seven (7) days after the Contract is awarded. Ensure compliance with federal wage rates, including checking certified payrolls and conducting compliance interviews in accordance with current (at time of bidding) federal requirements.
3. **Schedules:**
Review the progress schedule, schedule of Shop Drawing submissions, and schedule of values prepared by Contractor and consult with City's Representative concerning their acceptability.
4. **Conferences:**
Conduct and keep minutes of pre-construction conferences with contractors. Conduct weekly safety and coordination meetings, as required, maintain and circulate copies of minutes thereof.
5. **Liaison:**
Serve as the City's liaison with Contractor to implement the intent of the Contract Documents. Provide a public relation program to keep adjacent property owners advised of the commencement and progress of construction.
6. **Shop Drawings and Samples:**
 - a. Receive samples and shop drawing furnished by contractor, record date of their receipt, and track progress of approvals. Advise City's Representative of their acceptability.
 - b. Identify any Work that is unsatisfactory, faulty, or defective or does not conform to the Contract Documents, or does not meet the requirements of any inspections, tests, or approvals or that has been damaged. Advise the Contractor and the City's Representative when work should be corrected, rejected, uncovered for observation, or in need of special testing, inspection, or approval.
 - c. Verify that tests, equipment and systems startup, and operation and maintenance instructions are conducted as required by the Contract documents and that Contractor maintains adequate records thereof; observe, record, and report to the City's Representative appropriate details relative to the test procedures and startups.
 - d. Accompany visitors representing public and other agencies having jurisdiction over the Project; record the outcome of these inspections and report to the City's Representative.
7. **Interpretation of Contract Documents:**
Upon request of City Representative, consult with and advise the City Representative on necessary clarifications and interpretations of the Contract Documents as appropriate to the orderly completion of Contractor's work. Transmit to Contractor the City's clarifications and interpretations of the Contract Documents.
8. **Modifications:**
Consider and evaluate any changes or modifications in Drawings or Specifications and report them with recommendations to the City Representative for approval prior to authorizing any such changes or charges to Force Account or other items. Administer Change Orders, Contract Modifications and extra Payment Requests.
9. **Records:**
 - a. Maintain at the job site orderly files for correspondence, reports of job conferences, Shop Drawings and samples submissions, and reproductions of original contract Documents

REQUEST FOR PROPOSALS No.: RFP-12-16

PROJECT NAME: Lower Capitol Basin Sediment Trap/Wetland

including all addenda, change orders, field orders, additional Drawings issued subsequent to the execution of the Contract, City's Representative clarifications and interpretations of the Contract Documents, progress reports, and other Project related documents.

- b. Maintain in a secure location orderly files documenting certified payrolls and results of all compliance interviews. Provide periodic status reports to the City's Representative summarizing current status of federal wage rate compliance.
- c. Keep a diary or log book, recording hours on the job site, weather conditions, data relative to questions of extras or deductions, list of visiting officials and representatives of manufacturers, fabricators, suppliers and distributors, daily activities, decisions, observations in general and specific observations in more detail as in the case of observing test procedures.
- d. Record names, addresses, and telephone numbers of all Contractors, subcontractors, and major suppliers of materials, equipment, date, time and condition of such.
- e. The City or any of their duly authorized representatives, shall have access to any documents, books, papers, and records of the Consultant award # for the purpose of making an audit, examination, excerpts, and transcriptions. The City shall require the Consultant to maintain all required records for at least three years after the City makes final payment and all pending matters are closed.

10. Reports:

- a. Furnish the City's Representative weekly summary reports including Contractor's compliance, the approved progress schedule, and schedule of Shop Drawing submissions.
- b. Consult the City's Representative in advance of scheduled major tests, inspections, or start of important phases of the Work.
- c. Report immediately to the City's Representative upon the occurrence of any accident and furnish complete report shortly afterward.
- d. Furnish the City's Representative with all State and/or Federal reporting requirements including quarterly reports.

11. Payment Requisitions:

Review and certify applications using forms provided by the City for compliance with the established procedure for their submission and forward them with recommendations to the City's Representative noting particularly their relation to the schedule of values, work completed, and materials and equipment delivered or paid for at the site, but not incorporated in the Work.

12. Certificates, Maintenance, and Operation Manuals:

During the course of the Work, verify that contractor has supplied all certificates, maintenance, and operation manuals and other data needed to operate machinery and equipment actually installed and collect all such materials for presentation to City with final construction report.

13. Completion:

- a. Submit to Contractor a punch list of observed items requiring completion or correction.
- b. Conduct final inspection in the company of the City's Representative, Contractor, and utility companies involved and prepare a final list of items to be completed or corrected.
- c. Verify that all items on the final list have been completed or corrected, make recommendations to the City concerning acceptance and prepare a certificate of Substantial Completion.
- d. Conduct eleventh (11th) and twenty-third (23rd) month inspections with the City

REQUEST FOR PROPOSALS No.: RFP-12-16

PROJECT NAME: Lower Capitol Basin Sediment Trap/Wetland

Representative and Contractor's Representatives and report findings to the City's Representative immediately prior to the expiration of the two year or other warranty, as applicable.

- e. After completion of the project, prepare and submit to City certified "Record Drawing" plans showing final accepted conditions of the project. One set of prints, one set of as-builts on CD in CAD, and one set of as-builts on CD in PDF are required. A certified final construction report of conditions, maintenance items, inspection reports, contract modifications, correspondence, field orders and any other pertinent information shall be furnished by the project engineer to the owner.
- f. Assist City staff to implement the final quantities and payments of this project to the City's computerized job accumulation program.

A.7 Limitations of Authority of the Engineer

Except upon written instructions of City's Representative, Engineer:

- 1. Shall not authorize any deviation from the Contract Documents or costs, or approve any substitute materials or equipment, without consulting the Owner.
- 2. Shall not exceed limitations on the City's authority as set forth in the Contract Documents.
- 3. Shall not undertake any of the responsibilities of the Contractor, subcontractors, utility companies or responsibilities of their superintendent, to expedite the work.
- 4. Shall not advise on or issue directions relative to any aspect of the means, methods, techniques, sequences or procedures of construction unless such is specifically called for in the Contract Documents.

REQUEST FOR PROPOSALS No.: RFP-12-16

PROJECT NAME: Lower Capitol Basin Sediment Trap/Wetland

EXHIBIT "B"
COMPENSATION SCHEDULE

The Consultant shall prepare and submit proposed estimated compensation, bearing in mind that it should be a unit-based contract with a negotiated not-to-exceed amount. A Payment Schedule based upon the following shall be submitted with payment for a specific phase not exceeding the scheduled amount prior to completion of that phase:

Preliminary Design Phase	\$ _____
Final Design Phase	\$ _____
Engineering Design Subtotal	\$ _____
Bidding and Construction Phase	\$ _____
Total Contract Amount	\$ _____

REQUEST FOR PROPOSALS No.: RFP-12-16**PROJECT NAME: Lower Capitol Basin Sediment Trap/Wetland****EXHIBIT "C"**
PROJECT SCHEDULE

The key milestones in the West Edge District Civic Center Commons and Pump House Park project including the storm sewer construction are shown below. **The proposer should complete the following schedule as part of the proposal and include any other key dates as needed.**

Date	Schedule Item
<u>June 15, 2016</u>	Notice-to-proceed issued by City
<u> </u>	Hold scoping meeting with City Representatives
<u> </u>	Submit preliminary design and report
<u> </u>	Submit 65% design documents
<u> </u>	Submit 95% design documents
<u> </u>	Meet with City to discuss review of design
<u> </u>	Final design submittal and review
<u> </u>	Bid Opening
<u>Fall 2016</u>	Begin Construction
<u>March 2017</u>	Substantially Complete
<u>May 2017</u>	Final Project Completion



SMITH ENVIRONMENTAL & ENGINEERING

Sustainable Environmental Solutions

TECHNICAL PROPOSAL TO PROVIDE CONSTRUCTION DOCUMENTS, BIDDING ASSISTANCE AND CONSTRUCTION ADMINISTRATION SERVICES FOR LOWER CAPITOL BASIN SEDIMENT TRAP/WETLAND AKA "PUMP HOUSE PARK"

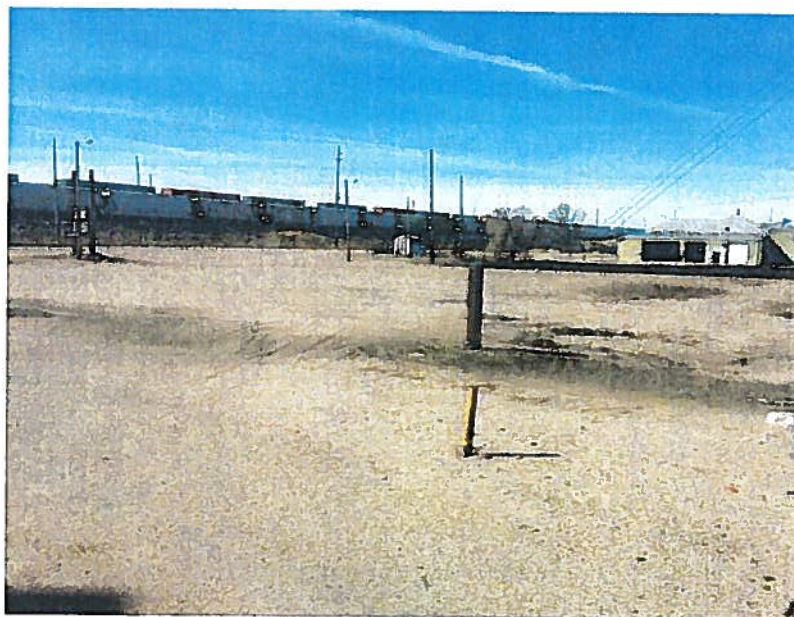
RFP-12-16



*Prepared for the City of Cheyenne
by Smith Environmental & Engineering*



April 22, 2016



EXHIBIT

tabbles

B

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1.0 COVER LETTER

April 22, 2016

Sara Vasquez, Purchasing Manager
City Purchasing Office, Room 104
2101 O'Neil Avenue
Cheyenne, WY 82001

RE: Proposal to Provide Construction Documents, Bidding Assistance and Construction Administration Services for Lower Capitol Basin Sediment Trap/Wetland aka "Pump House Park"

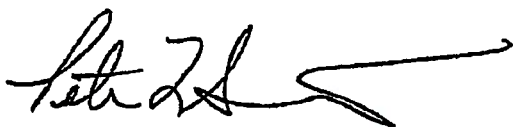
Dear Ms. Vasquez:

It is my pleasure to submit the Smith Environmental & Engineering (SMITH) Proposal to Provide Construction Documents, Bidding Assistance and Construction Administration Services for Lower Capitol Basin Sediment Trap/Wetland aka "Pump House Park" to the City of Cheyenne (the City), in response to RFP-12-16.

SMITH is a full-service environmental engineering consulting company established in June of 2000. We are a licensed business in WY and have a WY registered agent. Located in Westminster, Colorado, we provide design, design-build (DB), design-bid-build (DBB), and construction services in wetland, riparian, and lacustrine environments throughout Colorado and the western United States.

I hereby certify that the information and data contained in the attached proposal are true to the best of my knowledge. If you have any questions regarding our proposal, please contact me at (303) 551-7972, by email at petersmith@smithdelivers.com, or at the Westminster address below.

Respectfully submitted,
Smith Environmental & Engineering



Peter L. Smith, CPSS, PWS, CPESC
Vice President

Primary Point of Contact:
Peter Smith, Principal in Charge
Smith Environmental & Engineering
1490 W. 121st Ave., Suite 101
Westminster, CO 80234
Phone: (720) 887-4928
Direct: (303) 551-7972
Fax: (720) 887-4680
Email: petersmith@smithdelivers.com
Website: www.smithdelivers.com

2.0 PROJECT TEAM

Smith Environmental & Engineering (SMITH) is a full-service environmental engineering consulting company established in June of 2000. We are a licensed business in WY and have a WY registered agent. SMITH has extensive experience on projects coordinating multi-disciplinary solutions – integrated engineering and biological system solutions. We have extensive experience coordinating landscape requirements with stormwater quality treatment activities and integrating them into an urban setting. We have successfully managed numerous state and federal funded projects. We have extensive environmental remediation experience at sites with petroleum hydrocarbon and heavy metal contamination and especially integrating remediation solutions using treatment wetland technologies.



Located in Westminster, Colorado, we provide design, design-build (DB), design-bid-build (DBB), and construction services throughout the Rocky Mountain Region. Our company has provided public and private sector clients with expertise on more than 1,100 such projects in the last 15 years. SMITH is comprised of a 32-person staff that includes Professional Engineers (PE) registered in Wyoming and Colorado, Professional Wetland Scientists (PWS), Certified Asbestos Building Inspectors (CABI), Registered Environmental Managers (REM), cultural resource professionals certified with the Wyoming and Colorado State Historic Preservation Offices, Certified Safety, Environmental and Emergency Managers (CSEM), Certified Professional Soil Scientists (CPSS), Certified Professionals in Erosion and Sediment Control and public information specialists. SMITH also provides expertise in AutoCAD and Geographic Information System (GIS) analyses and database development.

SMITH has a comprehensive understanding of stormwater treatment in wetland environments. This comes from an extensive knowledge of soil-water-flora-fauna relationships in riverine, upland and wetland environments, and the design and the construction process associated with creating and maintaining treatment wetland sites in urban or rural environments. Over the last 15 years, SMITH has brought the following types of expertise to the analyses and design of water quality treatment wetlands:

- Stormwater quality assessment for contaminants
- Large scale system hydraulic and numeric modeling
- EPA SWMM hydraulic modeling
- Design and construction management of treatment wetlands
- Multifunctional park design
- Sustainable ecological design
- Environmental cleanup expertise and EPA Region 8 crosswalk linkage between SAP and QAPP
- Concept through final design of stormwater treatment wetlands - site design, construction drawings, and technical specifications
- Phase I and II environmental site assessments of petroleum and heavy-metal contaminated sites
- Development and implementation of remediation plans at petroleum hydrocarbon and heavy-metal contaminated sites having cultural resources constraints
- Contaminant and pollutant degradation modeling
- Human health and ecological risk assessments

- Voluntary Cleanup (Remediation) Plan development and agency coordination in Brownfield areas
- Integration of trails and urban landscaping with treatment wetland and riparian areas

SMITH will partner with five other firms in order to provide the services required under this contract. These sub consultants are landscape architecture firm LLG International LLC (LLG), Terracon Consultants Inc. (Terracon), Geocal Inc. (Geocal), Colorado State University Soil and Water Testing Laboratory, and Cheyenne-based Steil Surveying Services, LLC (Steil).

Geocal was established in 1989 and maintains a staff of engineers, environmental scientists, professional geologists, construction quality control specialists, and support staff who have an uncompromising commitment and dedication to quality and service. Geocal is a Wyoming-registered DBE. SMITH has worked with Geocal on several projects. On this project they will assist in geotechnical and environmental remediation design efforts to more than fulfil the City of Cheyenne's (City) DBE goal on the project.

The Colorado State University (CSU) Soil and Water Testing Laboratory will provide any necessary laboratory analyses. We will use the CSU laboratory only because the UW soil testing lab is no longer open.

LLG was founded in 2009 in response to the need for idea-driven design to integrate living city infrastructure with developing cities. The LLG staff has culturally diverse experience in landscape ecology, urban design and landscape architecture. With a combined 50 years of planning and design practice, LLG's members have worked together since 2005 on projects in the Rocky Mountain Region and throughout the United States, as well as in the Middle East and China.

Terracon will provide expertise in geotechnical investigations, assessing slope stability along the railroad tracks, W Lincolnway, and Bent Avenue, as well as providing materials testing as needed during construction. Terracon's Brent Wilkins, who is joining the team for this effort, is a Wyoming-registered P.E.

Established in 1982 in Cheyenne, Steil is a full service land surveying firm offering a variety of surveying services to both public and private sector entities. Steil is a recognized leader in all types of surveying and development planning and consulting, with over 150 years of combined field experience. They also offer consulting, project coordination, and representation for individuals, corporations, and public-sector clients.

2.1 PROPOSED PROJECT TEAM AND ORGANIZATION

SMITH Personnel

Jonathan Diller, P.E., Project Manager/Senior Water Resources Engineer – Mr. Diller has been working in the field of stormwater quality management and EPA regulations of stormwater quality for over 25 years. Mr. Diller has also been developing the advocacy of constructed wetlands BMP for the management of stormwater quality and quantity for over 10 years. Mr. Diller is currently working on a Master's thesis at the University of Florida that involves the modeling of the National BMP Database in a series of EPASWMM models. This work includes the modeling of the performance of BMP

wetlands and the removal of *E. coli*. Mr. Diller has also taken graduate work in Wetlands and Water Quality, and Urban Soils and Water Systems as part of his master's program of Soil and Water Science (24 of 30 hours complete.) Mr. Diller has experience in designing wetlands systems for stormwater quality improvement in an urban setting and the design of wastewater treatment wetlands subject to wide swings in water flow rate including periods of no flow. This includes the design of stormwater treatment wetlands in Northeastern Illinois which has extensive cold weather periods in the winter, similar to Cheyenne. He has developed concept through final design of stormwater treatment wetlands - site design, construction drawings, and technical specifications.

Peter Smith, PWS, CPSC, CPESC, Principal/Hydrologist/Wetland Scientist – Mr. Smith is a Principal/Vice-President of SMITH. His qualifications as a Principal, Project Manager, landscape designer, wetland scientist, soil scientist and hydrologist include more than 35 years of experience on over 1,200 projects managing project teams and budgets, preparing Project Management Plans (PMP) and QA Project Manuals, conducting studies, preparing concept through final designs, obtaining permits, preparing construction drawings and specifications, and providing construction services on DB and DBB projects. He has developed stormwater treatment wetlands plan sets for projects throughout the Rocky Mountain Region. Mr. Smith's qualifications include a comprehensive understanding of stormwater treatment wetlands, which comes from an extensive knowledge of soil-water-flora-fauna relationships in riverine, upland and wetland environments, and the design and the construction process associated with creating and maintaining wetland mitigation and stream restoration sites. This understanding comes from completing more than 700 wetland mitigation, risk assessment, remediation, stream restoration, soil survey, irrigation suitability, fluvial geomorphic, hydrologic, geochemical, and vegetation studies and treatment wetland designs. He has been Principal and Project Manager for more than 1,200 environmental projects and managed project budgets in excess of \$1 million. As a PWS, Mr. Smith serves as the QA Manager on all treatment wetland projects.

Darrin Masters, CWB, Assistant Project Manager – Mr. Masters has 26 years of professional experience in wildlife, environmental, and wetlands sciences and has served as Project Manager for more than 55 design, DB, and DBB projects where he managed project teams, prepared PMPs and QA manuals. His qualifications include extensive experience in project management, vegetation inventories, water quality analysis, wetland delineation, wetland mitigation design, preparing wetland mitigation construction documents, wetland function analysis, other natural resource inventory and assessments, noxious weed management, and Section 404 permitting. Mr. Masters has a broad range of other technical skills, which include AutoCAD drafting, spatial analyses using ArcGIS, resource use analysis using Compositional Analysis, data management, and radio telemetry. He is certified by Trimble Inc. for mapping grade GPS. He serves as the QA Manager for GIS on all SMITH projects.

Andrew S. Ricker, CSEM, REM, Senior Environmental Scientist – Mr. Ricker has more than 26 years of experience as a Senior Environmental Scientist and Health and Safety Manager. He has substantial environmental cleanup expertise and extensive experience preparing EPA Region 8 crosswalk linkage between SAP and QAPP. He has completed numerous Phase I and II environmental site assessments of petroleum and heavy-metal contaminated sites and developed and implemented remediation plans at petroleum hydrocarbon and heavy-metal contaminated sites, some having cultural resources constraints. He has completed contaminant and pollutant degradation modeling and many human health and ecological risk assessments. Mr. Ricker has served as the HSO on many Brownfield and transportation

projects and has authored numerous Health and Safety Plans (HASP), QA Plans (QAP), Sampling and Analysis Plans (SAP), No Further Action (NFA) Requests, Site Characterization Reports (SCR), Corrective Action Plans (CAP) and groundwater monitoring reports.

Rebecca L. Hannon, Qualified Supervisor (Q.S.), Wetland Scientist II – Ms. Hannon has more than six years of experience as a wetland scientist and wildlife biologist specializing in environmental permitting. Her experience includes wetland delineations, habitat suitability assessments for T&E species, small mammal trapping, bird surveys, MBTA compliance, and noxious weed surveys and removal. She has performed more than 30 wetland investigations and functional assessments and has prepared NWP and IP permit applications. She is highly experienced in the use of mapping grade GPS equipment and in the spatial analyses of resource layers using ArcGIS. She has authored many technical reports – BA, wetlands, wetland mitigation, riparian restoration, noxious weed management and wildlife management. She has also provided management and planning services for numerous wildlife projects.

Randall Ritchey, P.E., Quality Control and Structural Engineer – Mr. Ritchey has more than 45 years of experience in the planning, management, design, and construction of storm water, bridge and transportation projects. Projects have included wetland mitigation areas, detention ponds, interstate highways, light and heavy rail, pedestrian/bike trails and drainage structures. Mr. Ritchey's experience includes the management of projects, departments, and offices. Several of his projects have received awards for creative and aesthetically pleasing designs. Mr. Ritchey serves as the QA Manager for Construction Drawings on all SMITH design projects.

Greg Bradford, P.E., Civil/Environmental Engineer – Mr. Bradford has more than 30 years of experience in civil and environmental investigation, design, permitting and construction. Investigations have been for collection of environmental, geologic and geotechnical data on soils, surface waters and ground waters, remediation and construction. Design and permitting work has included landfills, impoundments, stream channel restoration, remedial actions, tank systems, pump and piping systems, small dams, surface water monitoring, diversion and collection systems, and ground-water monitoring and control systems. Mr. Bradford has worked as a surveyor, superintendent, cost estimator, contract administrator, field engineer/inspector, design engineer, project engineer and Project Manager.

Dr. Deborah Dunn, Environmental Scientist II/Designer/Water Quality Specialist – Dr. Dunn has more than 20 years of professional experience in environmental sciences, with an emphasis on plant ecology, wetland design, and waste water analysis. Her qualifications include plant survey and species inventories, plant community mapping, wetland design, and community waste water analysis specifically for *E. coliform*.

Jason Futey, Public Information Manager – Mr. Futey has more than 10 years of experience in Public Information and Engagement. As SMITH's Public Information Manager, he is highly skilled in public outreach including planning and executing public meetings and maintaining telephone hotlines and email databases for communication with stakeholders during projects.

LLG Personnel

Jeff Lakey, Principal, Registered Landscape Architect – Mr. Lakey practices ecological landscape architecture and community design based on the deep interrelatedness of people and nature, conjoining ecological integrity with artistry in design. His experience is strong in master planning and insightful

urban or site design. He is outstanding at making memorable places of clear and strong character, from the biggest ideas to the smallest construction details, he makes places economically and ecologically effective, and appealing to the senses.

Terracon Personnel

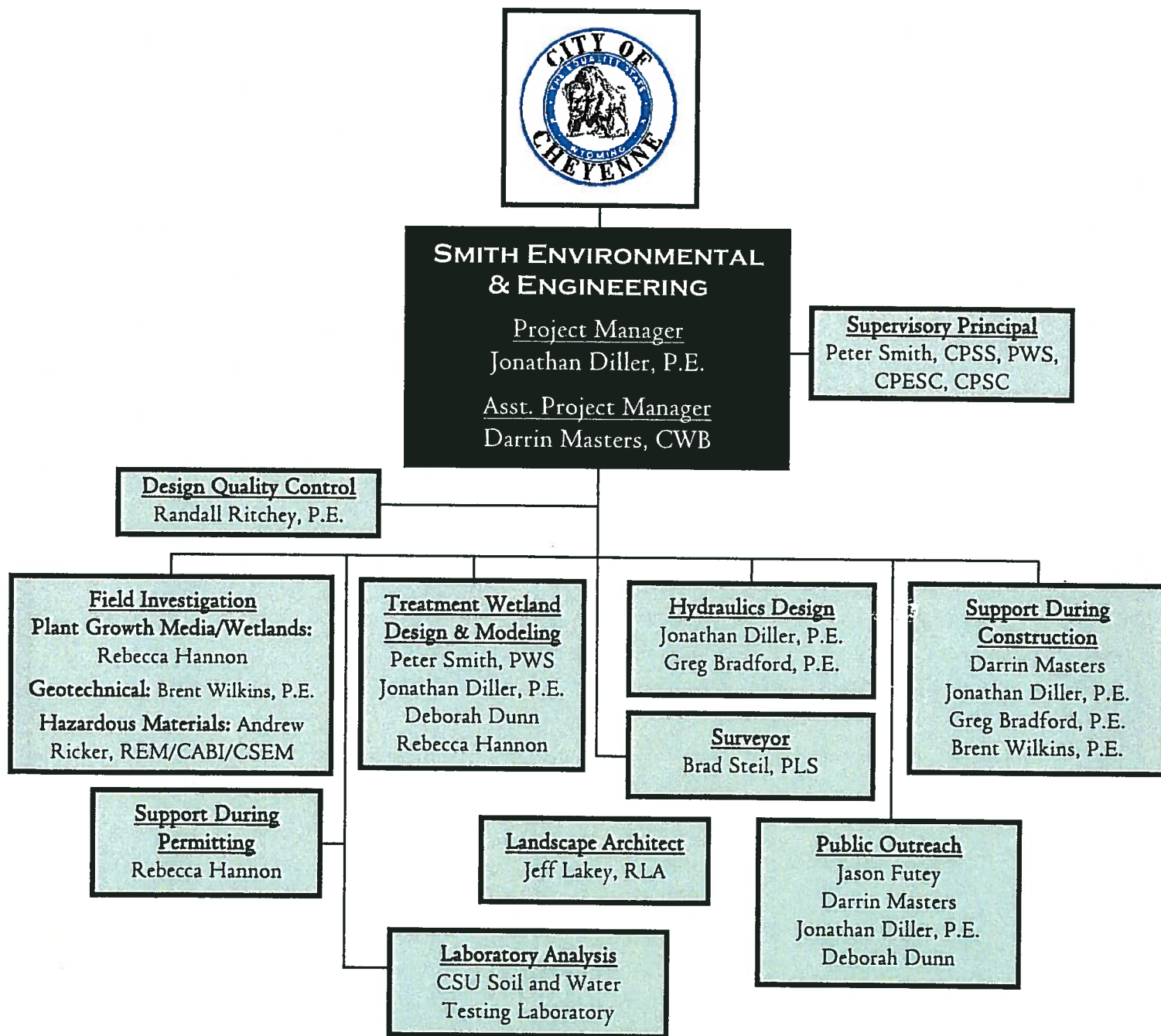
Brent Wilkins, P.E., Geotechnical Engineer – Mr. Wilkins is the manager of the geotechnical engineering department in Terracon's Cheyenne, Wyoming office. He has more than 29 years of experience in geotechnical and civil engineering and construction quality management. Mr. Wilkins supervises the geotechnical operations of the office and provides engineering consultation on projects including foundations, roadways and construction quality control and materials testing.

Steil Personnel

Brad Steil, PLS – Mr. Steil is a Professional Land Surveyor with Cheyenne-based Steil Surveying Services, LLC.

Geocal Personnel

2.2 ORGANIZATIONAL CHART



2.3 RESUMES

Full resumes for key personnel are included on the following pages in the order shown below.

Jonathan Diller, Project Manager/Senior Water Resources Engineer

Peter Smith, Principal/Hydrologist/Wetland Scientist

Darrin Masters, Project Manager/Senior Wetland Scientist

Andrew S. Ricker, Health & Safety Manager

Rebecca L. Hannon, Wetland Scientist II

Randall Ritchey, P.E., Quality Control Engineer

Greg Bradford, P.E., Civil/Environmental Engineer

Deborah Dunn, Environmental Scientist II/Designer/Water Quality Specialist

Jason Futey, Public Information Manager

Jeff Lakey, Principal Landscape Architect

Brent Wilkins, P.E., Geotechnical Engineer



SMITH ENVIRONMENTAL & ENGINEERING

Sustainable Environmental Solutions

Education

M.S. Soil and Water Science, University of Florida, (in progress)

B.S. Chemical Engineering, Rose-Hulman Institute of Technology, 1983

Employment History

Smith Environmental & Engineering

Jonathan M. Diller, P.E.

Flood Rebuild Advocate – Lyons Emergency Assistance Fund

Floyd Browne Group

BA Engineers, Inc.

Hanson Professional Service Inc.

Robinson Engineering, Ltd.

Ruettiger, Tonelli & Associates Paragon Professional Services

Resource International, Ltd.

Berkley, Howell & Associates

Springstead Engineering

Rose-Hulman Institute of Technology

Selected Publications

USEPA Office of Water NPDES Storm Water Phase II Rule presented to the Illinois Association for Floodplain and Stormwater Management Annual Meeting March 2001, revised and represented for the March 2002 Annual Meeting

Point/Nonpoint: An Introduction to the NPDES Phase II Program Storms and Floods, Fall 2005, Written by Jonathan M. Diller, PE, CFM and Tom Tomaszewski, CFM with Troy Golem

Professional Registrations, Certificates, and Memberships

Registered Professional Engineer: Colorado and Wyoming; NCEES record

Certified Floodplain Manager (CFM)

Colorado Association of Stormwater and Floodplain Managers

Will County Stormwater Management Committee (Member 1999-2004, Secretary 2005-2007)

Editorial Committee Member for Storms and Floods (IAFSM) (2005-2007)

Past Chairman National Membership Committee American Institute of Chemical Engineers

Jonathan Diller, P.E., CFM

Project Manager/Senior Water Resources Engineer

Qualifications Summary

Mr. Diller has been working in the field of stormwater quality management and EPA regulations of stormwater quality for over 25 years. Mr. Diller has also been developing the advocacy of constructed wetlands BMP for the management of stormwater quality and quantity for over 10 years. Mr. Diller is currently working on a master's thesis at the University of Florida that involves the modeling of the National BMP Database in a series of EPASWMM models. This work includes the modeling of the performance of BMP wetlands and the removal of e. coli. Mr. Diller has also taken graduate work in Wetlands and Water Quality, and Urban Soils and Water Systems as part of his master's program of Soil and Water Science (24 of 30 hours complete.) Mr. Diller has experience in designing wetlands systems for stormwater quality improvement in an urban setting and the design of wastewater treatment wetlands subject to wide swings in water flow rate including periods of no flow. This includes the design of a stormwater treatment wetlands in Northeastern Illinois which has extensive cold weather periods in the winter.

Experience Summary

Some of Mr. Diller's previous experience is described below.

- Pinello Ranch Wetland Mitigation Site Design, Colorado Springs, CO – see the project brief in section 6.0
- Consultant to Town of Lyons, Lyons, CO – Tasks include review of internal and external floodplain develop permits including HEC-RAS models, and review of land development projects.
- Flood Rebuild Advocate – Lyons Emergency Assistance Fund, Lyons, CO – Provides assistance to homeowners in obtaining floodplain development permits, coordinates design issues with insurance companies including NFIP and private underwriters, and meets with homeowners on site to discuss specific issues with proposed rebuild and/or flood protection activities.
- BA Engineers, Inc., Louisville, KY – Senior Project Manager and Operations Director. Served as PM for Logan Street combined sewer overflow diversion and storage project and Shadow Wood Sewer Diversion Project and Treatment Plant decommissioning project; provided expert witness evaluation for potential flooding lawsuit.
- Hanson Professional Service Inc., Lakewood Ranch, FL – Senior Project Manager; Project work included: preparation of master drainage plans, stormwater discharge models, and water and sewer drawings for regional and international airports; assessment of flooding impacts for rainfall events that impacted railroads in MO and KS; and SWPPP training.
- Robinson Engineering, Ltd., Frankfort, Minooka & Ottawa, IL – Senior Water Resources Engineer for watershed planning, flood studies and stormwater management system design and development plan reviews; Technical Director for Stormwater NPDES compliance activities; Technical Manager for Floodplain development and stormwater best management projects; Coordinator for permitting and compliance for all design projects that impact wetlands; Designer of multiple bridge projects including both new and replacement bridges for both roadway and pedestrian uses including projects within regulatory floodways.
- Resource International, Ltd., Ashland, VA – Project Engineer; Project work included: subdivision, commercial, industrial and institutional site plan design; stormwater NPDES Phase I permitting; regional stormwater management consulting services; engineering services including BMP program development, stormwater modeling, Chesapeake Bay Preservation Act Compliance and environmental impact assessments; stormwater runoff BMP design; regional landfill design projects; environmental permitting for scrap tire recycling facilities; groundwater remediation consulting services; wastewater treatment plant design for nutrient removal; and oil spill management programs.



SMITH ENVIRONMENTAL & ENGINEERING

Sustainable Environmental Solutions

Education

B. S., *Watershed Science (Hydrology)*, Colorado State University, 1976

Graduate Coursework, *Soil and Range Sciences*, Colorado State University, 1978

Employment History

Smith Environmental & Engineering
Sugnet and Associates
Stoneman-Landers, Inc.
Camp Dresser and McKee
Woodward-Clyde Consultants
Soil Conservation Service, USDA

Professional Registrations/Certifications

Certified Professional Soil Classifier, #1785, Soil Science Society of America
Certified Professional Erosion and Sediment Control Specialist, #512, Soil and Water Conservation Society
Registered Professional Soil Scientist, #49, National Society of Consulting Soil Scientists
Professional Wetland Scientist, #1273, Society of Wetland Scientists
Federal Fish & Wildlife Permit – ULTO, CBP and PMJM

Professional Membership

Soil and Water Conservation Society of America
National Society of Consulting Soil Scientists
Society of Wetland Scientists

Selected Publications

Smith, P. L., 2015. *Wetland Mitigation Banks: A Revenue Generation Method in Colorado Open Space Areas*, 2015 Colorado Open Space Alliance Meeting, Snowmass, Colorado.

Smith, P. L. and P. Wayland. 1999. *Wetland Mitigation Sites – Attenuating Solids, Salts and Metals in Irrigation Water*. In *Wetlands and Bioremediation, An International Conference*, Conference Proceedings. Salt Lake City, Utah.

Smith, P. L., Redente, E., and Hopper, E. 1987. *Soil Organic Matter*. In: *Reclaiming Mine Soils and Overburden in the Western United States: Analytical Parameters and Procedures*. Soil Conservation Society of America. Ankeny, Iowa.

Peter Smith, PWS, CPECC

Principal/Hydrologist/Wetland Scientist

Qualifications Summary

Mr. Smith's qualifications as a Principal, Project Manager, hydrologist, stormwater quality specialist, soil scientist, wetlands scientist, and treatment wetland designer include more than 1,250 completed projects over the last 38 years. On these projects he has managed, planned, designed and built water quality treatment systems for: 1) urban settings to remove sediments, N, P, metals, PCE, and *E. coli*; 2) mines with acid drainage to remove sediment and metals using wetlands; 3) petroleum hydrocarbon contaminated sites to remove PAH, DRO, GRO, and other VOC's; and 4) Individual Sewage Disposal Systems (ISDS) for on-site residential sewage disposal. He has conducted treatment wetland permitting and mitigation site design; built wetland treatment and mitigation areas; restored streams; performed ecological and human health risk assessments; prepared SWMPs and conducted soil erosion modeling; conducted wetland, soil and vegetation surveys; prepared reclamation plans and performed site reclamation; prepared biological and ecological assessments; designed threatened and endangered (T&E) species habitat; prepared NEPA documents (EAs and EISs); conducted remedial investigations at hazardous waste sites; conducted HEC RAS and HMS modeling for floodplain analyses; and managed cultural resource studies and developed Section 106 mitigation. He has authored or coauthored over 2,500 plan sets and/or technical reports as a consultant, made 15 professional presentations, co-authored a book on disturbed land reclamation, published over 14 professional papers, and holds 5 professional registrations/certifications. As Principal and Project Manager he has supervised and managed hundreds of federally-funded projects, including 319 grant funded projects, and many project teams and budgets greater than \$1 million. He was the Principal in Charge for all the projects shown in Section 6.0.

Experience Summary

Some of Mr. Smith's relevant project experience is described below.

- Stone Creek Ranch, Near Parker, CO – Design of treatment wetlands (2 ac) to remove P, other nutrients, *E. coli*, and sediment from stormwater flow before it enters Cherry Creek for this 300 ac residential development.
- Banning Lewis ranch, Colorado Springs, CO – Environmental studies and treatment wetlands design (4 ac) to remove salts and sediment from stormwater flow before it enters Sand Creek -10 sq mi development.
- Durango Wetland Mitigation Bank, Durango, CO – An assessment of the mitigation bank's performance attenuating solids, salts and metals in the water moving through it.
- Sanctuary on the Park, Loveland, CO – Concept design of a urban stormwater system using treatment wetlands to removed sediments and salts.
- Denver Water Quality Treatment Manual, Denver, CO – Co-authored the CCD's Stormwater Guidance and Planning Manual.
- Water Quality Treatment Pond - 52nd and Emerson, Denver, CO – Planning and design of treatment wetlands to remove sediment, fecal coliform, *E. coli*, and metals from flow entering this 19 ac urban detention pond.
- Argo Mine, Boulder County, CO – Design, permitting and build services at the Argo Mine site – a VCUP site. The purpose of this project was to: 1) reduce metals leaching and erosion of mine wastes into Little James Creek, 2) improve water quality in Little James Creek, and 3) decrease the interaction of mine wastes with precipitation and surface run-off.
- Jessie Mine/Mill Site, Summit County, CO – Design and construction of cleanup measures in accordance with the VCUP permit. Design and construction of a new 1,000-ft long, fish-compatible channel and wetlands (and other features) to divert runoff from the old tailings along the channel; design including HEC RAS modeling to determine the for the 100-yr flood event.
- Several ISDS, CO – ISDS design/modeling *E. coli* and fecal coliform removal.
- Cornerstar Development, Centennial, CO – Design of treatment wetlands (3 ac) to remove P, *E. coli* and sediment from stormwater flow before it enters Cherry Creek at this 30 ac commercial development.



**SMITH ENVIRONMENTAL &
ENGINEERING**

Sustainable Environmental Solutions

Education

*M. S. Wildlife Ecology and Conservation,
The University of Florida-Gainesville, 2002*

*B. S. Environmental Biology,
Heidelberg College-Tiffin, OH, 1988*

Employment History

*Smith Environmental & Engineering
Hoopa Valley Indian Tribe
Florida Fish and Wildlife Conservation Commission
The University of Florida
Anchorage Water and Wastewater Utility*

Selected Publications

*Dobey, S., Masters, D., Scheick, B., Clark, J. D.,
Pelton, M., Sunquist, M. Ecology of Florida
Black Bears in the Okefenokee-Osceola
Ecosystem. Wildlife Monographs, no. 158,
January 2005.*

*Nelson, S., Masters, D., Humphrey S., Kunz T. Fruit
Choice and Calcium Block Use by Tongan Fruit
Bats in American Samoa. Journal of
Mammalogy, December 2005.*

Selected Presentations

*Department of the Army Permitting Requirements
and Regulations. Session speaker, Colorado
Airport Operators Association Environmental
workshop, Denver International Airport.
January 2015.*

*Navigating the Migratory Bird Treaty Act. Session
speaker, Colorado Open Space Alliance Annual
Conference, Crested Butte, Colorado.
September, 2013*

*Bear-Human Conflicts on the Hoopa Reservation in
Northwestern California: What's the Problem?
Poster presented at the Fifteenth International
Conference on Bear Research and
Management. February 8-13, 2004. San
Diego, California.*

Professional Registrations and Affiliations

*2011 Certified Wildlife Biologist – The Wildlife
Society*

2015 The Wildlife Society - National Membership

*2015 The Wildlife Society - Colorado Chapter
Membership*

Darrin Masters, CWB

Assistant Project Manager/Senior Wetland Scientist

Qualifications Summary

Mr. Masters has 26 years of professional experience in watershed assessment, water quality analysis, microbiology, and wetlands sciences. His qualifications include extensive experience in project management, water quality analysis for total and fecal coliform bacteria, wetland delineation, riverine and wetland mitigation area design, wetland function analysis, vegetation inventories, ecological restoration planning and design, NEPA document preparation, public involvement meetings and coordination, and Department of the Army permitting. Mr. Masters has nine years of experience managing an EPA certified microbiology laboratory to meet standards for testing drinking water and wastewater.

Mr. Masters has a broad range of technical skills, which includes water chemistry and laboratory analysis based on standard methods, design and oversight of wetland mitigation area and secondary stormwater treatment wetland construction, watershed planning, coordination with federal permitting, and federal grant management.

Experience Summary

Some of Mr. Masters' relevant project experience is described below.

- Pinello Ranch Wetland Mitigation Area Design, El Paso County, CO – Project Manager. Managed a team to design a 9.81 ac wetland as compensatory mitigation for impacts to Waters of the U.S. During design, we evaluated ground and surface water hydrology, soils, and plant assemblages to ensure a high standard of wetland function when the project is built.
- Stone Creek Ranch Development, Douglas County, CO – Project Manager. Developed secondary stormwater treatment wetlands to reduce phosphorus loading into Cherry Creek from a proposed development. Developed plant assemblages that maximize phosphorus adsorption and designed wetlands to retain surface water for the duration necessary to take full advantage of treatment.
- Royal Tiger Mine and Mill Restoration Project, Summit County, CO – Evaluated existing wetland function and hydrology for placement and design of proposed constructed wetlands; designed wetland restoration areas to facilitate the reduction of heavy metal contamination from mine waste rock into the watershed; restoration included creating sustainable and functioning wetlands
- Jessie Mine and Mill Restoration Project, Summit County, CO - Designed wetland restoration areas to facilitate the reduction of heavy metal contamination from mine waste rock into the local ponded ecosystem and ultimately into the watershed; consulted on design of a diversion channel to ensure sustainable wetland hydrology.
- Falcon Drainage Basin Planning Study, El Paso County, CO – project manager; created a GIS map of environmental resources in the Falcon Watershed in El Paso County; evaluated wetland hydrology and riparian extent, shallow ground water, and highly erodible soils; included elk, pronghorn, black bear, and mountain lion range and distribution, and raptor habitat.
- South Platte River Vision, Denver County, Denver, CO – Consulted with the Denver Natural Resources Department on wetland design and placement. Reviewed design drawings to ensure that proposed wetland areas meet hydrological requirements.
- Goelein Gulch Wetland mitigation Construction, Durango, CO – Performed a soil survey to identified areas with unsuitable soils and made recommendations for soil salvage and importation. Recommended plant assemblages to sustain healthy wetlands. Inspected construction activities to ensure that final grade provided the hydrology needed for the proposed wetland plant groups.



**SMITH ENVIRONMENTAL &
ENGINEERING**

Sustainable Environmental Solutions

Education

B. S. Geology, University of Maryland, 1989

Employment History

Smith Environmental & Engineering

Adecco Technical, Inc.

Savci Environmental Technologies, LLC.

PMT & Associates, Inc.

**Professional Registrations and
Certifications**

CDLE OPS Individual Listed Consultant #6183

REM Registered Environmental Manager,
National Registry of Environmental
Professionals

CSEEM Certified Safety, Environmental, and
Emergency Manager, National Registry of
Environmental Professionals

OSHA Hazardous Materials Operations, 40-Hour
Safety Training Course and Refresher
Courses

CDPHE, Air Pollution Control Division Asbestos
Building Inspector/ Certification No: 12747

CDPHE, Air Pollution Control Division Asbestos
Management Planner/ Certification No:
12747

CDPHE, Air Pollution Control Division Asbestos
Project Designer Certification No: 12747

Lead Inspector, L II 878 07-08, Misers Inspection
and Training Inc., Englewood, CO

Lead Risk Assessor LRAI 878-07-08, Misers
Inspection and Training Inc., Englewood, CO

Continuing Education

Bioremediation Workshop, University of
Maryland, 1995

Fracture Trace and Lineament Analysis, National
Ground Water Association, State College, PA,
1994

Ground Water Monitoring and Sampling, National
Ground Water Association, Harrisburg, PA,
1992

Geophysics Applied to Environmental, Engineering
and Ground Water Investigations,
Blackhawk Geosciences, Inc., College Park,
MD, 1991

Site Characterization for Subsurface
Remediation's, USEPA, Boston, MA, 1990

Andrew S. Ricker, CSEEM, REM

Senior Environmental Scientist

Qualifications Summary

Mr. Ricker has more than 26 years of experience as a Project Manager and Environmental Scientist. He has substantial environmental cleanup expertise and extensive experience preparing EPA Region 8 crosswalk linkage between SAP and QAPP. He has completed numerous Phase I and II environmental site assessments of petroleum and heavy-metal contaminated sites and developed and implemented remediation plans at petroleum hydrocarbon and heavy-metal contaminated sites, some having cultural resources constraints. He has completed contaminant and pollutant degradation modeling and many human health and ecological risk assessments. Mr. Ricker has served as the HSO on many Brownfield and transportation projects and has authored numerous Health and Safety Plans (HASP), QA Plans (QAP), Sampling and Analysis Plans (SAP), No Further Action (NFA) Requests, Site Characterization Reports (SCR), Corrective Action Plans (CAP), groundwater monitoring reports, and Work Plans. Field operation experience includes operating and maintaining pump and treat groundwater remediation systems, overseeing vacuum truck free phase petroleum product and soil vapor removal operations, overseeing chemical oxidation injection remediation, installing monitoring and production wells, and removing underground storage tanks, hollow-stem auger drilling with split-spoon sampling, mud-rotary drilling, well gauging and sampling, direct-push investigations, geophysical surveys, and aquifer testing.

Experience Summary

As SMITH's Safety Manager, Mr. Ricker has provided safety management services on many projects. Some are described below.

- Adams St. College LUST Remediation, Alamosa, CO – Performed in-situ remediation services at this LUST site, which included a feasibility analysis to select an appropriate remediation technology. Three remediation technologies were evaluated including excavation with disposal, excavation and land "farming", and in-situ remediation using chemical ORC. ORC injection of a pilot area was conducted and groundwater monitoring was conducted. Conducted risk-based analysis that led to the site receiving regulatory closure from CDLE OPS.
- Burt Toyota, Englewood, CO – Conducted a subsurface soil and groundwater investigation – LUST site. Site Characterization Report, Corrective Action Plan, and Quarterly monitoring reports, ORC injection remediation, asbestos abatement, risk-based analysis and Phase II reports.
- I-25 and Santa Fe, Denver, CO – Served as HSO both on site and in generating the HASP and MMP associated with the construction of a new interchange in an area with numerous contaminant sources.
- CDOT, Numerous throughout Colorado – Served as the HSO, monitoring the implementation of site specific HASPs and MMPs.
- Former Circle K #3875, Westminster, CO – Generated a VCUP application for the project and received a No Action Determination Letter from CDPHE.
- Sanctuary on the Park, Loveland, CO – Conducted Phase I and II ESAs of the site. Generated a VCUP application for the project and obtained a No Action Determination Letter from CDPHE.
- North Carroll Shopping Center, Hampstead, MD – Conducted extensive subsurface soil and groundwater investigation for PCE contamination, including supervising the installation of a groundwater pump and treat system, conducting aquifer testing and analysis, risk based analysis, the operation and management of the treatment system, and conducting quarterly monitoring of 11 groundwater wells over a three year period, resulting in the eventual regulatory closure from MDE of the case.
- Argo Mine, Boulder County, Colorado – Design-Build Waste Rock and Acid Mine drainage remediation - under a federally-funded Brownfield grant. Prepared the VCUP application and the ABCA report for the site, and provided construction oversight of SMITH and subcontractor construction teams.
- 92nd and Lowell Plume, Westminster, CO – Generated a VCUP application for the project and obtained a No Action Determination Letter from CDPHE



SMITH ENVIRONMENTAL & ENGINEERING

Sustainable Environmental Solutions

Education

B. S. Natural Resources Management, Minors in Spanish and Conservation Biology, Colorado State University, 2009

Employment History

Smith Environmental & Engineering
Missouri River Communities Network
Colorado State University Office of Admissions
Johnson County Parks and Recreation District

Certifications

OSHA – 40-Hour HAZWOPER, 2012
OSHA – 10-Hour Training for the Construction Industry, 2011
Certified Operator, Forest, Rangeland, Aquatic, Industrial and Right-of-Way, Public Health and Turf Control, Colorado Department of Agriculture, #26278, 2012
Qualified Supervisor, Outdoor Vertebrate Control, Colorado Department of Agriculture, #26278, 2012
Adult, Child, Infant CPR/AED, 2015
Standard First Aid, 2015

Trainings

Railroad Safety Training, UPRR and BNSF, 2015
Colorado Native Plant Master, 2014
Missouri Stream Team Program, 2009-2010
City of Columbia TreeKeepers Program, 2010
City of Columbia Aquatic Restoration Program, 2010

Rebecca L. Hannon

Wetland Scientist II

Qualifications Summary

Ms. Hannon has more than five years of experience as a wetland scientist and wildlife biologist specializing in environmental permitting. Her experience includes wetland delineations, T&E species HSA, small mammal trapping, bird surveys, Migratory Bird Treaty Act compliance, and noxious weed surveys and removal. She has performed more than 30 wetland investigations and functional assessments and has prepared 404 permit applications for the U. S. Army Corps of Engineers. She is highly experienced in the use of both recreational and mapping grade GPS equipment, and in the spatial analyses of natural resource layers using ArcGIS. She has coauthored technical reports including Biological Assessments, and has conducted several independent research projects. She has provided management and planning services for projects involving black-tailed prairie dogs (*Cynomys ludovicianus*).

Experience Summary

A select list of some of Ms. Hannon's project experience is described below.

- Sunset Bridge, Longmont, CO – Completed a Waters of the US delineation, habitat suitability assessment for threatened and endangered species and noxious weed survey for a bridge that was washed out by flooding on St. Vrain Creek. Assisted on the Section 4(f) and 6(f) evaluation, and completed an SB40 evaluation for CPWV.
- DIA Wetlands, Denver, CO - Performed wetlands delineation along 3 mi of Third Creek and 2.8 mi of Second Creek at the Denver International Airport. Mapped all wetlands with a Trimble sub-meter GPS, processed data into ArcGIS features with associated metadata.
- Tower Landfill, Commerce City, CO – Performed a Functional Assessment of Colorado Wetlands (FACWet) analysis for wetlands along Second Creek on property owned by the Tower Landfill near the Denver International Airport. Provided the results and scores for inclusion in a comprehensive wetland report and Section 404 permit application.
- Sandstone Ranch, Larkspur, CO – Updated the wetland delineation for entire 2,000 ac property in 2014. Prepared a report of the findings for submission to the USACE to extend the existing Section 404 permit. Performs annual point-intercept vegetation surveys to assess Preble's Meadow Jumping Mouse habitat in proposed housing development and authors annual report.
- Weld County Road 1 Bridge, Longmont, CO – Completed a Waters of the US delineation and habitat suitability assessment for threatened and endangered species for a bridge that was washed out by flooding on St. Vrain Creek. Identified an active Bald Eagle nest and prepared a technical memo to guide planners in staying compliant with the Bald and Golden Eagle Protection Act. Performed weekly migratory bird surveys prior to construction, monitored the eagles during construction activities and served as Project ECS.
- Stapleton Road Extension, El Paso County, CO – Completed a Waters of the US report for a new road alignment. A Wetland Finding Report and functional assessment were also completed as required by CDOT when wetland impacts within the right-of-way are significant.
- Stone Creek Ranch, Douglas County, CO – Completed several environmental surveys required for a proposed housing development. Prepared a bird protection plan, weed management plan, Section 404 permit application and mitigation plan, and riparian enhancement plan. Also prepared a Biological Assessment for the Preble's Meadow Jumping Mouse.
- SWIF Homes - Categorical Exclusions, San Juan County and McKinley County, NM, and Apache County, AZ – Assisted with Categorical Exclusion assessments at nine home sites and utility corridors in the Navajo Nation; produced maps in ArcGIS with baseline data and aerial imagery to assist field crew in identifying the rural sites; created shapefiles of the field collected data for the final CE.



**SMITH ENVIRONMENTAL &
ENGINEERING**

Sustainable Environmental Solutions

Education

B. S. Civil Engineering, Structural Emphasis, University of Colorado, 1967

Registration

Professional Engineer: Colorado 10472

Employment History

Smith Environmental & Engineering
Ritchey Engineering
Huitt-Zollars, Inc.
Washington Group International
Washington Infrastructure Services
MK/Centennial
Centennial Engineering, Inc.
Southern Colorado Prestress
URS/The Ken R. White Co.
HNTB

Continuing Education (Selected)

2000, Ethics in Engineering, ACEC Colorado, 4 PDH
2001, Risk Management and Ethics, ACEC Colorado, 4 PDH
2003, Risk Management and Ethics, ACEC Colorado, 4 PDH
2003, Project Management Training, Washington Group International, 21 PDH
2003, Practical and Legal Aspects of Using Email, ACEC Colorado, 4 PDH
2004, The Deposition, Expert Witness Practice Seminar, ACEC Colorado, 4 PDH
2006, Project Management Training, Huitt-Zollars, 16 PDH
2006, Liability IQ for Architects and Engineers, XL Insurance, 4 PDH
2007, Law for Design Professionals, Jackson Kelly Attorneys at Law, 2.5 PDH
2007, Steel Bridge Workshop, Highway Bridge Services, 16 PDH
2007, Bridge Inspection Refresher Training, National Highway Institute, 18 PDH
2007, The Psychology of Ethics, ACEC Colorado, 4 PDH
2008, Welding Seminar, American Institute of Steel Construction, 8 PDH
2008, Building Code Seminar, American Concrete Institute, 7.5 PDH
2008, Bridge Girder Erection, Colorado Contractors Association, 12 PDH
2010, Annual Expert Witness Practice Seminar, ACEC Colorado, 4 PDH
2010, Applying Expertise as an Expert Witness, ACEC, 11 PDH
2012, Navigating Peer Review, ACEC Colorado, 4 PDH
2012, Administration and Practice Workshop, Dispute Resolution Board Foundation, 6.5 PDH

Randall Ritchey, P.E.

Structural and Quality Control Engineer

Qualifications Summary

Mr. Ritchey has more than 45 years of experience in the planning, management, design, and construction of bridge, transportation, and structural projects. Projects have included interstate highways, city streets, rural roads, light and heavy rail, airports, transit, parking structures, pedestrian/bike trails and drainage structures. Mr. Ritchey's experience includes the management of projects, departments, and offices.

Clients have included CDOT and other state departments of transportation, federal agencies, transit districts, toll road authorities, cities, and counties. Several of his projects have received awards for creative and aesthetically pleasing designs.

Experience Summary

Some of Mr. Ritchey's experience is highlighted below.

- Madison Avenue Bridge over Greeley-Loveland Canal, Loveland, CO – Provided an independent design check of the plans and specifications prepared by another consultant. Presently, he is providing construction phase services. The project includes a two-cell nonstandard concrete box culvert and retaining walls for a trail underneath Madison Avenue. The services also included a load rating analysis. All services were provided in accordance with CDOT requirements.
- Monument Creek Pedestrian Bridges, Colorado Springs, CO – Chief Bridge Engineer for the design of six pre-engineered steel truss bridges. This was a federal-aid project subject to the review and approval of CDOT.
- Golf Cart Bridges, Arrowhead at Vail, CO – Chief bridge engineer for the design of two prestressed concrete double-tee bridges over the Eagle River. The bridges had two spans each with a pier in the Eagle River.
- Vermont Street Pedestrian Underpass, Fort Collins, CO – Project Manager for the design of a pedestrian and bicycle underpass under the Union Pacific Railroad. The underpass was constructed of precast concrete box sections.
- Broadway and College Pedestrian Underpass, Boulder, CO – Supervised the structural design of a cast-in-place concrete box pedestrian underpass. The architectural treatment of the 24-foot wide by eight-foot high structure consists of stone-faced entrance walls, specially designed form-liners for artwork on the interior surfaces, and a skylight in the median of the roadway above. Seven retaining walls on curvilinear and stepped alignments were provided.
- 47th Street over Bikepath near Foothills Parkway, Boulder, CO – Project consisted of the design of a bikepath undercrossing that was constructed with a cast-in-place concrete box section. The project included recessed lighting and an anti-graffiti epoxy coating on the exposed surfaces. Mr. Ritchey was responsible for the supervision and quality assurance of the design.
- Table Mesa Pedestrian Bridge, Boulder, CO – Project manager and senior structural engineer for the preliminary and final design of a new pedestrian bridge that provides access across a six-lane highway from an existing parking structure to a proposed bus stop. The structure is 234 feet long, 12 feet wide and is enclosed with roofing and glass sides. The project features elevator and stair towers on each end. The client is the Regional Transportation District and coordination with CDOT and the City of Boulder was required.



SMITH ENVIRONMENTAL & ENGINEERING

Sustainable Environmental Solutions

Education

B.S. Civil Engineering, Washington State University

Graduate Coursework in Geology & Hydrology,
Boise State University

Graduate Coursework in Hydrology & Earth
Construction, University of Colorado at Denver

Employment History

Smith Environmental & Engineering

Veolia ES

Consulting Engineering

Dogan Construction Company

Hamon Contracting, Inc.

Registrations and Training

Professional Engineer: Colorado

40-hour HAZWOPER training for work at
hazardous waste sites,

8-hour updates current through April 2017

12-hour training for work in confined spaces.

8-hour training for work as a Supervisor on
hazardous waste projects.

24-hour training as Competent Person for
excavations

Greg Bradford, P.E.

Civil/Environmental Engineer II

Qualifications Summary

Mr. Bradford has more than 30 years of experience in Civil and Environmental investigation, design, permitting and construction. Investigations have been for collection of environmental, geologic and geotechnical data on soils, surface waters and ground waters for assessment, management, remediation and construction. Design and permitting work has included landfills, impoundments, stream channel restoration, remedial actions (heavy metals, petroleum, volatiles and semi-volatiles, pesticides, and radioactive wastes for in-situ and onsite treatment and removal for offsite disposal), tank systems, pump and piping systems, small dams, surface water monitoring, diversion and collection systems, and ground-water monitoring and control systems. Mr. Bradford has worked as a surveyor, superintendent, cost estimator, contract administrator, field engineer/inspector, design engineer, project engineer and project manager.

Experience Summary

Some of Mr. Bradford's engineering and materials experience is described below.

- I-25 and Santa Fe, Denver, CO – Served as HSO both on site and in generating the HASP and MMP associated with the construction of a new interchange in an area with numerous contaminant sources.
- Civil/Utility Construction Projects. Projects included land development, road construction, and airport construction. Responsibilities included: working as laborer, grade checker, pipe layer, heavy equipment operator, foreman, superintendent, contract administrator, design engineer, project engineer and project manager; designing sewer and water systems to meet flow and regulatory requirements; preparing drawings and specifications; inspecting excavations, inspecting grading and paving and water and sewer construction, including soils sampling/analyses and testing, for compliance with the specifications; inspecting concrete structures; working with the contractors, developers, town and sewer districts to coordinate construction practices and resolve field problems; reviewing and correcting drawings and specifications; preparing field reports, problems and resolutions, and compliance with the specifications; and, preparing final reports for regulatory review and approval.
- Remediation/Reconstruction. Projects have included volatiles, semi-volatiles, metals, pesticides, PCBs, petroleum products (refineries), LUSTs and low-level radioactive wastes. Responsibilities have included: sampling soils, surface water and ground water; using sampling data to determine type, pathways and extent of contamination; determining types and locations for migration controls; determining remediation methods; negotiating and obtaining approval for remediations; preparing designs, drawing and specifications; hands-on implementation of cleanup, including waste management and consolidation; QA/QC of cleanups; project engineer; and, project manager.
- Municipal, Industrial and Hazardous Landfill Design, Permitting and Construction. Determined volume requirements and designed landfills to fulfill those requirements. Designs included liner and cap systems, leachate control systems, surface water controls, surface- and ground-water monitoring systems, and roads. Determined permitting requirements, and then collected and prepared information to fulfill those requirements (geologic, hydrologic, meteorological, water rights, surveying, etc.). Prepared specifications for landfill construction. Worked as Quality Control/Quality Assurance officer for landfill construction, including: collecting soil samples, interpreting soils analyses, selecting soils for construction, density testing, and observing construction methods; observing synthetic liner installation, testing seaming and patching for leaks, observing leak testing, and observing placement of liner cover materials; installation of gas collection and venting systems; and, inspecting cap construction. Also worked as laborer, grade checker and heavy equipment operator for the excavation and construction (liner and leachate systems installation, materials placement, cap construction) of landfills.



SMITH ENVIRONMENTAL & ENGINEERING

Sustainable Environmental Solutions

Education

Doctorate of Arts Biology, Idaho State University

M.S. Botany, University of Northern Colorado

B.A. Biology Education, University of Northern Colorado

A.S. Survey and GIS, Kansas State University – Salina, KS

Employment History

Smith Environmental & Engineering

Spartan Engineering

LW Survey, Engineering & Design

McPherson County Public Health Department

Bucher, Willis & Ratliff

Bethany College

Oglala Lakota College

Department of Forestry, Dominica, British West Indies

Selected Publications

Dunn, D. Edwards, J. 1980. Forest of Morne Trois Pitons National Park, Dominica, British West Indies, Department of Forestry, Roseau, Dominica.

Dunn, D. 1977. Probable Salt Tolerance Mechanisms of *Sporobolus airoides*. Master's Thesis, University of Northern Colorado, Greeley, CO.

Deborah Dunn, D.A.

Environmental Scientist II/Designer/Water Quality Specialist

Qualifications Summary

Dr. Dunn has more than 20 years of professional experience in environmental sciences, with an emphasis on plant ecology, wetland design, and waste water analysis. Her qualifications include plant survey and species inventories, plant community mapping, wetland design, and community waste water analysis specifically for *E. coli*. Her experiences also extend to public presentations, and project management and reporting.

Dr. Dunn has a broad range of technical skills, including vegetation inventories, interpreting aerial photography, using ArcGIS and AutoCAD Civil to generate maps and project designs, and managing grants and projects.

Experience Summary

A selection of Dr. Dunn's relevant projects experiences are described below.

- Pinello Ranch Wetland Mitigation – Worked closely with project manager to accurately represent the existing plant communities, surface features and ground water data as a computer model in AutoCAD, ensuring that the proposed wetland communities were placed according to their environmental requirements. Produced all of the landscape and culvert design drawings for the 60% and 100% submittal package.
- Forest type inventory of Morne Trois Pitons National Park, Department of Forestry, Roseau, Dominica, British West Indies (a Peace Corps/Smithsonian Institute supported project) – Combined field work with local park rangers and aerial photography with abiotic criteria (elevation, mean temperature, and annual precipitation) to map the locations of all forest types of the 30 square mile park for the forestry department's master design plan. All regions were field verified by either line transects, or in areas that were not readily accessible, with the aid of experienced tree spotters using binoculars. Assisted the designer to designate those areas of the park suitable for tourism and areas which contained species needing special protection. Started inventory list of tree, fern and orchid species in the national park.
- Analysis of salt tolerance in *Sporobolus airoides*, Greeley, CO – Specimens collected from different floodplains in Colorado were grown for six months in hydroponic solutions to which either NaCl, KCl or MgPO₄ had been added. Changes in the concentrations of Na⁺, K⁺ and Mg²⁺ in both the leaves and roots were determined to atomic absorption spectrometry and flame spectrometry. Tissue sections of the leaves were examined for anatomical adaptations to hypersaline conditions.
- Environmental Lab Technician, Bucher Willis and Ratliff, Salina, KS – Analyzed community wastewater effluent to surface waters for total solids, total volatile solids, nitrogen-nitrate and ammonia levels, and *E. coli* count
- Science Research – A Minority Student College Science Grant, Oglala Lakota College, SD – Supervised original student research projects that were presented at the annual meeting of Native American Science Students at Haskell Indian Nations University. Projects included germination requirements for timsila (*Psoralea esculenta*) to determine feasibility of cultivating this important native food that was decreasing population, and the effectiveness of nitrogen treated zeolite (an abundant mineral resource located on the reservation) as a soil additive for wheat cultivation.



SMITH ENVIRONMENTAL & ENGINEERING
Sustainable Environmental Solutions

Education

B.S. Journalism, Ohio University, 1999

Employment History

Smith Environmental & Engineering

Woodward & McDowell, Los Angeles, CA

Horace Heidt's Magnolia Estate Apartments,
Sherman Oaks, CA

SDI Media USA, Culver City, CA

Jason Futey

Public Information Manager

Qualifications Summary

Mr. Futey has more than 10 years of experience in Public Information and Engagement. As SMITH's Public Information Manager, he is highly skilled in public outreach including planning and executing public meetings and maintaining telephone hotlines and email databases for communication with stakeholders during projects. Mr. Futey's skills include information distribution, documentation design and layout, and collaboration with project managers to provide the most accurate and up-to-date information to the public.

Experience Summary

Some of Mr. Futey's public engagement experience is described below:

- SH 119 at Jay Road and Niwot Intersections – For this CDOT project, performed public information services including maintaining a project hotline and logbook of calls. Created and distributed weekly email updates on closures and other project developments. Created and distributed construction update flyers to stakeholders. Performed public outreach to optimize public satisfaction with the project.
- I-25 and US 24 Bridge Repairs – For this CDOT project, set up a phone hotline and email address, created a public information plan, updated the voice message alerting stakeholders of closures. Answered any and all questions and concerns regarding the construction. Created and sent flyers to stakeholders for each phase of construction.
- Havana Street and Parker Road – For this CDOT project, attended a preconstruction meeting, sent out monthly construction update flyers to stakeholders, set up a phone line and email account and responded to any and all questions. Maintained a logbook of stakeholder interactions.
- SH 177 University and Bellevue – For this CDOT project, attended regular preconstruction and project progress meetings to discuss public information approach in concern with project. Managed public dissatisfaction, maintained hotline and email addresses, responded to any and all questions and concerns about the project. Developed and distributed construction update flyers to more than 800 stakeholders.
- SH 350 Historic Bridge over Purgatoire River – For this CDOT project, attended a preconstruction meeting, sent out regular construction update flyers to stakeholders, set up a phone line and email account and responded to any and all questions. Maintained a logbook of stakeholder interactions.
- SH 157 over US 36 (Foothills) – Set up a phone hotline and email address, created a public information plan, updated the voice message alerting stakeholders of closures for this CDOT project. Answered any and all questions and concerns regarding the construction. Created and sent regular flyers to stakeholders.
- SH 44 over Bull Seep and South Platte – For this CDOT project, maintained a project hotline and logbook of calls. Created and distributed weekly email updates on closures and other project developments. Created and distributed construction update flyers to stakeholders. Performed public outreach to optimize public satisfaction with the project.
- SH 287 Federal Signal Improvements – Created and maintained hotline and email for public questions and concerns. Maintained a logbook of all stakeholder interactions. Coordinated with Project Managers to address stakeholder concerns.
- US 50 in Canyon City – For this CDOT project, attended preconstruction meeting, created and maintained project hotline and email address to address stakeholder concerns, coordinated with Project Manager to update stakeholders with regular flyers.

Education

Master of Landscape Architecture, Harvard Graduate School of Design

B.S. Landscape Architecture, Oregon State University

Employment History

LLG International, LLC (LLG)

RNL (RNL)

City of Boulder (COB)

Fort Collins Park Planning Division (FCPP)

Colorado State University (CSU)

The Perron Collaborative (RP)

Professional & Civic Affiliations

Alumni Council, Harvard University Graduate School of Design

Professional Registrations, Certificates, and Memberships

PLA#209 – Colorado

LA-0180 – Wyoming

Honors

00 AIA Colorado – Contribution to the Built Environment Award, Colorado State University Institute for the Built Environment (Associate Director) (CSU IBE)

95 Environmental Stewardship Award – WREN Pit Reclamation, Larimer County Board of Commissioners, Colorado

95 Outstanding Reclamation and Regional Cooperation Awards – WREN Pit Reclamation, Colorado Mined Land Reclamation Board, Colorado

93 Honorable Mention for Ecological Design – United States Botanical Garden Design Competition, Washington, DC

Jeff Lakey, ASLA MLA CLARB

Principal Landscape Architect – LLG International, LLC

Qualifications Summary

Mr. Lakey is the co-founder, lead designer, and strategist of LLG International, LLC. He is exploring how to integrate a systems view of life into ecological city planning, urban design, landscape architecture, and community development. He wants to rebuild the balance of human environments with natural environments. To do this, he is integrating ecological systems into cities with green infrastructure projects. More broadly, he aims to integrate living biological support infrastructure into landscapes and cities to produce fossil free energy, clean water, organic food, and recirculating waste. Humanity is confronting its greatest challenge – that of rebalancing its relationship with nature in order to cultivate a habitable planet.

Experience Summary

Some of Mr. Lakey's previous experience is described below.

Ecological Design

12-13 Media City Ecological Concept Master Plan – Beijing PRC (LLG)

12 Tian Fu Green River Village Urban Waterways Edge Design – (LLG)

08-09 NREL RSF Building – Golden, CO (RNL)

07 Rock Ridge Spa Resort Ecological Design – Morrison, CO (RNL)

06 Beavers Village Ecological Development Strategy – Fraser River, CO (RNL)

Urban Design & Architectural Planning

07-08 FOSS Community Urban Design – Abu Dhabi, UAE (RNL)

08 Horizon City Urban Design – Aurora, CO (RNL)

07 Ghantoot Green City Urban Design – Abu Dhabi, UAE (RNL)

05 Cairo Heights Urban Design – Cairo, Egypt (RNL)

88 Whitworth College Track & Fields Master Plan – Spokane, WA (RP)

Resort/Corporate/Campus Parks & Open Space Planning

12 Resort Highway Entries at Yunling Golf Resort – Kunming PRC (LLG)

12-14 Dali Miwan Resort Landscape Master Plan – Dali, Yunnan (LLG)

11 Clubhouse at Yunling Mountain Golf Resort – Kunming, PRC (LLG)

08-09 One Steamboat Place Plaza – Steamboat Springs, CO (RNL)

07-08 Forest of Seih Sederah Parks & OS Plan – Abu Dhabi, UAE (RNL)

Municipal Land Systems Planning

05 Asses Management System Pilot Program – Boulder, CO (COB)

06 Boulder Parks and Recreation Master Plan – Boulder, CO (COB)

04 & 05 City of Boulder Parks and Recreation Capital Improvement Plan – Boulder, CO (COB)

BRENT F. WILKINS, P.E.

GEOTECHNICAL DEPARTMENT MANAGER



PROFESSIONAL EXPERIENCE

Mr. Wilkins is the manager of the geotechnical engineering department in Terracon's Cheyenne, Wyoming office. He has over 29 years of experience in geotechnical and civil engineering and construction quality management. He supervises the geotechnical operations of the office and provides engineering consultation on projects including foundations, roadways and construction quality control and materials testing.

Mr. Wilkins' primary experience is in subsurface exploration and geotechnical engineering evaluation for building and bridge foundations, roadways, landfills and foundation forensic investigations. He provides engineering analysis and recommendations for foundation settlement, soil bearing capacity, soil lateral resistance, liquefaction potential and soil seismic parameters. He is experienced in the evaluation of shallow and deep foundations, slope stability and ground water control. He has prepared engineered designs and construction specifications for retaining walls, stabilized slopes, large pipe-arch culvert foundations and landfills.

Mr. Wilkins' experience also includes the management of geotechnical exploration activities and laboratory soil testing, engineering evaluation and design of pipe-arch structures, extensive construction QA/QC monitoring, landfill design and permitting and surface mine permitting.

PROJECT EXPERIENCE

Fiscal Years 2006 – 2012 & 2016 Water and Sewer Rehabs – Cheyenne Board of Public Utilities

Project manager for the geotechnical exploration for the reconstruction of yearly water and sewer improvements throughout Cheyenne.

Pioneer & Carey Avenues Reconstruction – City of Cheyenne

Project manager and senior engineer for the geotechnical exploration, testing and analysis of the reconstruction of eleven blocks of roadway and utility improvements. In addition to standard subgrade profiles, existing pavement thickness and condition profiles were provided at multiple sections within each block. Pavement design recommendations were provided for alternative combinations of rehabilitation and reconstruction.

2009 and 2010 Water, Sewer and Street Rehabilitation – City of Laramie

Project manager for the geotechnical exploration on the reconstruction of yearly water, sewer and street improvements throughout the City of Laramie. Provided recommendations for subgrade remediation, including fly ash treatment and geo-grid reinforced fill for street reconstruction projects.

12th Street Reconstruction – City of Cheyenne

Project manager for geotechnical exploration and laboratory testing for reconstruction of 12th Street between Ridge Road and Taft Avenue. Provided engineering recommendations included pavement thickness design and subgrade preparation.

Education

*Master of Science,
Civil Engineering,
South Dakota School of Mines and
Technology, 1995*

*Bachelor of Science,
Civil Engineering,
University of Missouri-Rolla, 1986*

Registrations

*Professional Engineer:
Wyoming, No.9459*

Affiliations

*National Society of Professional
Engineers (NSPE)*

American Society of Civil Engineers

Work History

*Terracon Consultants, Inc.,
Department Manager,
2001-present*

*Bowser-Morner Associates, Inc.,
Senior Engineer,
1996-2001*

*Golder Associates, Inc.,
Staff Engineer,
1989-1994*

*Crockett Engineering Consultants,
Staff Engineer,
1987-1989*

Waste Water Treatment Plants – City of Cheyenne

Project manager for geotechnical exploration and laboratory testing for substantial remediation and additions to the Dry Creek and Crow Creek wastewater treatment plants. Provided engineering design recommendations for shallow and deep spread footing foundations, underground basins, lateral earth pressures, dewatering, trench stability, and construction issues.

30" Water Main Extension – Board of Public Utilities

Project manager for the geotechnical exploration and site evaluation for the extension of the Western Hills main to the Round Top Water Treatment Plant.

Rob Roy and Hog Park Dam Safety – Board of Public Utilities

Project manager for the annual monitoring of piezometers and biannual monitoring of inclinometers at two earthen embankments. Provided recommendations for safety enhancement and maintenance of the embankments based on analysis of water levels within the dams and inclinometer movements.

Sewage Lift Station – City of Cheyenne

Project manager for the geotechnical exploration and site evaluation for North Park Soccer Complex sewer extension and lift station. Provided spread footing foundation recommendations for the lift station and trenching/backfill recommendations for the sewer extension.

Missile Drive Bridge – City of Cheyenne

Project manager for geotechnical exploration and site evaluation for proposed reconstruction of approach embankments. Provided engineering recommendations for embankment settlement, lateral earth pressures and earthworks.

ADDITIONAL INFORMATION

Co-author and presenter, "Performance of Cast-In-Place Pile Installed with a Fixed Mast Drilling Platform in Claystone", 34th Annual Conference on Deep Foundations, Deep Foundation Institute, Kansas City, Missouri.

Foundation Vibrations, Terracon Distant Learning Short Course, Cheyenne, Wyoming.

Training course in Geotechnical and Foundation Engineering - Soil Slope and Embankment Design, Federal Highway Administration, Cheyenne, Wyoming.

Concrete Pavement Design, Wyoming Technology Transfer, Casper, Wyoming.

Micropiles, 20th Annual Geotechnical Seminar, ASCE GEO-Omaha, Omaha, Nebraska.

Enhanced Geotechnical Site Characterization For Evaluating Foundations And Ground Modification, ASCE Geo-Institute, Short Course at GeoOdyssey, Blacksburg, Virginia.

Soiling Nailing and Reinforced Soil Walls, ASCE Seminar, Seattle, Washington.

3.0 SCHEDULE

SMITH has prepared a schedule for completing the design services under this contract – see Exhibit C on the following page. SMITH's support during bidding and construction will follow the schedule described in the RFP.

The schedule is fairly aggressive but doable and will provide the Construction Documents to the City by October 17, three weeks before Construction Bids are due. We have developed schedule streamlining strategies and will implement them as the project progress. These strategies are presented below:

1. Field investigations conducted concurrently with hydraulic analyses and SWMM modeling.
2. Update the project schedule every two weeks during the first four months of the project.
3. Continually be looking for and implementing schedule streamlining activities.

REQUEST FOR PROPOSALS No.: RFP-12-16

PROJECT NAME: Lower Capitol Basin Sediment Trap/Wetland

EXHIBIT "C"
PROJECT SCHEDULE

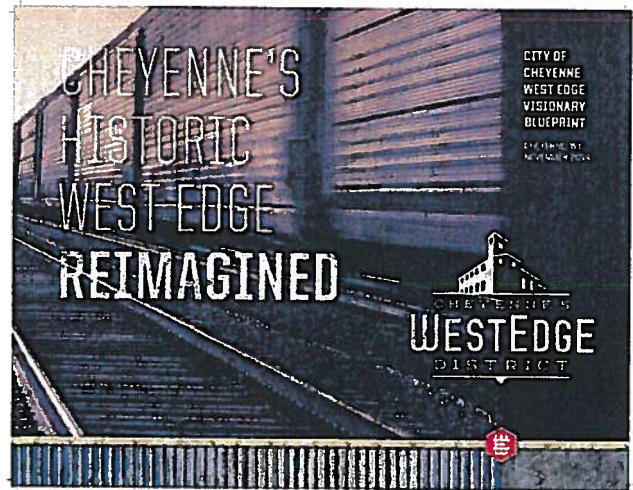
The key milestones in the West Edge District Civic Center Commons and Pump House Park project including the storm sewer construction are shown below. The proposer should complete the following schedule as part of the proposal and include any other key dates as needed.

Date	Schedule Item
<u>June 15, 2016</u>	Notice-to-proceed issued by City
<u>June 17, 2016</u>	Hold scoping meeting with City Representatives
<u>July 22, 2016</u>	Submit preliminary design and report
<u>August 30, 2016</u>	Submit 65% design documents
<u>September 26, 2016</u>	Submit 95% design documents
<u>September 30, 2016</u>	Meet with City to discuss review of design
<u>October 17, 2016</u>	Final design submittal and review
<u>November 7, 2016</u>	Bid Opening
<u>Fall 2016</u>	Begin Construction
<u>March 2017</u>	Substantially Complete
<u>May 2017</u>	Final Project Completion

4.0 PROJECT AREA AND VISION FOR THE PROJECT

Cheyenne is a community of choice which means it's a place where a wide variety of people can find what they are looking for in a hometown, while providing an exemplary standard of living that is second to none. It's clear that Cheyenne's people yearn for a downtown that bustles with economic, cultural and social activity: a pleasant place to live, work, shop and dine, and a capital city that Wyoming deserves.

The West Edge of the downtown Cheyenne has abandoned, idle and/or underused properties that are available for reuse. The proposed Project Site is one of those locations. The West Edge Compass Approach involves flood control, Brownfields redevelopment, public spaces, and reinvestment. The West Edge area between the downtown core, railroad tracks, and existing residential neighborhoods received a \$1 million brownfield EPA coalition grant in 2011. It helped to position the community to start redevelopment of the West Edge to revitalize the downtown with a vision that incorporates surface stormwater drainage features along with improved streetscape and downtown parks. The proposed sediment removal and *E. coli* reduction area (Project Site) is between Ames Avenue and the UPRR railroad tracks and spur on City-owned land. This water quality improvement project will also assist in closing the gap associated with cleaning up some of the Brownfields in the West Edge.



The West Edge drainage basin is a developed area located within the central portion of Cheyenne. During rain events storm water flows across the area, but has a tendency to pool because of the relatively flat topography; the large amount of impervious surface; the barrier created by the Union Pacific railroad yard to the south; and the limited points where water can flow out of the basin.

Environmental contamination is an issue at the Project Site as documented by Ayres (2014, 2015). Results of the environmental assessment indicate that PAH, VOC, GRO, and heavy metal concentrations exceed Wyoming direct contact residential SCL's, or migration to groundwater SCL's, in one or more of the soil samples collected at the site. Laboratory analysis of groundwater samples collected from the three monitoring wells did not detect significant concentrations of PAH, DRO, or heavy metals concentrations above detection limits. However, concentrations of 1, 2, 4-trimethylbenzene and DRO exceeded the ground water cleanup standard in samples collected from one monitoring well. We will not be disturbing soil or groundwater, however, at this well location.

Water level data collected in October 2013 indicate that depth to water range from 6.51 feet to 11.65 feet below ground surface. Average hydraulic conductivity values for the water table observation wells screened in the unconsolidated deposits was 0.0023 cm per second. Groundwater flow is generally east southeast at an average horizontal hydraulic gradient of .005 feet per foot (one half of 1% slope). The average horizontal groundwater flow velocity in the unconfined aquifer is estimated to be 0.13 feet per day.

A glaring omission of the Ayres reports is the characterization of the soil horizons that can be used as plant growth media – the soil material that will facilitate the growth of the wetland and upland (landscaped) plant communities. We will address this deficiency in our scope of work.

The City was successful in getting a 319 Grant from the EPA for this project. **Congratulations, that is a significant accomplishment!** The grant will allow the City to attack the pollutant issues of excess sediment and *E. coli* in the storm water flows from the West Edge area.

4.1 VISION FOR THE PROJECT

The design and construction of the sediment removal and treatment wetland site is a significant and demonstrable indication to the public of the City's serious intent to implement the West Edge vision. We are excited as a project team to be partnering with the City on the completion of the sediment removal/treatment wetland project.

A constructed treatment wetland will be built to naturally treat stormwater runoff from impervious pavement surfaces of the downtown area. A wetland allows natural processes to filter and restore water instead of using chemical treatment. The treatment wetlands will be excellent habitat for wildlife, especially song birds, waterfowl, and small raptors. The entire area will be stepped down in terraces from street level to the lowest point in the natural park – the forebay and treatment wetland areas. The guiding principles of this stormwater treatment wetland include:

- Stormwater will flow initially through a forebay where sediment will be allowed to settle, and periodically removed, and then through a constructed treatment wetland where *E. coli* will experience mortality, petroleum contaminated soils will experience bioremediation and heavy metals will be chelated by the organic fraction of the hydric topsoil.
- Contaminated soil will be kept on site as much as possible to keep costs down. It will be covered with clean topsoil and landscaped.
- Features of the treatment wetland area will accomplish multiple goals – public recreation/open space, public awareness and education, stormwater handling, brownfield mitigation, aesthetic improvement, and sediment and *E. coli* removal.
- Formal (trails and more formally landscaped) areas and natural features (treatment wetlands) will be mixed to create a complete system that is consistent with the vision of the downtown area. This will provide a greenway path opportunity for connection to the Greater Greenway System along Crow Creek in the future.
- Trail connectivity – new trails will be connected to nearby trails in and around the area.

Our design will result in a greater than a 50% (the RFP calls for 30-50%) sediment reduction, as recommended in the draft TMDL guidelines for Crow Creek. The proposed treatment wetlands will increase detention time and provide substrates for biofilms that host bacteria predators, and coupled with the wetland waters' exposure to UV light, will result in a significant reduction of *E. coli* in the treatment wetland. Research of Weber and Legge (2008) and SMITH projects where we have implemented these methodologies bear this out. The proposed sediment trap (fore bay) will be designed for easy and routine maintenance to remove the accumulated sediments on both a regular basis and on an as-needed basis after major storm events. The combination of both sediment removal and *E. coli* reduction will result in substantially improved water quality and aquatic habitat within Crow Creek.

5.0 SCOPE OF WORK

The goal of the design is to create a bioengineered forebay and treatment wetland in a park-like setting within the approximate two acre Project Site, where there currently is none, such that it will remove sediment and *E. coli*, a pathogenic bacterium in stormwater coming from Cheyenne's West Edge. The proposed sediment removal and *E. coli* treatment wetland will occur between Ames Avenue and the UPRR railroad tracks and the 15th Street ROW. *E. coli* reduction is accomplished as residence time of flow in a treatment wetland is increased. The final design will be stamped by SMITH's Wyoming P.E. SMITH's Professional Wetland Scientist (PWS) will stamp the treatment wetland drawings (only). We believe it is very important for a PWS to stamp the treatment wetland drawings, in addition to SMITH's WY PE because the treatment wetland features will be jointly designed by engineers and wetland scientists. It also shows SMITH's commitment from the highest level of the company – Principal in Charge. Mr. Peter Smith, Principal in Charge, is a Professional Wetland Scientist (#01273).



SMITH will complete the following project milestones to accomplish project objectives:

- Hydrologic and hydraulic data gathering and analyses
- Update the Lower Capital Basin EPA SWMM Model to accurately reflect the sediment trap/treatment wetland.
- Design a forebay and treatment wetlands
- Integrate trail and formal design features with the forebay and treatment wetlands that is consistent with the vision of the downtown area.
- Provide construction management services during construction
- Make all task scheduling decisions based on the schedule due dates

5.1 E. COLI DEGRADATION AND MODELLING

E. coli are a pathogenic bacteria and reduction is accomplished as residence time of flow is increased. The presence of plants in treatment wetlands has a beneficial effect on *E. coli* reduction. It is believed the plants exude antibiotics in the rhizosphere that quite literally kill *E. coli*.

We are proposing to use Surface Flow (SF) and Horizontal Subsurface Flow (HSSF) treatment wetlands for *E. coli* reduction. Our experience and literature (Kadlec and Wallace 2009) suggest that a HSSF is the most efficient treatment wetland for removal of *E. coli*. Reasons why HSSF treatment wetlands have greater effectiveness include:

- A more uniform water level in the root zone which keeps the predator populations alive during time of drying
- Bedding material of the appropriate size increases the volume of biofilm where predators live
- Reduced problems with mosquitos and algal blooms that can cause an odor issue and reduce O₂ concentration in the water

- Protection from freezing in winter

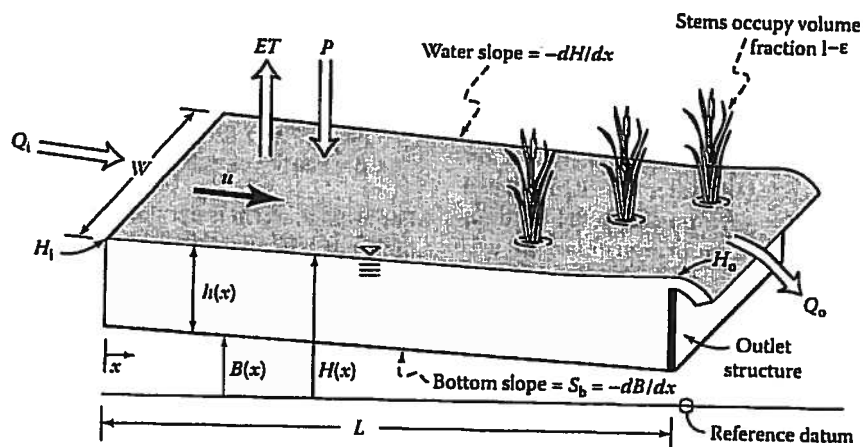
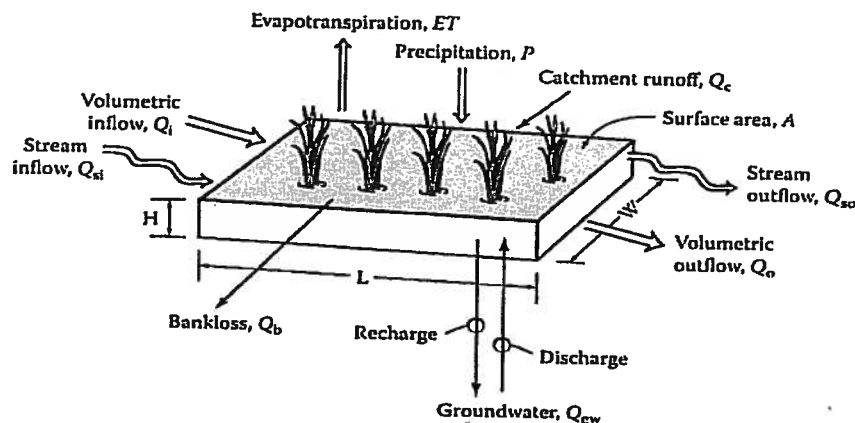
Free water system (FWS) wetlands are efficient at dramatically improving water quality and causing *E. coli* mortality. The components in a FWS wetland that relate to the water budget and *E. coli* removal are shown to the right (from Kadlec and Knight 1996).

The primary factors that achieve effective sediment reduction and *E. coli* removal are detention time and hydraulic loading of the wetland. Detention time (DT) defines how long water remains in the wetland and is defined by the equation below (from Kadlec and Wallace 2009).

$$DT = \text{Volume of Wetland containing water in active flow (V)} / \text{Flow Rate (FR)}$$

The hydraulic loading rate is inversely proportional to DT for a given wetland depth.

The Manning formula, which has traditionally been used to describes the turbulent flow characteristics of an open ditch cannot be used to characterize a wetland surface water flow for a couple of reasons. First, Manning's equation is a correlation for turbulent flows, whereas FWS wetlands are nearly always in a laminar or more transitional flow regime. Second, FWS plant stem density impacts the frictional effects on surface flow. FWS bed hydraulics can be shown in the picture below from Kadlec and Knight (1996).



The presence of ample water in wetlands supports a variety of microbial and plant species and creates hydric soils (soils with redoximorphic features). The emergent plants provide diverse, oxygen rich habitats for abundant zooplankton and bacterial populations, resulting in complex energy and nutrient pathways. Bacteria are involved in the degradation of various pollutants and are competitors of *E. coli*. Zooplankton species are the primary predators of *E. coli* (Weber and Legge 2008). Recent research (Weber and Legge 2008) indicates that some plants roots release enzymes that also kill *E. coli*. These species will be incorporated into our treatment wetland species list. Wetland type or class is defined by their hydro zones and volume and duration of flooding or proximity to water table. The incorrect understanding of hydro zone and wetland plant species relationships is a frequent cause of vegetation problems and failure in constructed (treatment) wetlands. Hydro period curves summarize water level data, length of and depth of flooding, and depth to ground water over a period of time. The direction and depth of flooding also affects plant physiology because of soil oxygen concentration, soil pH, dissolved and chelated macro and micro nutrients, and toxic chemical concentrations. We will develop hydro period curves based on our EPA SWMM modeling data and include them in our Concept Design and Report.

Plant selection is important in design considerations of treatment wetlands. Plants need to be selected not only for defined hydro zone and tolerance to potential pollutants but also for their effectiveness facilitating chemical transformations leading to the killing of *E. coli* and support of biofilms.

5.2 PRE-DESIGN ACTIVITIES

SMITH will complete the following pre-design activities at the Project Site to gather necessary data to complete design:

- Hydrologic and hydraulic data gathering and analyses
- Water budget characterization of the contributing watershed to the treatment wetland
- Update the Lower Capital Basin EPA SWMM Model
- Hydro zone curves for the Project Site
- Characterize the depth and horizontal extent of high organic matter soil horizons at the Site – two to four soil borings to identify soil horizons with high organic matter content that, when collected, will serve as plant growth media (topsoil) during construction.
- As part of soil organic matter characterization effort, we will record groundwater depth data at points in between the two depth-to-groundwater transects generated by Ayres (2014).

Soil and groundwater contaminant characterization by Ayres (2014, 2015) is sufficient for design and requires no further investigation.

5.3 DESIGN OVERVIEW

Sediment reduction will be achieved through the use of a forebay, which will be designed to have easy entrance and exit for equipment involved in the regular removal (maintenance) of sediment from the concrete bottom. It will be designed according to the standards of the WDEQ (2013).

After the scoping meeting, SMITH will complete the following design tasks in the following order. SMITH will utilize the City of Cheyenne and Board of Public Utilities Construction Specifications and Standard Drawings, dated 2014, in preparing drawings and specifications during design.

- Concept Plan - Design and Report
- Review by the City and WDEQ and Meeting between SMITH and the City to review Concept Plan comments
- Meet with WDEQ to review the City's progress by their other consultant on the VRP process (SMITH understands it will not be taking this project through the VRP process, we understand this will be done by Ayres.)
- Preliminary (65%) Drawings, Specifications, and Cost Estimate: forebay (sediment trap), treatment wetlands, trails and formal design features – per the requirements of RFP (Section A.3)
- Review by the City and WDEQ
- Meeting between SMITH and the City to review comments
- Final (95%) Drawings, Specifications, and Cost Estimate: forebay (sediment trap), treatment wetlands, trails and formal design features – per the requirements of RFP (Section A.4)
- Final Review by the City and WDEQ
- Obtain written approval from the City and WDEQ of the Final Design
- Prepare Construction Documents and Project Manual: forebay (sediment trap), treatment wetlands, trails and formal design features – per the requirements of RFP (Section A.4). Provide them three weeks before the Construction bids are due.

5.4 PUBLIC OUTREACH

SMITH will conduct two public outreach meetings for this project. The first public meeting will occur late in late summer/early fall (before preliminary final design but after preliminary design) and explain the purpose of the project, its benefits to the Crow Creek watershed, and explain storm water quality information.

The next public meeting will include a project tour at the end of project construction. We will coordinate with the City's team currently engaged in the VRP process. Public notice and media releases, securing a meeting site, preparing exhibits and handouts, and correlation of public comments will be completed by the SMITH public outreach and design team. We will use innovative outreach methods that we have used in the past, similar to those used during the completion of the West Edge planning initiative.

5.5 SURVEYING

SMITH understands that the Project Site will be entirely on City-owned property. We do not anticipate nor include cost for the acquisition of new permanent right away. We do, however, expect to obtain a temporary construction easement adjacent to Union Pacific railroad right-of-way.

The SMITH team will field survey horizontal and vertical locations of utilities critical to the alignment and location of the treatment wetlands. All surveying and temporary control points shall be referenced into the cities control network. The vertical datum shall be the NAVD 1988.

5.6 UTILITIES

The SMITH team will investigate and coordinate designs with all known utilities. We will resolve any conflicts with other utilities either active or abandoned. We will conduct an additional utility investigation of the two-acre site to provide additional utility locates and locates of potential utility conflicts. During design we will resolve utility conflicts as necessary. Our cost estimate assumes few to no utility conflicts. We will utilize the geophysical investigation data compiled by Olsen Associates (2015) in completing these utilities related tasks. We will obtain utility information possessed by the City of the site and review it as our starting point for this utilities investigation.

5.7 COORDINATION WITH OTHER AGENCIES AND PROJECTS.

We anticipate extensive coordination with the WDEQ and to obtain DEQ permits to construct the proposed treatment wetlands and sewer improvements. At this point in time we do not propose changing or making sewer improvements outside of the two-acre treatment wetland area.

5.8 MODELING

Based on available literature and established models, SMITH will develop a model to identify optimum residence time for anticipated flow characteristics to remove the maximum amount of *E. coli*. It is expected that *E. coli* concentrations fluctuate throughout the year so the design will take into consideration when higher levels of *E. coli* occur when designing and determining the optimum residence time for the flow variation. First flush stormwater events will have the highest *E. coli* concentrations.

5.9 VEGETATION SELECTION AND LANDSCAPE DESIGN

SMITH's wetland scientists will develop the wetland types that will be shown in the treatment wetland design drawings. Wetland types are expected to include Emergent Marsh, Permanent Wetlands, Seasonal Wetlands, and possibly Riparian. They will show native species known for exuding enzymes and facilitating chemical reactions that reduce *E. coli*. There will also be an upland plant community zone (formal landscape zone) comprised of trees, shrubs and grasses that a pedestrian trail will meander through. The team's Landscape Architect will play an important role in the integration of upland features (pedestrian trail, park-like vegetation and features) with the treatment wetlands, which will tend to look and feel more like a natural and non-anthropogenic landscape. Treatment wetland ecosystem interpretative signs will be an important component of this area. Our Landscape Architect will ensure that the Greenway Path connecting to the Greater Greenway System along Crow Creek is incorporated into the design.

5.10 HYDROLOGIC AND HYDRAULIC ANALYSIS

SMITH will prepare a watershed characterization and update the Lower Capitol Basin EPA SWMM model – hydrologic and hydraulic analysis and water budget characterization of the contributing flows to the Project Site. We will prepare a report detailing the findings. We will meet with the City to present the report. The report will also be submitted to the WDEQ so that we can obtain their written approval.

5.11 PRELIMINARY DESIGN – CONCEPT AND THE 65% DESIGN

After consultation with the City at the scoping meeting we will move ahead with preparing the Concept Plan and Report, and then 65% design. The 65% design will include drawings, specifications, and an opinion of probable construction cost (OPCC). The OPCC will include allowances for contingencies.

5.12 FINAL DESIGN AND CONSTRUCTION DOCUMENTS

After the City reviews the 65% design documents and upon written authorization from the City to proceed, SMITH will complete the Final design (95%). The first public meeting will occur after Preliminary Design but before the start of Final Design. Final design will incorporate changes and recommendations of the City and the WDEQ from the Preliminary Design as well as input from the public meeting. SMITH will submit final drawings, specifications, and opinion of probable cost (OPCC) to the City for their final review.

After the City has completed their review and gives SMITH written authorization to proceed with preparation of the Construction Documents (CD), SMITH will prepare the CD and Project Manual. The CD will be stamped by SMITH's WY PE. SMITH's PWS will also stamp the treatment wetlands drawings and specifications. We believe it is very important for a PWS to stamp the treatment wetland drawings since the treatment wetlands are at the core of this water quality improvement project and have any understanding of treatment wetlands that engineers, in general, do not have. The CDs will include the items listed on pages 33 and 34 of the RFP. SMITH will provide five paper copies of the CDs and Project Manual and one electronic copy. We will provide three copies and one reproducible set of construction contract documents to the City.

5.13 SUPPORT DURING BIDDING AND CONSTRUCTION

SMITH will assist the City through the bidding process including: 1) organizing and conducting a pre-bid conference for perspective contractors, 2) answering questions pertaining to the drawings and specifications, 3) prepare and publish any necessary addenda to the contract documents, and 4) give recommendations to the City in selecting a contractor.

After the Contractor is selected and is under contract, SMITH will conduct a Pre-Construction meeting and prepare minutes of the meeting. Prior to the Pre-Construction meeting we will: 1) prepare an Inspection Plan, 2) prepare a Traffic Control Plan, and 3) Prepare a Testing Plan. During actual construction we will complete all tasks described on pages 35 and 36 of the RFP with the 1,000 hour budget mandated in the RFP.

5.14 REFERENCES

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6.0 REFERENCES

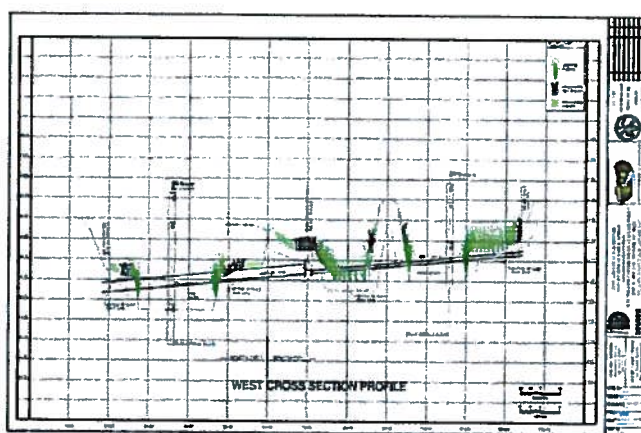
6.1 SMITH: WATER QUALITY TREATMENT WETLAND – 52ND AND EMERSON

Denver, Colorado

City and County of Denver, Wastewater Capital Projects Management,
Selena Klosowski, PE, (303) 446-3520, selena.klosowski@denvergov.org;
Olsson Associates, David Krickbaum, PE, (303) 374-3133,
dkrickbaum@oaconsulting.com



The City and County of Denver (CCD) wanted to create a Regional Detention Pond that would be designed and managed as a natural area that will be sustained by the intermittent surface flow of stormwater – an area that could support native vegetation, create desirable edaphic features, provide habitat for indigenous wildlife or plant species, and provide passive recreational opportunities. So SMITH (and Olsson Associates) provided innovative storm water quality treatment planning and design services to the CCD. The 19-ac site had substantial lead (Pb) contamination from the former nearby Asarco Smelter, a Superfund Site. The purpose of these services was to provide a sustainable, aesthetically pleasing natural area that also achieved water treatment goals, minimized long-term maintenance by the CCD, and acted as a natural area park. SMITH provided an evaluation of plant community and pedologic dynamics and its effect on stormwater treatment; grading and detailed design plans for the wetland treatment areas; a detailed Wetland Planting Plan; expert water quality treatment advice on the best outlet to minimize maintenance and maximize the effectiveness of the stormwater quality treatment; an evaluation of surface and ground water interaction during stormwater events and whether ground water quality would be affected; and an evaluation of water rights issues as it related to the use of ground water for supporting treatment wetlands. SMITH co-authored the detailed Maintenance Plan Document, which included a maintenance schedule for the pond and wetland treatment areas; an estimated cost for each maintenance item; and a cost estimate for installing the detailed design features (including plantings). SMITH prepared a planting plan for Palustrine emergent (PE), Palustrine scrub-shrub (PSS), and Palustrine Forested (PF) treatment wetlands recommending specific species known for bioaccumulation of metals and salts, and microbial populations for breaking down petroleum hydrocarbons and fecal matter, including



E. coli. After a joint public presentation by SMITH and the CCD where the design plans and intent of the treatment wetlands were described to local neighbors and businesses in the watershed, the Project Manager for the CCD, Selena Klosowski, said "Thank you so much for your presentation today. I am impressed in the quality of work you... have provided." And later after funding was stopped and the project was put on indefinite hold in 2013, she said "Thank you all for your patience and hard work."

Project dates: 2011-2013

Original cost: \$60,000

Final cost: \$60,000

6.2 SMITH: STONE CREEK RANCH SECONDARY STORMWATER TREATMENT WETLANDS

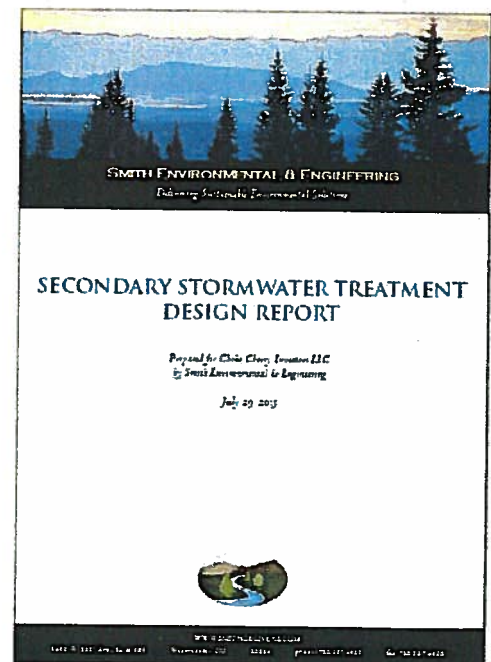
Parker, Colorado

Choke Cherry LLC, Michael Sanders, (310) 460-6324, landdad@gmail.com

Choke Cherry, LLC proposes to develop a 300-ac property, which straddles Cherry Creek in Douglas County. The Cherry Creek Basin Water Quality Authority (CCBWQA) requires secondary treatment of stormwater before it enters Cherry Creek because of phosphorus loading issues in the creek and Cherry Creek Reservoir. Furthermore, the proposed project lies in Preble's meadow jumping mouse (PMJM) habitat – the Cherry Creek riparian area. Additionally the site contains cultural resources that were determined to be eligible for listing on the National Register of Historic Places. SMITH's original scope was to complete a wetlands delineation and report and habitat suitability assessment for threatened and endangered species. However, because of CCBWQA phosphorus loading requirements, PMJM habitat impacts, wetland impacts, and Douglas County

Intergovernmental Agreement and Comprehensive Plan (IGACP) constraints, additional environmental studies and design were needed for the project including:

- Siting, design and design report for secondary stormwater quality treatment areas to meet the CCBWQA's requirements – secondary treatment to remove phosphorus from stormwater before it enters Cherry Creek. SMITH described different methods to achieve this goal and treatment wetlands were selected because of: 1) their effectiveness at removing phosphorus, 2) they are self-sustaining, and 3) the long-term cost savings.
- Historical/archaeological (cultural) resources survey, report, Section 106 clearance, and recordation mitigation
- Noxious weed survey and management plan
- Migratory bird nest survey and management plan
- Biological Assessment to address impacts to PMJM



habitat and provide mitigation for these impacts. SMITH obtained a Take permit from the USFWS for the project. SMITH developed a fairly inexpensive PMJM mitigation plan that provides a 4:1 ratio of mitigation area to impacted area.

- Wetland mitigation plan that will compensate for 0.35 ac of wetland impacts. SMITH also obtained a Section 404 permit from the USACE, which also included cultural resource mitigation (recordation) and PMJM compensatory mitigation.

SMITH has continued to work with Choke Cherry, LLC throughout the design process to ensure the project's compliance with applicable County, SHPO, USACE, USFWS regulations, including, but not limited to, the County's Riparian Conservation Zone management plan, Section 404 of the Clean Water Act, the Endangered Species Act, and the National Historic Preservation Act.

Project dates: 2013-present

Original cost: \$40,000

Final cost: \$160,000

What accounts for any change in cost? Client and regulators required additional studies and design.

6.3 SMITH: ARGO MINE – DESIGN-BUILD MINE REMEDIATION AND STREAM RESTORATION

Boulder County, Colorado

Boulder County Parks and Open Space, Barry Shook, Project Coordinator, (303) 678-6200, bshook@co.boulder.co.us

Going through CDPHE's Voluntary Cleanup Program (VCUP), SMITH developed an innovative and cost-effective approach to reclaim nine waste rock piles at the Argo Mine site near Jamestown, CO, and completed reclamation activities according to this innovative approach. Runoff and acid mine drainage from the Argo Mine was adversely impacting Little James Creek, a drinking water supply creek, and contaminating groundwater flowing into the creek. The purpose of this project was to: 1) reduce leaching and erosion of mine waste, high in metals, into Little James Creek, 2) improve water quality in Little James Creek, and 3) decrease the interaction of mine wastes with precipitation and surface run-off. SMITH accomplished these goals under the general guidelines of a Brownfield Cooperative Agreement between Boulder County and the CDPHE/EPA.



Approximately 2,500 cubic yards of waste rock in nine different piles from past mining activity occurred on site and was part of the source of the sediment and metals in Little James Creek. The innovative approach involved removing mine waste from nine

piles around the site, mixing it with fly ash, water, and cement, and filling a large stope (hazard)



Stope before filling



Stope immediately after filling, topsoiling, and application of erosion control blanket

on site, rather than creating one large pyramid-type pile above ground that would be subject to future erosion.

SMITH prepared several documents for the project. The first, the Analysis Brownfields Cleanup Alternatives (ABCA) report, was prepared according to EPA and CDPHE ABCA guidelines for the project. The ABCA analysis identified and evaluated cleanup alternatives that would mitigate risks to human health and the environment associated with contamination at the site, as required by an EPA Brownfield Grant awarded to the Boulder County Parks and Open Space (BCPOS). SMITH conducted public meetings in Jamestown to present and receive comments on the proposed cleanup alternatives and also conducted a public meeting for the Boulder County Planning Department (BCPD).



The former Stope area after two growing seasons (grey pipe marks the center of the former stope opening)

SMITH also developed a Phase II Feasibility Study according to EPA and CDPHE guidelines for the project. This document presented an application to the CDPHE and CDRMS for the Argo Mine site on behalf of BCPOS. Completed in 2008, this report described the removing of the waste rock and mixing it with cement, fly ash and water and stowing it in the large 2,400 cu. yd. stope on site as the preferred alternative. After this, SMITH obtained a 1041 clearance from the Boulder County Planning Department and the Boulder County Board of Commissioners.

SMITH completed the backfill design/construction methodology report in 2009 for the BCPOS and CDRMS. This report included a thorough analysis of paste backfill and dry backfill methodologies. As part of this document SMITH also developed the construction methodologies that were followed during the back-filling of the stope, including quality control procedures and a health and safety plan. The field investigations, ABCA and VCUP reports, design drawings, and construction methodologies report were prepared under the direct supervision of a SMITH PE and a SMITH Certified Professional Soil Scientist.

Following the development of its innovative approach, the SMITH team (SMITH and Lawrence Construction personnel) excavated mine waste rock, mixed it with fly ash, water, and cement, and consolidated it into the stope (large adit to the underground mine) east of the mine structure. Stope backfilling with waste rock began in September 2010 and was completed in November 2010. Pre-stope filling and post-filling pictures are shown to the right. Approximately 2,500 cu yds. of waste rock was removed, the site was covered with about 12 inches of topsoil and seeded with a native seed mix by SMITH. Steeper topsoiled and seeded areas were then covered with erosion control blanket. A stormwater runoff collection system was installed at the beginning of construction and continued through August 2011. Monthly runoff samples were collected when runoff occurred, and analyzed for RCRA metals concentrations. Quarterly stormwater runoff monitoring reports were prepared and submitted the CDPHE. Total Design/Build Cost: \$381,000.

Project dates: 2010-2011

Original cost: \$381,000

Final cost: \$381,000

6.4 SMITH: CORNERSTAR SHOPPING CENTER – SECONDARY STORMWATER TREATMENT

Parker, Colorado

Alberta Development Partners, LLC, Jerry B. Richmond III, Principal, Phone: (303) 771-4004, Fax: (303) 771-4086; SEM Architects, Gary Rhodes, (303) 220-8900, studio@semarchitects.com; DHM Design, Joshua Rupert, (303) 892-5566, jrupert@dhmdesign.com

SMITH was contracted by Alberta Company because of our innovative approach to remove phosphorus (P) from storm water on both a short-term and long-term basis at this 40-ac shopping center, which is in the Cherry Creek watershed. The CCBWQA has set stringent P

Acronyms

DP = Dissolved Phosphorus

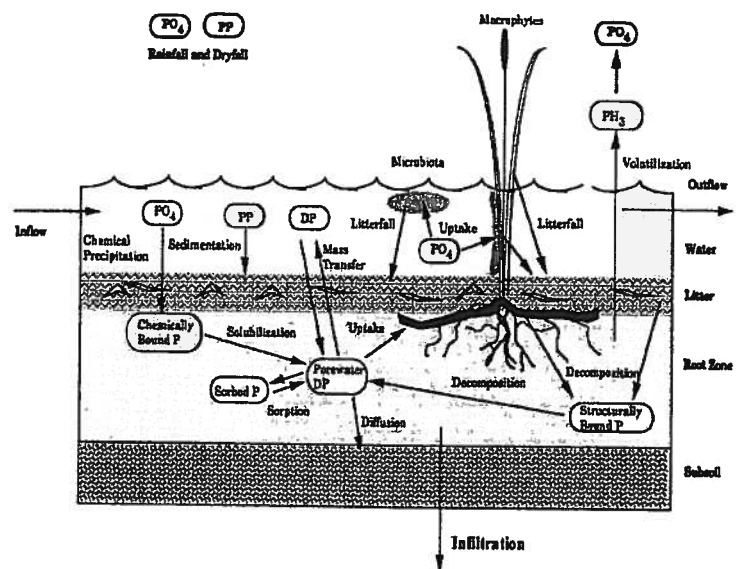
PH₃ = Phosphine

PO₄ = orthophosphate

PP = Particulate Phosphorus

loading limits in the creek and phosphorus is common in storm water runoff. A phosphorus loading assessment was conducted

by SMITH of the expected runoff from the parking lots. P is also nutrient required for plant growth and is frequently a limiting factor for vegetative productivity. Our innovative approach was to create treatment wetlands to remove P from the storm water runoff before it entered Cherry Creek. Because wetlands are a proven method for adsorbing and utilizing dissolved nitrogen and total P, SMITH recommended wetlands as its method of stormwater treatment and designed a secondary stormwater treatment (wetland) area. P is utilized in a wetland in a complicated biogeochemical cycle, which involves many pathways and diverse temporary and permanent sinks and sources. When a constructed wetland is subjected to a new supply of water and phosphorus, it responds by readjusting storages, pathways, and structure. The constructed wetland strategy was adopted by the project design team and SMITH designed the detention ponds and treatment wetland to remove P before it entered Cherry Creek. The project was built in 2009-2010. The newly constructed treatment wetland required a successional period to adapt to the intended inputs, therefore a period of adjustment was incorporated into the design of the Cornerstar treatment wetlands. After a couple of years of edaphic and plant community adjustment, the Cornerstar treatment wetlands began functioning in their long-term sustained mode. The treatment wetlands were designed for a largely-



impermeable 12-ac area where storm water, after flowing through a detention pond (first treatment), passes through a Palustrine emergent wetland (secondary treatment). These treatment wetlands have been functioning properly and removing P from storm water runoff for five years, and are shown in the figure above (west of the bike path and detention pond, but east of Cherry Creek – top of picture is north).

Project dates: 2008

Original cost: \$39,000

Final cost: \$39,000

6.5 SMITH: SANCTUARY ON THE PARK DEVELOPMENT – STORMWATER TREATMENT

Loveland, Colorado

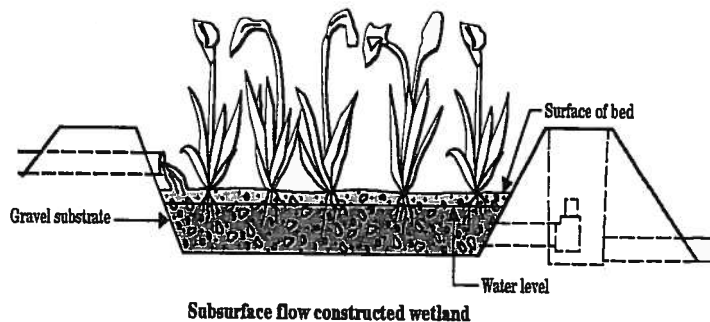
City of Loveland, Kevin Gingery, (970) 962-3571, gingek@ci.loveland.co.us; Stantec Consulting, Inc., Mike Carr, (303) 285-4558, mike.carr@stantec.com

Stormwater leaving this 100-ac site and was causing nutrient (N, P, K) loading in Silver Lake, approximately 0.5 mi downstream of the property. This issue was of great concern to the City of Loveland. A two ac pond (shown to the right) lies in the north end of the property (the upstream end of the property) and receives nutrient-laden water from up gradient of the site. The City proposed to have the developer create a five ac detention pond (DP) in the south end of the property for water quality treatment purposes.



SMITH was brought in to create an innovative, cost-effective stormwater quality improvement plan. In preparation for creating this plan SMITH completed water quality sampling, had the samples analyzed, and developed a conceptual water quality (WQ) treatment methods for the stormwater flows to reduce nutrient loading into Silver Lake. Water samples were collected at two locations, one at the north end of the pond where stormwater entered the pond and at the south end of the pond where water exited it and went into a swale that drained water to the south and off the property. Based on laboratory data, the primary nutrients of concern in the pond and storm water flows were dissolved nitrogen (DN) and total phosphorus (TP). Because wetlands are a proven method for adsorbing and utilizing dissolved DN and TP, SMITH recommended wetlands as its method of stormwater treatment and designed stormwater wetland treatment areas.

SMITH's design included: 1) a forebay at the entrance of the WQ treatment area to trap sediment, 2) a vegetated (wetland) swale through the bottom of the proposed DP, and 3) low wetland benches (subsurface flow wetlands) along the perimeter of the proposed DP that would receive flow and treat water as shown to the right. SMITH also designed a sinuous



wetland swale between the DP and the south end of the property to maximize stormwater treatment before flow exited the property.

Project dates: 2009-2010

Original cost: \$17,000

Final cost: \$21,000

What accounts for any change in cost? Client requested two additional monitoring wells be installed.

6.6 SMITH: GOLD RUN GULCH WATER QUALITY TREATMENT AT THE JESSIE MINE AND MILL SITE

Summit County, Colorado

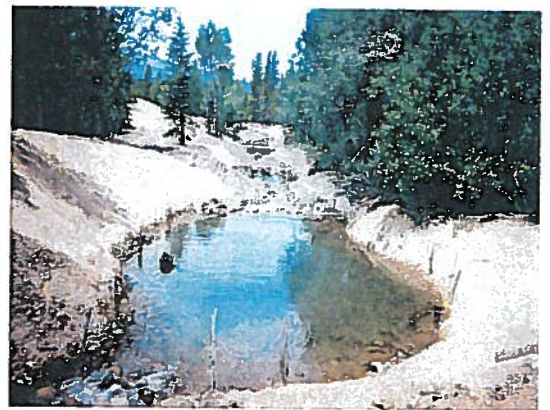
Summit County Government, Brian Lorch, (970) 668-4067; Colorado Department of Public Health and Environment, Kevin Mackey, (303) 692-3381

The SMITH team designed and constructed cleanup measures in accordance with the VCUP permit issued by the CDPHE to Summit County. The abandoned Jessie Mine and Mill site is owned by Summit County and Town of Breckenridge, and includes an access road, the partial remains of an old mill site, a one- to two- GPM discharge of acid mine drainage from collapsed adits into Gold Run Gulch (a tributary to the Swan River), several beaver dams and ponds in Gold Run Gulch that border the tailings piles, a three-acre waste-rock dump, and several uncapped tailings piles (collectively more than two acres) bordering Gold Run Gulch. Metals that adversely affect the water quality in Gold Run Gulch come from the waste-rock dump, tailings piles, and the acid mine drainage. The cleanup measures that SMITH designed and the team constructed include 1) a relocated (new) 1,300-ft long, fish-compatible Gold Run Gulch channel that consists of a series of riffles, pools and runs to provide fish and aquatic habitat, 2) a permanent dewatering/diversion trench to divert Gold Run Gulch flow into the new channel, 3) a new road south of the new Gold Run Gulch channel, 4) a new berm to divert runoff from the old mill site away from the

new Gold Run Gulch channel, 5) buck and rail fencing to provide controlled access to the tailings area, 6) the identification, salvaging, and application of topsoil, 8) wetlands, wildlife habitat and a riparian ecosystem along the new channel, and 9) an upland revegetation plan and cut and fill slopes of the new road. The new road and site amenities are compatible with the Nordic trail that runs along the edge of the area. To complete the design drawings, hydrologic, cultural resources, fish habitat, wildlife, plant growth media, and wetlands, and surveying were completed. HEC HMS analysis was completed to create a stream with riffles and pools that would pass a 100-year flood event. The design process included a schematic design, conceptual design, preliminary final design, and construction drawings that were submitted to Summit County, the CDPHE, and the Town of Breckenridge for approval in accordance with VCUP requirements. The construction was finished July 30, 2007. Because of SMITH's hydrologic, wildlife, and plant growth



After channel relocation, seeding, and willow cutting installation – 2007



After channel relocation and seeding/plantings – 2007

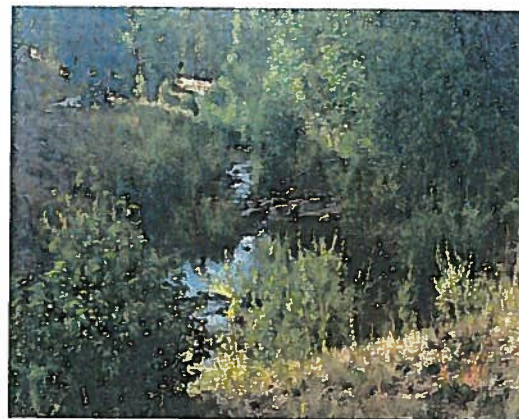
media studies and other site planning, SMITH was able to reduce construction costs (and quantities) from its original proposal amount by \$39,000. After SMITH attained the "no-further-action" determination from the CDPHE, Brian Lorch, a Project Manager for Summit County's Open Space and Trails department, said to SMITH: "Thank you all for your assistance in making these projects successful." Total Design/Build Cost: \$386,000. Three years of monitoring and maintenance activity were completed from 2007-2009. The project successfully closed in early 2010.

Project dates: 2007-2010

Original cost: \$425,000

Final cost: \$386,000

What accounts for any change in cost? SMITH was able to reduce construction costs.



Channel and Wetlands - 2015

6.7 SMITH: COOPER SLOUGH

Fort Collins, Colorado

City of Fort Collins, Matt Fater, PE, (970) 224-6023

The City of Fort Collins was considering routing surface flows from a largely agricultural watershed into Cooper Slough, which flowed into an irrigation canal. This largely agricultural watershed also receives runoff from the Anheuser-Busch facility near I-25. The City was concerned about impact to waterfowl when these flows were diverted into Cooper Slough. Significant development is planned for the area north of the irrigation canal. Water quality in the watershed to the north will change when the development is completed. At present the City has little to no surface or groundwater quality data from the slough or the watershed to the north. SMITH evaluated the effects of adding runoff from the watershed to the north on waterfowl in the slough. SMITH completed a report summarizing literature reviewed; anticipated water quality changes in the slough; potential detrimental effect to waterfowl; and water quality treatment measures to minimize adverse water quality change if waterfowl would be adversely affected by the diversion.



Project dates: 2008

Original cost: \$25,000

Final cost: \$25,000

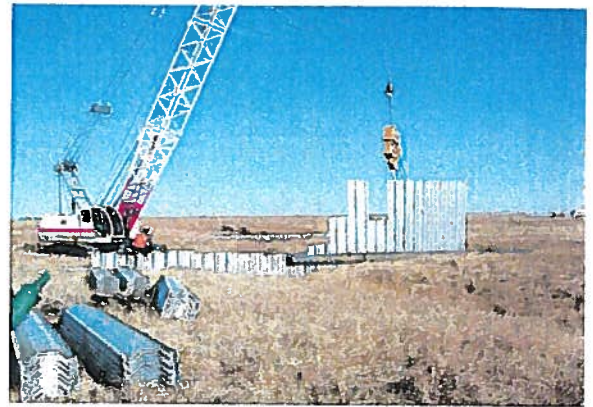
6.8 SMITH: BANNING LEWIS RANCH DEVELOPMENT

Colorado Springs, Colorado

Banning Lewis Ranch, Greg Barbuto, former Director of Construction, BLRMC, (719) 955-7100

SMITH is providing a wide range of environmental planning, design, permitting, and construction services on this project, both with the former owner, Banning Lewis Ranch Management Company (BLRMC), and the new owner, Oakwood Homes, since 2014. When working for BLRMC (2008-2013), SMITH participated in the weekly project team meetings with engineers and planners and providing input on subdivision, road, park, stream, stormwater treatment, and open space design and planning decisions. SMITH's environmental services include groundwater investigations, groundwater well permitting evaluations, stormwater discharge permitting, stormwater treatment, groundwater cut-off wall concept design, several acres of wetland mitigation site design, wetland mitigation site construction management, wetland permit revisions, and cultural resources evaluations. SMITH evaluated water rights on the ranch and the need for a permanent dewatering well permit and a water augmentation plan. We obtained the temporary dewatering well permit for the first two phases of the project and designed pipeline systems and groundwater cutoff walls to maintain groundwater levels in wetland mitigation areas. SMITH planned and evaluated the restoration of some wetlands and the creation of others (mitigation wetlands) in Village I and II (the first phases of the development). The wetland creation area, shown in the picture below, also serves as a stormwater treatment area, before flows enter the nearby creek. SMITH designed trail and boardwalks to provide pedestrian access through the wetlands areas; planned, field-staked, and constructed the wetland mitigation areas and pedestrian crusher-fine trail network; and prepared three years of annual monitoring reports for the U.S. Army Corps of Engineers (USACE) in fulfillment of the project's Individual Permit. For over two years SMITH met weekly with the BLR project team to coordinate on all environmental issues. Upon review of SMITH's wetland mitigation site construction work, Mr. Greg Barbuto, Director of Construction for the BLRMC, said: "Well done, well done, well done. Thank you!" Total Design/Build Cost: \$756,000.

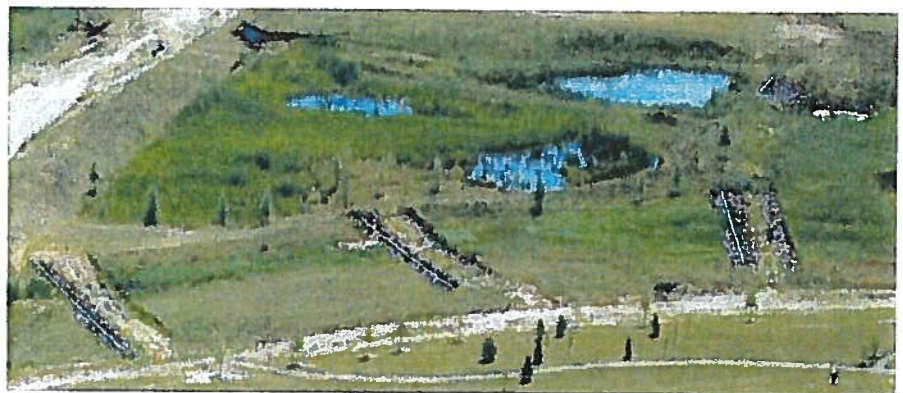
Project dates: 2006-present
Original cost: \$756,000
Final cost: \$756,000



SMITH team installing sheet pile for a water diversion pond.



Functioning water diversion ponds with wetland vegetation



6.9 SMITH: PINELLO RANCH WETLAND MITIGATION AREAS

El Paso County, Colorado

Colorado Springs Utilities, Gayle Sturdivant, PE, PMP, (719) 649-4274, 121 S. Tejon St., Colorado Springs, CO 80903, gsturdivant@csu.org

The Southern Delivery System (SDS) is a regional water delivery program that will transport water from the Pueblo Reservoir to the City of Colorado Springs and will serve the future water needs of the City of Colorado Springs, City of Fountain, Security Water District, and the Pueblo West Metropolitan District. SMITH was selected by UTILITIES over three other design firms because of its outstanding wetland mitigation design experience. The



purpose in creating the Pinello Ranch Mitigation Wetlands (PRMW) is to mitigate SDS Project impacts to palustrine emergent (PE) and palustrine scrub-shrub (PSS) wetlands. On this two year project SMITH performed design services, permitting support services, bid phase services, and wetlands consultant services during construction of the PRMW areas for Colorado Springs Utilities. Drawings were stamped by a SMITH Professional Wetland Scientist and signed off on by a SMITH PE. The mitigation wetlands will provide similar function and value as the wetlands that have been impacted by the SDS pipeline project. Our effort during design included a design handbook, drawings, specifications, and construction cost estimate; it created 9.04 acres of groundwater-fed wetlands at the three different wetland mitigations sites. SMITH evaluated groundwater and hydric soil data to determine the optimum depths to excavate the land at each site so as to create four different soil-hydrologic zones – the riparian/wetland types: Emergent marsh, Permanent palustrine emergent, Seasonal palustrine emergent and Riparian grass/shrub land. Our design enhanced wildlife habitat within the Fountain Creek riparian corridor. Floodway/floodplain modeling was completed to provide a basis for requesting a No Rise Certificate from the El Paso County Floodplain Administrator. Development of the wetland mitigation areas will require the removal of three to six ft of soil material to lower the plant root zone into the capillary fringe of the underlining water table. The existing A horizon in the proposed mitigation areas will be salvaged and stockpiled and used as topsoil (suitable plant growth media). The mitigation wetlands will be constructed and planted/seeded in 2016-2017.

Project dates: 2015-present

Original cost: \$292,000

Final cost: \$292,000

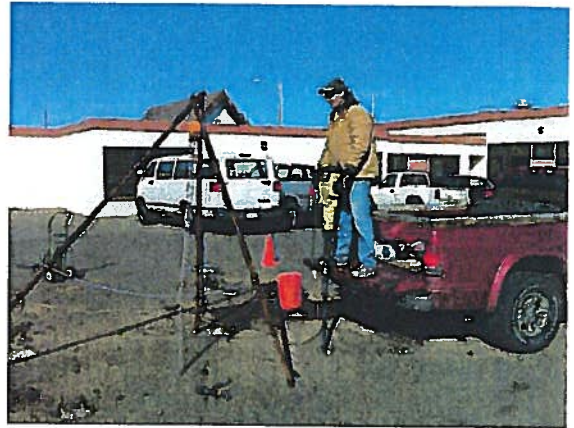
6.10 SMITH: ADAMS STATE COLLEGE LUST REMEDIATION (EVENT 3783)

Alamosa, Colorado

Adams State College, Erik van de Boogaard, (719) 587-7951

Adams State College had a gasoline release at its fueling area within its maintenance yard area. SMITH developed the most cost-effective and innovative approach in its proposal as compared to the many other proposals submitted and was selected by Adams State College to complete Phase III investigation and remediation activities starting in 2009. SMITH provided in-situ bioremediation services in

coordination with CDLE-DOPS guidelines. A feasibility analysis was conducted to select an appropriate remediation technology. A BTEX (benzene, toluene, ethylbenzene, and xylenes) plume existed from a former LUST that was removed from the maintenance facility. Several remediation contractors have worked at the site with varying degrees of success. SMITH's innovative approach involved injection of oxygen reducing compound (ORC) to complete the oxidation of the Light Non-Aqueous Phase Liquid (LNAPL) BTEX. SMITH used the risk-based modeling software program RISC4 to aid in the evaluation of existing human health risks associated with benzene concentrations in the groundwater down gradient from the source area. SMITH evaluated three remediation technologies for this project including excavation with disposal at an appropriate landfill, excavation and land "farming," and in-situ remediation using ORC. The advantages and disadvantages of these three remediation technologies were considered and provided the basis for the selection of ORC injection as the most feasible remediation technology for the site. The ORC injection of a pilot area was conducted in March 2009. SMITH's groundwater monitoring of the pilot injection area began in May 2009 and ended in August 2012. This quarterly monitoring entailed collecting groundwater samples from the 10 monitoring wells on site on a quarterly basis. ORC injections in the pilot test area provided beneficial effect in the BTEX contaminated area. Monitoring was conducted during this period and quarterly monitoring reports were provided to the CDLE-DOPS. Adams State College received its letter of No Further Action (NFA) needed in September 2012. SMITH completed the Reimbursement Fund application for the college to receive reimbursement from the State Reimbursement Fund by the summer of 2013.



Project dates: 2009-2012

Original cost: \$67,000

Final cost: \$67,000

6.11 SMITH: BURT SUBARU AND BURT TOYOTA – PHASE I AND II ENVIRONMENTAL SITE ASSESSMENT, AND REMEDIATION

Englewood, Colorado

Burt Dealerships, Hank Held, (303) 748-0001, bheld@burt.com

SMITH conducted a Phase II ESA and provided innovative site remediation services at the Burt Subaru facility in Englewood, CO starting in late 2005 and lasting through 2010. The purpose of the investigation was to address potential RECs and other concerns identified in a previous Phase I ESA of the property and develop a preliminary Remedial Action Plan (RAP). The first step of the RAP was to complete a Phase II ESA, which included a subsurface soil investigation, ACM survey, an investigation



of the fluorescent lighting for potential PCB-containing ballasts, and a records research to determine the disposition of a 5,000-gallon gasoline UST at the property. Results of the Phase II ESA identified a LUST release from the neighboring Toyota facility onto the Subaru Property. The RAP was revised based on these data and a remediation strategy developed – injection of Oxygen Reducing Compounds (ORC) and natural attenuation. In preparation for drilling and installing monitoring wells and injection wells, SMITH prepared a HASP. SMITH prepared a SAP and obtained temporary monitoring well permits from the Colorado Department of Water Resources. SMITH conducted a subsurface investigation of the two properties that included installing 13 monitoring wells and drilling 11 additional soil borings. Drilling included collecting split barrel soil samples and completing bore hole log descriptions. SMITH prepared a Site Characterization Report (SCR). Based on these data, SMITH developed an innovative, cost-effective Corrective Action Plan (CAP) for the Burt Subaru property. The innovative approach involved conducting a pilot test in the fall of 2007 and then full remediation starting in late 2007 that included the injection of ORC into the petroleum hydrocarbon plume to accelerate chemical breakdown of benzene, toluene, ethyl-benzene, and xylene in conjunction with a monitoring and natural attenuation (MNA). SMITH personnel used the accumulated site data collected during remediation to conduct risk-based modeling of the site to assess the long term potential for contaminant migration versus natural attenuation of the contaminant plume on site. SMITH used the risk-based modeling software program RISC4 to aid in the evaluation of human health risks associated with benzene concentrations in the groundwater down gradient from the source area. SMITH also completed the Reimbursement Fund application to obtain reimbursement for Burt Subaru from the State Reimbursement Fund. Burt Automotive Group sold the property to Summit Automotive in 2010. The MNA program was handed off to another consultant in 2010 when Burt sold the dealership to Western Summit Automotive.

Project dates: 2005-2010

Original cost: \$176,000

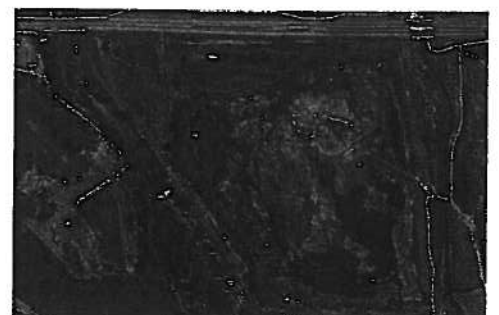
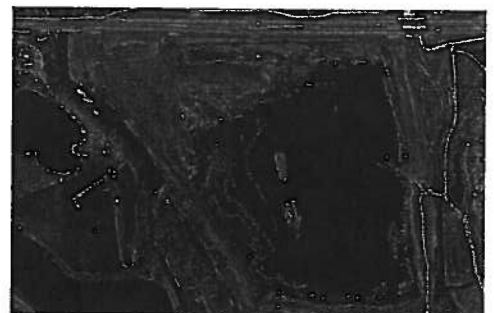
Final cost: \$176,000

6.12 LLG: FORMER RADIO WREN TOWER SITE – WESTERN MOBILE GRAVEL PIT/USFWS ECOLOGICAL RESTORATION

Fort Collins, Colorado

La Farge West, Inc., (Was Western Mobile, Inc.) Judy Schmell, (970) 223-2556

The original WREN tower site was later developed as an aggregate mine, and yet later restored by Western Mobile Corporation with strategies developed by a local team of research scientists and Jeff Lakey, Colorado State University landscape architecture professor. The primary strategy proposed seasonal management of water levels in the mined area to mimic natural ecological revegetation processes, such as managing for moist surface soils at key locations during the time of cottonwood seed rain to facilitate the potential return of cottonwood gallery forests near the Cache la Poudre River.



Project dates: 1993

Original cost: (Proprietary to client)

Final cost: (Proprietary to client)

6.13 BRENT WILKINS/TERRACON EXPERIENCE

Terracon has provided several geotechnical investigations and reports for the City of Cheyenne. Relevant project briefs for Brent Wilkins/Terracon are included on his resume (pgs. 19-20).

7.0 CONFLICT OF INTEREST

Neither SMITH nor its subconsultants is currently performing professional services for, or has an ongoing professional relationship with, any other person or legal entity whose interests may be adverse to the City of Cheyenne with respect to the scope of work to be performed, pursuant to this RFP, nor are there any other facts or circumstances present that create or may appear to create a conflict of interest.

8.0 OTHER INFORMATION

Attachment B-2, Acknowledgment of Addenda, and Attachment B-3, Use of Subconsultants, are included on the following pages.

REQUEST FOR PROPOSALS No.: RFP-12-16

PROJECT NAME: Lower Capitol Basin Sediment Trap/Wetland

"ATTACHMENT B-2"
ACKNOWLEDGMENT OF RECEIPT
OF RFP PACKET, SAMPLE AGREEMENT AND ADDENDA

The undersigned hereby acknowledges receipt of the City of Cheyenne Request for Proposals (RFP) packet for Lower Capitol Basin Sediment Trap/Wetland, RFP-12-16 including: Sample Professional Services Agreement, Scope of Work, Compensation Schedule, Project Schedule, and the following addenda issued during the procurement process.

Addendum #1:  dated 4/15/16

Addendum #2:  dated 4/20/16

Addendum #3: _____ dated _____

Failure to acknowledge receipt of the proposal packet, and all addenda may cause the proposal to be considered non-responsive to this solicitation.

Name of Proposer: Peter L. Smith

Signature of Proposer: 

Date: April 21, 2016

REQUEST FOR PROPOSALS No.: RFP-12-16

PROJECT NAME: Lower Capitol Basin Sediment Trap/Wetland

"ATTACHMENT B-3"

USE OF SUB-CONSULTANTS LIST

Sub-consultant Name/City/State	% of Value of Work	Work Tasks to be Assigned
Terracon Cheyenne, WY	2%	Geotechnical
Steil Surveying Services, LLC Cheyenne, WY	3%	Surveying
LLG International, LLC Denver, CO	6%	Landscape Architecture
CO State University, Soil & Water Testing Lab - Ft. Collins, CO	0.3%	Laboratory Analysis
Geocal, Inc. Centennial, CO	2%	Geotechnical and Environmental
Total Value of Subcontracts		

Name of Proposer: Peter L. Smith

Signature of Proposer: 

Date: April 21, 2016

9.0 PROJECT SPECIFIC INFORMATION

SMITH's scope of work described herein is consistent with the RFP's Exhibit A – Scope of Work. SMITH understands that this document will become part of the Professional Services Agreement.

10.0 CONSULTANT ELIGIBILITY FORM

Attachment B-4 is included on the following page.

REQUEST FOR PROPOSALS No.: RFP-12-16

PROJECT NAME: Lower Capitol Basin Sediment Trap/Wetland

"ATTACHMENT B-4"
CERTIFICATION REGARDING DEBARMENT, SUSPENSION
AND OTHER RESPONSIBILITY MATTERS

This contract is a covered transaction for purposes of 49 CFR Part 29. As such, the consultant is required to verify that none of the consultant, its principals, as defined at 49 CFR 29.995, or affiliates, as defined at 49 CFR 29.905, are excluded or disqualified as defined at 49 CFR 29.940 and 29.945.

The consultant is required to comply with 49 CFR 29, Subpart C and must include the requirement to comply with 49 CFR 29, Subpart C in any lower tier covered transaction it enters into.

By signing and submitting its bid or proposal, the bidder or proposer certifies as follows:

The certification in this clause is a material representation of fact relied upon by the City of Cheyenne. If it is later determined that the bidder or proposer knowingly rendered an erroneous certification, in addition to remedies available to the City of Cheyenne, the Federal Government may pursue available remedies, including but not limited to suspension and/or debarment. The bidder or proposer agrees to comply with the requirements of 49 CFR 29, Subpart C while this offer is valid and throughout the period of any contract that may arise from this offer. The bidder or proposer further agrees to include a provision requiring such compliance in its lower tier covered transactions.

Date: April 21, 2016

Signature: _____



Title: Vice President

11.0 NON-COLLUSION CERTIFICATION

Attachment B-5 is included on the following page.

REQUEST FOR PROPOSALS No.: RFP-12-16

PROJECT NAME: Lower Capitol Basin Sediment Trap/Wetland

"ATTACHMENT B-5"
NON-COLLUSION AFFIDAVIT

The undersigned duly authorized official of the proposer hereby certifies, to the best of her/his knowledge and belief, that:

That I am an officer or employee of the Smith Environmental & Engineering (proposing entity) having the authority to sign on the behalf of the corporation; and,

That the prices in the attached proposal were arrived at independently by Smith Environmental & Engineering (proposing entity) without collusion, consultation, communication, or any agreement, for the purpose of restricting competition as to any matter relating to such prices with any other proposer or with any other competitor regarding an understanding, or planned common course of action with any other vendor of materials, supplies, equipment, or service described in the RFP designed to limit independent proposals or competition; and

That unless otherwise required by law, the contents and prices contained in the proposal have not been communicated by Smith Environmental & Engineering (proposing entity) or its employees or agents to any person not an employee or agent of Smith Environmental & Engineering (proposing entity), or its surety on any bond furnished with the proposal, and will not be communicated to any such person prior to the official opening of the proposal; and,

That I have fully informed myself regarding the accuracy of the statements made in this affidavit.

Proposer: Peter L. Smith

Signature: _____



Title: Vice President

Date: April 21, 2016

12.0 RESTRICTIONS ON LOBBYING CERTIFICATION

Attachment B-6 is included on the following pages.

REQUEST FOR PROPOSALS No.: RFP-12-16

PROJECT NAME: Lower Capitol Basin Sediment Trap/Wetland

"ATTACHMENT B-6"
RESTRICTIONS ON LOBBYING CERTIFICATION

Byrd Anti-Lobbying Amendment, 31 U.S.C. 1352, as amended by the Lobbying Disclosure Act of 1995, P.L. 104-65 [to be codified at 2 U.S.C. § 1601, et seq.] - Contractors who apply or bid for an award of \$100,000 or more shall file the certification required by 49 CFR part 20, "New Restrictions on Lobbying." Each tier certifies to the tier above that it will not and has not used Federal appropriated funds to pay any person or organization for influencing or attempting to influence an officer or employee of any agency, a member of Congress, officer or employee of Congress, or an employee of a member of Congress in connection with obtaining any Federal contract, grant or any other award covered by 31 U.S.C. 1352. Each tier shall also disclose the name of any registrant under the Lobbying Disclosure Act of 1995 who has made lobbying contacts on its behalf with non-Federal funds with respect to that Federal contract, grant or award covered by 31 U.S.C. 1352. Such disclosures are forwarded from tier to tier up to the recipient.

APPENDIX A, 49 CFR PART 20—CERTIFICATION REGARDING LOBBYING

Certification for Contracts, Grants, Loans, and Cooperative Agreements

Pursuant to United States Public Law 101-121, Section 319, the undersigned duly authorized official of the proposer hereby certifies, to the best of her/his knowledge and belief, that:

1. No Federal appropriated funds have been paid or will be paid, by or on behalf of the undersigned, to any person for the purpose of influencing or attempting to influence an officer or employee of any agency, a Member of Congress, an officer or employee of Congress, or an employee of a Member of Congress in connection with the awarding of any Federal contract, the making of any Federal grant, the making of any Federal loan, the entering into of any cooperative agreement, and the extension, continuation, renewal, amendment, or modification of any Federal contract, grant, loan, or cooperative agreement.
2. If any funds other than Federal appropriated funds have been paid or will be paid to any person for making lobbying contacts to an officer or employee of any agency, a Member of Congress, an officer or employee of Congress, or an employee of a Member of Congress, in connection with this Federal contract, grant, loan, or cooperative agreement, the undersigned shall complete and submit a Standard Form-LLL, "Disclosure Form to Report Lobbying", in accordance with its instructions [as amended by "Government wide Guidance for New Restrictions on Lobbying," 61 Fed Reg. 1413 (1/19/96). Note: Language in paragraph (2) herein has been modified in accordance with Section 10 of the Lobbying Disclosure Act of 1995 (P.L. 104-65, to be codified at 2 U.S.C. 1601, et seq.)]
3. The undersigned duly authorized official shall require that the language of this certification be included in the award documents for all subawards at all tiers (including subcontracts, subgrants, and contracts under grants, loans, and cooperative agreements) and that all subrecipients shall certify and disclose accordingly.

REQUEST FOR PROPOSALS No.: RFP-12-16

PROJECT NAME: Lower Capitol Basin Sediment Trap/Wetland

This Certification is a material representation of fact upon which reliance was placed when this transaction was made or entered into. Submission of this Certification is a prerequisite for making or entering into this transaction, imposed by 31, USC Section 1352 (as amended by the Lobbying Disclosure Act of 1995). Any proposer (person) who fails to file the required certification shall be subject to a civil penalty of not less than ten thousand dollars (\$10,000) and not more than one hundred thousand dollars (\$100,000) for each such failure to file.

[Note: Pursuant to 31, USC Section 1352(c)(1)-(2)(A), any person who makes a prohibited expenditure or fails to file or amend a required certification or disclosure shall be subject to a civil penalty of not less than ten thousand dollars (\$10,000) and not more than one hundred thousand dollars (\$100,000) for each such failure to file.

The Proposer, Smith Environmental & Engineering certifies or affirms the truthfulness and accuracy of each statement of its certification and disclosure, if any. In addition, the Proposer understands and agrees that the provisions of 31 USC A 3801, et seq., apply to this certification and disclosure, if any.

Proposer: Peter L. Smith

Signature: 

Title: Vice President

Date: April 21, 2016

APPENDIX 1: LETTER OF COMMENDATION

A letters of commendation SMITH received for a previous project is included on the following page.

I. Banning Lewis Ranch, from Greg Barbuto

March 21, 2013

Subject: Letter of Recommendation – Smith Environmental and Engineering

To Whom It May Concern,

Smith Environmental and Engineering (SMITH) has provided a wide range of environmental planning, design, permitting, and construction services at Banning Lewis Ranch (BLR) for the new owner, Oakwood Homes, and did so for the former owner, Banning Lewis Ranch Management Company (BLRMC). They started providing services in 2003 and have provided nearly \$1 million of consulting and construction services. I recently left the employ of Oakwood Homes where I was the Development Manager at BLR. Before Oakwood Homes I served as Development Manager at BLRMC. I utilized the services of SMITH for over six years while I was working at BLR.

The range of SMITH's services included threatened and endangered species (Preble's meadow jumping mouse, Colorado butterfly plant, and Ute ladies'-tresses orchid) habitat suitability assessments; wetland mitigation site (30 ac) planning and design; wetland permit revisions; trail and boardwalk design; wetland mitigation site construction; liaison services with the U.S. Army Corps of Engineers; preparation of annual wetland mitigation site monitoring reports to satisfy Individual Permit requirements; stormwater compliance advice; water rights evaluations; ground water investigations and permitting; cultural resources evaluations; gravel pit development feasibility assessments; and other general environmental planning services as needed for a large residential development. As part of the threatened and endangered species services, SMITH reevaluated the potential for Preble's meadow jumping mouse habitat to occur on BLR. This included reviewing previous studies and trapping efforts and SMITH's professional judgment as to whether or not PMJM habitat existed on BLR. They determined it was not, obtained the U.S. Fish and Wildlife Services' concurrence, and helped BLRMC avoid the cost of trapping to determine absence of PMJM.

For over two years SMITH met weekly with the BLR project team to coordinate on all environmental issues. During this time I had the good pleasure working closely with SMITH staff on many challenging environmental issues. I have appreciated their professionalism and high degree of understanding they brought regarding environmental permitting, design, studies, and construction in the Colorado Springs Area. Their services were provided on time and at a reasonable price. They met or exceeded regulatory agency and Banning Lewis Ranch quality standards. I would heartily recommend SMITH to any company or governmental agency in need of environmental planning, design, permitting and construction in the El Paso County area. They are a great firm to have working for you. We really appreciated all that they have done at BLR.

If you have any questions, please feel free to contact me at GBarbuto@taylormorrison.com.

Sincerely,



Greg Barbuto, P.E., C.C.M.
Land Development Director
Taylor Morrison (Previously with Oakwood Homes)

[illegible]



TASK 1: PRELIMINARY DESIGN PHASE SERVICES

[illegible]

TASK 3: BIDDING AND CONSTRUCTION PHASE SERVICES

	\$	12,160.00	\$	34,104.00	\$	7,644.00	\$	10,248.00	\$	8,936.00	\$	23,520.00	\$	19,956.00	\$	33,440.00	\$	3,300.00	\$	8,448.00	\$	522.00	\$	-	\$	-	\$	8,800.00				
Sub-Totals:																																
	\$	35,500.00	\$	85,750.00	\$	25,676.00	\$	45,528.00	\$	16,660.00	\$	23,520.00	\$	37,284.00	\$	25,536.00	\$	23,000.00	\$	32,768.00	\$	1,311.00	\$	-	\$	200.00	\$	1,750.00	\$	3,000.00	\$	22,000.00

Overall Project Total:

The following plans reflect the cost proposal in the attached spreadsheet dated 5-17-2016

Project Includes the Following:

1. Design of Treatment Wetlands
2. Native ecosystem only – 5 ecosystems -- three upland: tall grassed prairie, mixed grassed prairie and short grassed prairie; palustrine scrub shrub; and palustrine emergent: emergent marsh and permanent wetland.
3. Depth to groundwater investigation
4. Lowest channel into capillary fringe to maintain wetland hydrology
5. Planted forebay side slopes and possibly smaller forebay with retaining walls
6. Diversion structure
7. Bypass pipe connections
8. Topsoil investigation
9. All soil stays on site which includes the design of capping system
10. Includes up to 3 acres for soil pile
11. Includes 3 interpretive signs and seating benches and optional exercise stations
12. Includes multimodal trail
13. Connecting trail to Ames – no other improvements outside of the 2 acres site
14. Pedestrian bridge
15. Parking lot design
16. Sawtooth weir from forebay
17. Better estimate of construction services
18. Public ingress and egress
19. The design of a single phase recirculating pump with controls.

Project Does Not Include the Following:

1. Complex irrigation system design
2. Additional contaminant testing of soils